

The arithmetic geometry of elliptic curves (DRAFT)

Math 714, Fall 2026

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Contents

Preface	7
Prerequisites	7
Notation and conventions	7
A road map	8
Origin of these notes	8
Acknowledgements	8
AI disclosure	9
Chapter 1. Elliptic curves as Riemann surfaces 2 of 8 sections finished	11
1.1. Riemann surfaces	11
1.2. Complex tori	13
1.3. Isogenies DRAFT	15
1.4. Meromorphic functions DRAFT	18
1.5. Invariant differentials DRAFT	19
1.6. Projective plane curves as Riemann surfaces DRAFT	21
1.7. The uniformization theorem DRAFT	22
1.8. Isomorphism classes of complex tori DRAFT	22
Chapter 2. Algebraic varieties 0 of 4 sections finished	25
2.1. Varieties and their points DRAFT	25
2.2. Base extension and geometric properties DRAFT	28
2.3. Morphisms and rational maps DRAFT	28
2.4. Dimension, regularity and smoothness DRAFT	29
Chapter 3. Algebraic curves 0 of 7 sections finished	31
3.1. Closed points and local rings DRAFT	31
3.2. Morphisms of curves DRAFT	32
3.3. Divisors DRAFT	34
3.4. Differentials DRAFT	36
3.5. The Riemann–Roch theorem DRAFT	37
3.6. The Hurwitz formula DRAFT	38
3.7. Plane curves DRAFT	39
Chapter 4. Elliptic curves as algebraic curves 1 of 8 sections finished	43
4.1. Weierstrass equations	43
4.2. The group law DRAFT	49
4.3. Isogenies DRAFT	53
4.4. The invariant differential DRAFT	56
4.5. The Tate module DRAFT	58
4.6. The Weil pairing DRAFT	60
4.7. Endomorphisms and automorphisms DRAFT	62
4.8. Selmer elliptic curves DRAFT	65
Chapter 5. Elliptic curves over finite fields 0 of 5 sections finished	69
5.1. The Frobenius automorphism DRAFT	69

5.2.	Hasse's bound	DRAFT	69
5.3.	The Weil conjectures	DRAFT	70
5.4.	The endomorphism ring	DRAFT	76
5.5.	The Cartier operator and the Hasse invariant	DRAFT	76
Chapter 6.	Elliptic curves over local fields	In preparation	81
Chapter 7.	Galois cohomology	In preparation	83
Chapter 8.	The Mordell–Weil theorem	In preparation	85
Chapter 9.	Integral points	In preparation	87
Chapter 10.	Twisting, Selmer, and Tate–Shafarevich	In preparation	89
Chapter 11.	Complex multiplication	In preparation	91
Chapter 12.	The Tate curve	In preparation	93
Chapter 13.	The L -function and BSD	In preparation	95
Chapter 14.	The Sato–Tate conjecture	In preparation	97
Chapter 15.	Introduction to modular curves	In preparation	99
	Glossary		101
	References		105

About these notes. These notes accompany Math 714 and are a work in progress: sections are written and revised during the semester, and some are not yet finished.

Sections marked **DRAFT** are written but not yet proofread, and sections marked **TODO** still have parts to be written; beside each mark is the date the section last changed.

Last updated 24 Sep 2026.

Preface

These are the notes for Math 714, a first graduate course on the arithmetic of elliptic curves. The main reference is [Sil09], which we follow closely from Chapter 4 on.

Prerequisites

We assume the first-year graduate courses in algebra and complex analysis. More precisely, the reader should be comfortable with:

- **Algebra.** Finite and infinite Galois theory, including profinite groups and the Krull topology; modules over a principal ideal domain; basic commutative algebra (localization, integral extensions, discrete valuation rings).
- **Algebraic number theory.** Number fields and p -adic fields and their rings of integers; Dedekind domains; the decomposition of primes; and the finiteness of the class group and of the unit rank.
- **Complex analysis.** Holomorphic and meromorphic functions, Liouville's theorem, the residue theorem, and analytic continuation. Chapter 1 builds the theory of Riemann surfaces it needs.

No algebraic geometry is assumed. In the spirit of [Poo06], we use the language of scheme theory and explain how to think concretely about scheme-theoretic constructions in terms of commutative algebra. Chapters 2 and 3 collect what we use from the theory of algebraic varieties and algebraic curves.

Notation and conventions

Rings are commutative with 1. We write $\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$ and $\mathbb{C}$ as usual, $\mathbb{F}_q$ for the field with q elements, $\mathbb{Z}_p$ and $\mathbb{Q}_p$ for the p -adic integers and numbers, and $\#S$ or $|S|$ for the cardinality of a set S .

Fields. Throughout, k is an arbitrary field; in particular it need not be perfect. We fix an algebraic closure $\bar{k}$, and write k^{sep} for the separable closure of k in $\bar{k}$. We write $\text{Gal}_k := \text{Gal}(k^{\text{sep}}/k)$ for the absolute Galois group of k . The group Gal_k acts on $X(k^{\text{sep}})$; since restriction $\text{Aut}(\bar{k}/k) \rightarrow \text{Gal}_k$ is an isomorphism, this also determines the action on $X(\bar{k})$.

Varieties. A k -variety is a separated scheme of finite type over k . We do not assume varieties to be integral or reduced. A curve is an integral k -variety of dimension 1, and a curve is nice if it is smooth, projective, and geometrically irreducible.

Coordinates. Upper case letters are homogeneous coordinates on projective space, lower case letters are affine coordinates, and the two are related by dehomogenization. On the standard patch $U_0 = \{X_0 \neq 0\}$ of $\mathbb{P}^n$ we write $x_i = X_i/X_0$; for a plane curve in $\mathbb{P}^2$ with coordinates X, Y, Z the affine chart is $Z \neq 0$, with $x = X/Z$ and $y = Y/Z$. The same rule applies to the names of polynomials: F is homogeneous and f is its dehomogenization.

Elliptic curves. For an elliptic curve E over k and an integer m , $E[m]$ is the group scheme given by the kernel of multiplication by m , and $E[m](\bar{k})$ is the m -torsion subgroup of $E(\bar{k})$. Likewise, $\mu_m(\bar{k})$ is the group of m -th roots of unity in $\bar{k}^\times$. For a nice curve C we write $\text{Div } C$ and $\text{Pic } C$ for the divisor group and the Picard group, and $\text{Div}^0 C$, $\text{Pic}^0 C$ for their degree-zero parts. Isogenies in $\text{Hom}(E_1, E_2)$, $\text{End}(E)$ and $\text{Aut}(E)$ are defined over the ground field, and $\text{Hom}(E_{1,\bar{k}}, E_{2,\bar{k}})$ denotes those defined over $\bar{k}$. For N prime to the characteristic, $\bar{\rho}_{E,N}$ is the mod N Galois representation on $E[N](k^{\text{sep}})$, and $\rho_{E,\ell}$ is the ℓ -adic one on the Tate module $T_\ell(E)$.

Frobenius. In characteristic p , F is the relative Frobenius morphism and V is its dual, the Verschiebung; F^r is the p^r -th power Frobenius $E \rightarrow E^{(p^r)}$, and over $\mathbb{F}_q$ with $q = p^e$ the q -th power Frobenius endomorphism is F^e , also written F when the ground field is fixed.

The table below, based on [Poo06, Warning 1.9.6], compares our notation with Silverman's, for a nice curve C and elliptic curves E_1, E_2 over k .

Our notation	Notation in [Sil09]
k arbitrary field	k perfect ($k^{\text{sep}} = \bar{k}$)
$\text{Div } C$	$\text{Div}_k(C)$
$\text{Div } C_{k^{\text{sep}}}$	$\text{Div}(C)$
$\text{Pic } C$	No notation for this.
$\text{Pic } C_{k^{\text{sep}}}$	$\text{Pic}(C)$
$(\text{Pic } C_{k^{\text{sep}}})^{\text{Gal}_k}$	$\text{Pic}_k(C)$
$\text{Hom}(E_1, E_2)$	$\text{Hom}_k(E_1, E_2)$
$\text{Hom}(E_{1,k^{\text{sep}}}, E_{2,k^{\text{sep}}})$	$\text{Hom}(E_1, E_2)$
$\text{End}(E), \text{Aut}(E)$	$\text{End}_k(E), \text{Aut}_k(E)$
$\text{End}(E_{k^{\text{sep}}})$	$\text{End}(E)$
$E[m](k^{\text{sep}})$	$E[m]$
$\text{Gal}_k = \text{Gal}(k^{\text{sep}}/k)$	$G_{\bar{k}/k} = \text{Gal}(\bar{k}/k)$
F^r, V^r	$\phi_r, \hat{\phi}_r$

A road map

The notes fall into four parts.

- **Foundations (Chapters 1–4).** Chapter 1 treats elliptic curves over $\mathbb{C}$ as complex tori $\mathbb{C}/\Lambda$, as a source of geometric intuition for what follows. Chapters 2 and 3 develop varieties and curves over an arbitrary field, and Chapter 4 uses them to set up the general theory: Weierstrass equations, the group law, isogenies, the Tate module and the Weil pairing.
- **Local theory (Chapters 5 and 6).** Elliptic curves over finite fields, including the Hasse bound, and over local fields: minimal models, reduction and the formal group.
- **Global theory (Chapters 7–10).** Galois cohomology, the Mordell–Weil theorem, integral points; twists, Selmer groups, and the Tate–Shafarevich group.
- **Further topics (Chapters 11–15).** Complex multiplication, the Tate curve, the L -function and the conjecture of Birch and Swinnerton-Dyer, the Sato–Tate conjecture, and an introduction to modular curves.

A thread running through the notes is the comparison of an elliptic curve with the multiplicative group $\mathbb{G}_m$: over $\mathbb{C}$, the torus $\mathbb{C}/\Lambda$ against $\mathbb{C}^\times = \mathbb{C}/2\pi i\mathbb{Z}$; over an arbitrary field, their invariant differentials, torsion points and Tate modules; and in Chapter 12, the Tate curve, which realizes an elliptic curve over a p -adic field as a quotient of $\mathbb{G}_m$.

Chapter 1 can be read independently of Chapters 2 and 3, and Chapter 4 relies on Chapters 2 and 3. After Chapter 4, the main line runs through Chapters 5 to 8; Chapters 9 and 10 build on Chapter 8, and each of the further topics can be read after the chapters it cites. The worksheets for the problem sessions, with their solutions, are linked from the web page of the notes; each one says which sections it draws on.

Origin of these notes

I taught Math 714 (The Arithmetic of Elliptic Curves) at UMass in the fall of 2026. I prepared hand written notes ahead of each section, scanned them, and used Claude to transform them into this format.

Acknowledgements

This text owes an enormous intellectual debt to Silverman's book [Sil09] and Poonen's notes [Poo06]. I also benefited greatly from Milne's excellent book [Mil20].

AI disclosure

Throughout this project, we used Claude code for writing code, carrying out computations, formatting, and proofreading. All mathematical results (beyond the draft stage) in this manuscript, as well as the AI-generated code and computations verifying them, have been independently verified by the author, who takes full responsibility for the accuracy and content of this work.

CHAPTER 1

Elliptic curves as Riemann surfaces

Sources. This chapter follows [Mir95] (parts of chapters I, II and IV) and [DS05] (parts of Chapter I).

The main goal of this chapter is to introduce the category of **complex tori** and to develop their theory from the complex analytic point of view. Students without previous experience in algebraic geometry can use it as a source of geometric intuition when we turn to the algebraic theory later on.

1.1. Riemann surfaces

Intuitively, a Riemann surface is a topological space that looks like $\mathbb{C}$ near each of its points. We make this intuition more precise.

DEFINITION 1.1.1. Let X be a topological space.

(1) A **chart** in X is a homeomorphism $\phi: U \rightarrow V$ between open subsets $U \subseteq X$, $V \subseteq \mathbb{C}$.

(2) For any two charts ϕ_1 and ϕ_2 we obtain two **transition maps**

$$\phi_2 \circ \phi_1^{-1}: \phi_1(U_1 \cap U_2) \rightarrow \phi_2(U_1 \cap U_2), \quad \phi_1 \circ \phi_2^{-1}: \phi_2(U_1 \cap U_2) \rightarrow \phi_1(U_1 \cap U_2).$$

(3) Two charts ϕ_1 and ϕ_2 are **compatible** if either $U_1 \cap U_2 = \emptyset$, or the transition maps are holomorphic.

(4) A **complex structure** on X is a family $\{\phi_i\}_i$ of compatible charts such that $X = \bigcup_i U_i$.

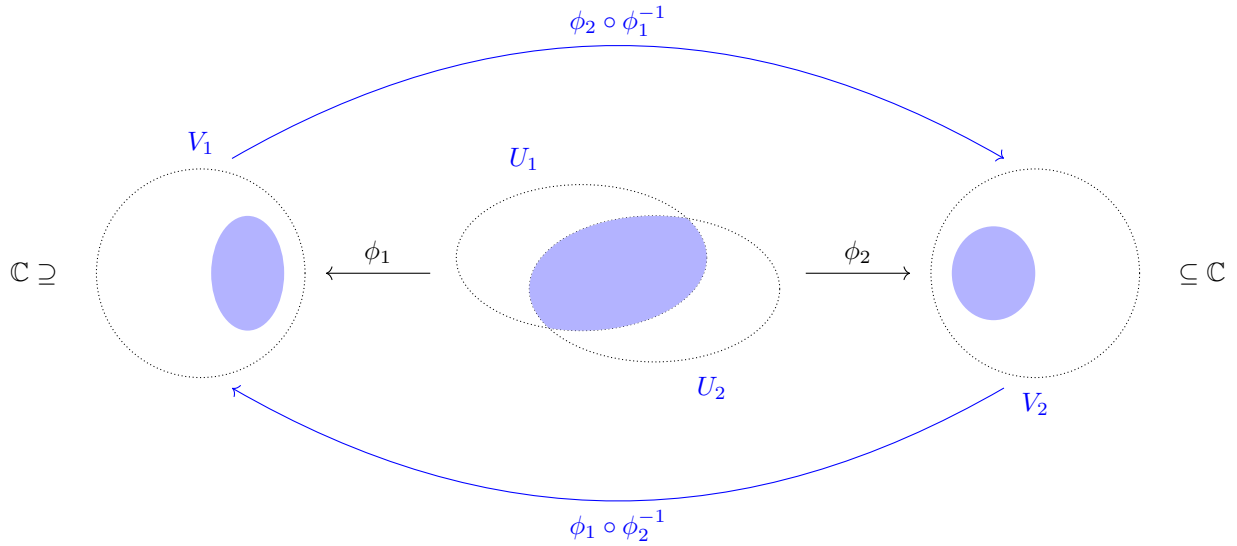


FIGURE 1.1.1. Two charts on overlapping open sets, and the transition maps between their images.

A topological space is **connected** if it is not the union of two disjoint non-empty open subsets, and **Hausdorff** if any two distinct points of the space have disjoint open neighborhoods.

DEFINITION 1.1.2. A **Riemann surface** is a connected Hausdorff topological space with a complex structure.

Recall that a topological space is [second countable](#) if it admits a countable basis for its topology. For Riemann surfaces this is automatic, by a theorem of Radó [For91, Theorem 23.3], but it is part of the definition of a manifold in general: the long line is connected, Hausdorff and locally homeomorphic to $\mathbb{R}$, and not second countable.

The simplest examples are open subsets of $\mathbb{C}$.

EXAMPLE 1.1.3. Every connected open $U \subseteq \mathbb{C}$ is a Riemann surface, with the complex structure consisting of the single chart $\text{id}: U \rightarrow U$.

More interestingly, the one-point compactification of $\mathbb{C}$ is a Riemann surface.

EXAMPLE 1.1.4. Define

$$\mathbb{P}^1(\mathbb{C}) := (\mathbb{C}^2 - \{(0, 0)\}) / \mathbb{C}^\times,$$

with the quotient topology. This is the [Riemann sphere](#), named after the sphere because it is isomorphic to the sphere S^2 with the complex structure given by stereographic projection (Example 1.1.9). A point of $\mathbb{P}^1(\mathbb{C})$ is a line through the origin:

$$(X_0 : X_1) := \{(\lambda X_0, \lambda X_1) : \lambda \in \mathbb{C}^\times\}.$$

The subsets $U_i := \{(X_0 : X_1) \in \mathbb{P}^1(\mathbb{C}) : X_i \neq 0\}$, for $i = 0, 1$, are open, and $\mathbb{P}^1(\mathbb{C}) = U_0 \cup U_1$. The maps

$$\phi_0: U_0 \rightarrow \mathbb{C}, \quad (X_0 : X_1) \mapsto X_1/X_0, \quad \phi_1: U_1 \rightarrow \mathbb{C}, \quad (X_0 : X_1) \mapsto X_0/X_1,$$

are homeomorphisms. They are compatible: $\phi_0(U_0 \cap U_1) = \mathbb{C}^\times$, and on it

$$(\phi_1 \circ \phi_0^{-1})(w) = \phi_1(1 : w) = 1/w$$

is holomorphic. We identify $\mathbb{C}$ with U_0 via ϕ_0 , and write $\infty := (0 : 1)$ for the one remaining point of $\mathbb{P}^1(\mathbb{C})$.

We use charts to extend the notion of a holomorphic map between open subsets of $\mathbb{C}$ to Riemann surfaces.

DEFINITION 1.1.5. A [holomorphic map](#) of Riemann surfaces is a continuous function $f: X \rightarrow Y$ such that for any two charts $\phi_1: U_1 \rightarrow V_1$ of X and $\psi_2: U_2 \rightarrow V_2$ of Y with $U_1 \cap f^{-1}(U_2) \neq \emptyset$, the map

$$\psi_2 \circ f \circ \phi_1^{-1} \Big|_{\phi_1(U_1 \cap f^{-1}(U_2))}$$

is holomorphic.

Figure 1.1.2 shows the maps in Definition 1.1.5.

DEFINITION 1.1.6. An [isomorphism](#) of Riemann surfaces is a bijective holomorphic map with holomorphic inverse.

EXAMPLE 1.1.7. The charts of a Riemann surface are isomorphisms onto their images.

EXAMPLE 1.1.8. Let $\mathbb{H} := \{z \in \mathbb{C} : \text{Im}(z) > 0\}$ denote the [upper half plane](#), and $\mathbb{D} := \{z \in \mathbb{C} : |z| < 1\}$ the [open unit disk](#). The map

$$\phi: \mathbb{H} \rightarrow \mathbb{D}, \quad z \mapsto \frac{z - i}{z + i},$$

is an isomorphism of Riemann surfaces.

EXAMPLE 1.1.9. Let $S^2 := \{x \in \mathbb{R}^3 : x_1^2 + x_2^2 + x_3^2 = 1\}$ be the unit sphere, with north pole $\mathbf{n} := (0, 0, 1)$ and south pole $\mathbf{s} := (0, 0, -1)$. The stereographic projections

$$\sigma_{\mathbf{n}}: S^2 - \{\mathbf{n}\} \rightarrow \mathbb{C}, \quad x \mapsto \frac{x_1 + ix_2}{1 - x_3}, \quad \sigma_{\mathbf{s}}: S^2 - \{\mathbf{s}\} \rightarrow \mathbb{C}, \quad x \mapsto \frac{x_1 - ix_2}{1 + x_3},$$

are homeomorphisms. Since $x_1^2 + x_2^2 = 1 - x_3^2$, we have $\sigma_{\mathbf{n}}(x)\sigma_{\mathbf{s}}(x) = 1$ for $x \neq \mathbf{n}, \mathbf{s}$. So $\sigma_{\mathbf{n}}(S^2 - \{\mathbf{n}, \mathbf{s}\}) = \mathbb{C}^\times$, and on it

$$(\sigma_{\mathbf{s}} \circ \sigma_{\mathbf{n}}^{-1})(w) = 1/w$$

is holomorphic. Thus $\{\sigma_{\mathbf{n}}, \sigma_{\mathbf{s}}\}$ is a complex structure, and S^2 , a connected subspace of $\mathbb{R}^3$, is a Riemann surface.

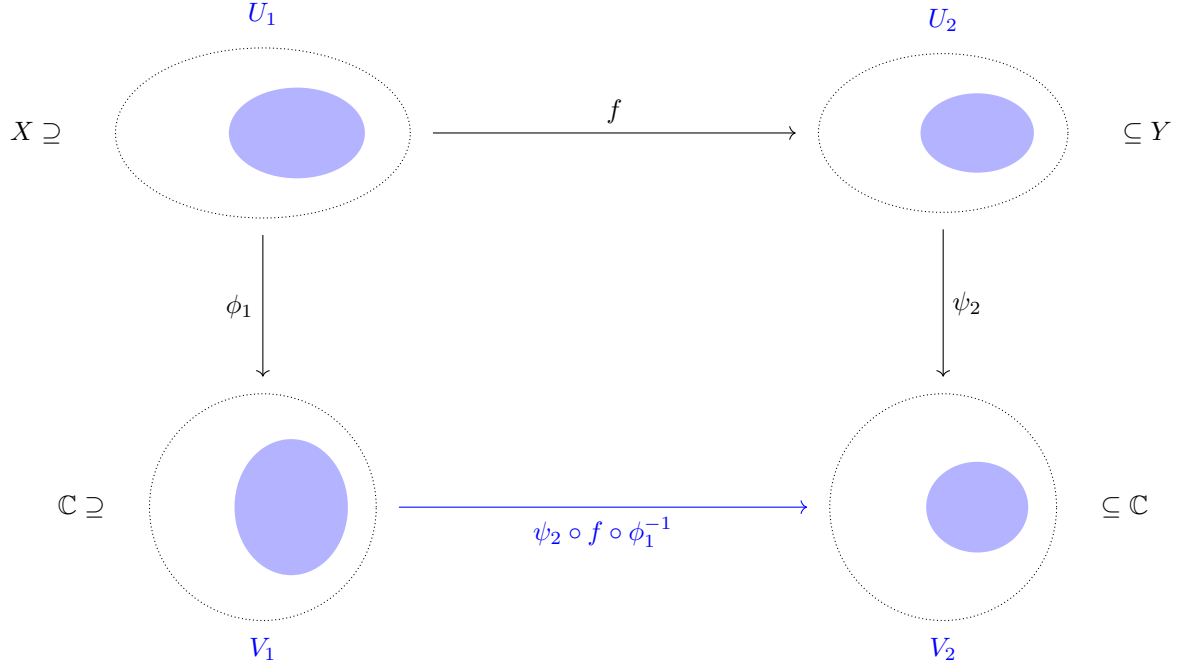


FIGURE 1.1.2. A holomorphic map read in charts: $\psi_2 \circ f \circ \phi_1^{-1}$ on $\phi_1(U_1 \cap f^{-1}(U_2))$.

Define $F: S^2 \rightarrow \mathbb{P}^1(\mathbb{C})$ by $F(x) := (1 : \sigma_{\mathbf{n}}(x))$ for $x \neq \mathbf{n}$, and $F(\mathbf{n}) := \infty$. It is a bijection, since $\sigma_{\mathbf{n}}$ is a bijection onto $\mathbb{C}$ and $U_0 = \mathbb{P}^1(\mathbb{C}) - \{\infty\}$. In the charts of Example 1.1.4,

$$\phi_0 \circ F \circ \sigma_{\mathbf{n}}^{-1} = \text{id}_{\mathbb{C}}, \quad \phi_1 \circ F \circ \sigma_{\mathbf{s}}^{-1} = \text{id}_{\mathbb{C}},$$

the second because $1/\sigma_{\mathbf{n}} = \sigma_{\mathbf{s}}$ away from $\mathbf{n}$ and $\mathbf{s}$, while $\sigma_{\mathbf{s}}(\mathbf{n}) = 0 = \phi_1(\infty)$. Hence F and F^{-1} are holomorphic, and F is an isomorphism of Riemann surfaces.

Recall that a connected topological space X is **simply connected** if, for any base point $x \in X$, the **fundamental group** $\pi_1(X, x)$ is trivial. In other words, simply connected spaces are those in which any loop can be continuously deformed to a point within X .

THEOREM 1.1.10 (Uniformization theorem). *Every simply connected Riemann surface is isomorphic to exactly one of $\mathbb{C}$, $\mathbb{P}^1(\mathbb{C})$, or $\mathbb{D}$ [For91, Section 27].*

So far, we have only seen examples of simply connected Riemann surfaces. One cheap way of ruining the simply connected property is by puncturing. Indeed, $\mathbb{C}^\times$ is a Riemann surface by Example 1.1.3, and $\pi_1(\mathbb{C}^\times, 1) \cong \mathbb{Z}$. We now turn to more interesting examples of surfaces that are not simply connected: donuts.

1.2. Complex tori

DEFINITION 1.2.1. A **lattice** in $\mathbb{C}$ is a subgroup $\Lambda \subseteq \mathbb{C}$, **free** of **rank** 2, that spans $\mathbb{C}$ as an $\mathbb{R}$ -vector space.

By definition, every lattice Λ admits a $\mathbb{Z}$ -basis that is also an $\mathbb{R}$ -basis of $\mathbb{C}$; that is, a pair of elements $\omega_1, \omega_2 \in \mathbb{C}$ such that $\Lambda = \mathbb{Z}\omega_1 \oplus \mathbb{Z}\omega_2$ and

$$\Lambda \otimes_{\mathbb{Z}} \mathbb{R} \cong \mathbb{R}\omega_1 \oplus \mathbb{R}\omega_2 = \mathbb{C}.$$

It follows that Λ is a discrete subset of $\mathbb{C}$.

EXAMPLE 1.2.2. $\Lambda_i := \mathbb{Z} \oplus i\mathbb{Z}$ is a lattice, the underlying group of the ring $\mathbb{Z}[i]$ of **Gaussian integers**.

EXAMPLE 1.2.3. $\Lambda_\omega := \mathbb{Z} \oplus \omega\mathbb{Z}$ for $\omega = -\frac{1}{2} + \frac{\sqrt{-3}}{2}$ a primitive cube root of 1 is a lattice, the underlying group of the ring $\mathbb{Z}[\omega]$ of Eisenstein integers.

Figure 1.2.1 shows both lattices.

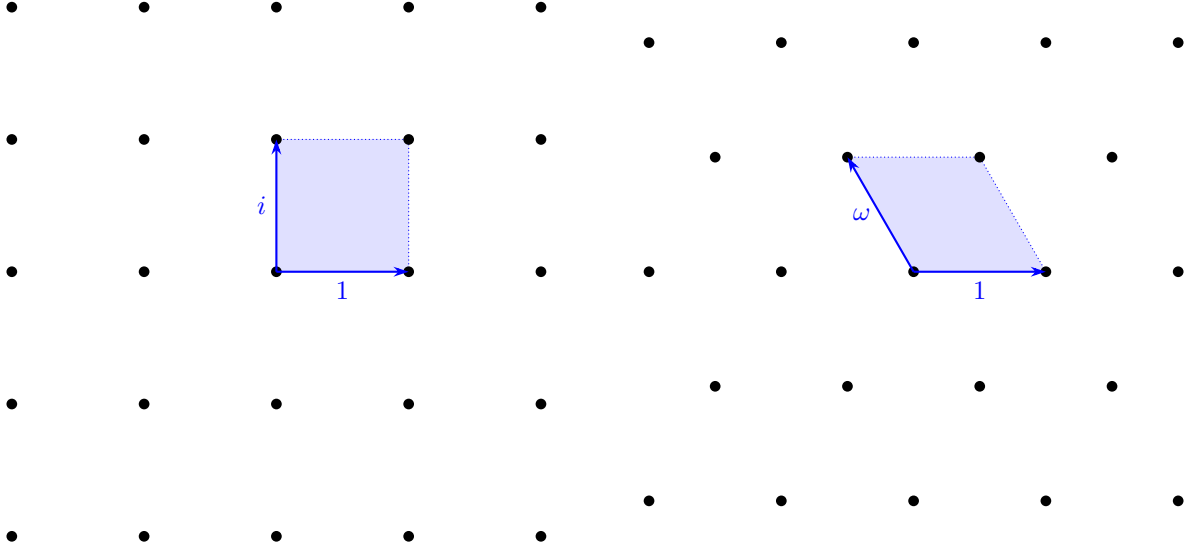


FIGURE 1.2.1. The lattices Λ_i (left) and Λ_ω (right), with the $\mathbb{Z}$ -bases $\{1, i\}$ and $\{1, \omega\}$ in blue and their fundamental parallelograms shaded.

Any lattice $\Lambda \subseteq \mathbb{C}$ gives rise to a Riemann surface, namely the quotient group

$$\mathbb{C}/\Lambda := \{z + \Lambda : z \in \mathbb{C}\}$$

with the quotient topology. We call $\mathbb{C}/\Lambda$ a **complex torus**.

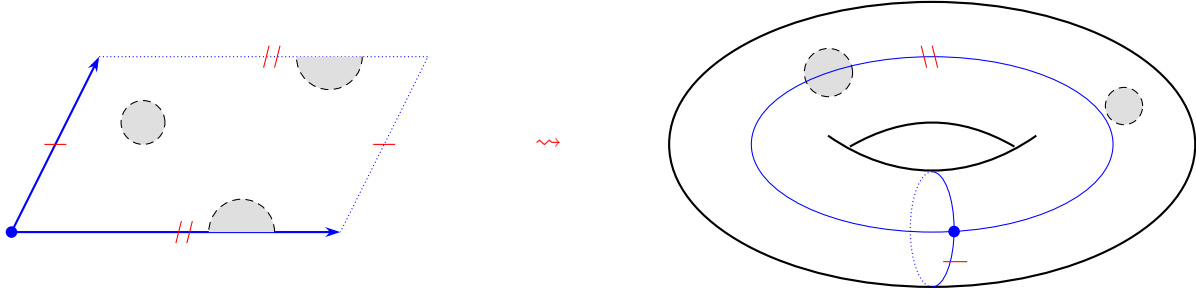


FIGURE 1.2.2. Gluing opposite sides of a fundamental parallelogram gives the torus $\mathbb{C}/\Lambda$: the two pairs of sides become the two circles marked with the same ticks, and disks in the parallelogram map to disks on the torus.

LEMMA 1.2.4. $\mathbb{C}/\Lambda$ is a compact Riemann surface.

PROOF. Consider the quotient map

$$\pi: \mathbb{C} \rightarrow \mathbb{C}/\Lambda.$$

It is continuous by the definition of the quotient topology, and open: for $U \subseteq \mathbb{C}$ open, $\pi^{-1}(\pi(U)) = \bigcup_{\lambda \in \Lambda} (U + \lambda)$ is open. Since π is continuous and surjective, $\mathbb{C}/\Lambda$ is connected, as $\mathbb{C}$ is. It is Hausdorff because Λ is discrete: if $z - w \notin \Lambda$, choose $\varepsilon > 0$ with $|z - w - \lambda| > 2\varepsilon$ for every $\lambda \in \Lambda$; then the images under

π of the ε -disks around z and w are disjoint open neighborhoods of $\pi(z)$ and $\pi(w)$. Let $\Lambda = \mathbb{Z}\omega_1 \oplus \mathbb{Z}\omega_2$, and consider the **fundamental parallelogram**

$$\mathcal{F}_\Lambda := \{t_1\omega_1 + t_2\omega_2 : 0 \leq t_1, t_2 < 1\}.$$

Its closure $\overline{\mathcal{F}_\Lambda}$ is compact, and $\mathbb{C}/\Lambda = \pi(\overline{\mathcal{F}_\Lambda})$, so $\mathbb{C}/\Lambda$ is compact.

For the complex structure, note that no two distinct points of the open parallelogram $\mathcal{F}_\Lambda^\circ$ differ by an element of Λ , so π restricts to a homeomorphism from $\mathcal{F}_\Lambda^\circ$ onto its image, and likewise on every translate $z + \mathcal{F}_\Lambda^\circ$. Hence every chart $\phi: U \rightarrow V$ with $U \subseteq z + \mathcal{F}_\Lambda^\circ$ descends to a chart $\tilde{\phi}: \pi(U) \rightarrow V$ with $\tilde{\phi} \circ \pi = \phi$ on U .

$$\begin{array}{ccc} U & \xrightarrow{\phi} & V \\ \pi \downarrow \wr & & \downarrow \tilde{\phi} \\ \pi(U) & \xrightarrow{\tilde{\phi}} & V \end{array}$$

The translates $\pi(z + \mathcal{F}_\Lambda^\circ)$ cover $\mathbb{C}/\Lambda$: translating is needed to reach the image of the boundary of the parallelogram. Two such charts are compatible because, on the overlap, $(\pi|_{U_2})^{-1} \circ \pi|_{U_1}$ sends u to $u + \lambda$ with $\lambda \in \Lambda$ depending continuously on u , hence locally constant; so the transition maps are locally $\phi_2 \circ (u \mapsto u + \lambda) \circ \phi_1^{-1}$, which are holomorphic. $\square$

DRAFT

21 Sep 2026

1.3. Isogenies

A complex torus is both a Riemann surface and an abelian group. Holomorphic maps between complex tori are affine linear, and those respecting the group structure are linear.

PROPOSITION 1.3.1. *Suppose $\varphi: \mathbb{C}/\Lambda_1 \rightarrow \mathbb{C}/\Lambda_2$ is holomorphic. Then there exist $\alpha, \beta \in \mathbb{C}$ with $\alpha\Lambda_1 \subseteq \Lambda_2$ such that*

$$\varphi(z + \Lambda_1) = \alpha z + \beta + \Lambda_2$$

for every $z \in \mathbb{C}$. The map φ is an isomorphism if and only if $\alpha\Lambda_1 = \Lambda_2$.

PROOF. Let $\pi_j: \mathbb{C} \rightarrow \mathbb{C}/\Lambda_j$ be the quotient maps. Since $\mathbb{C}$ is simply connected and π_2 is a covering map, $\varphi \circ \pi_1$ lifts to a continuous map $\tilde{\varphi}: \mathbb{C} \rightarrow \mathbb{C}$ with $\pi_2 \circ \tilde{\varphi} = \varphi \circ \pi_1$. As π_2 is a local isomorphism, $\tilde{\varphi}$ is holomorphic. For any $\lambda \in \Lambda_1$ we have $\tilde{\varphi}(z + \lambda) + \Lambda_2 = \tilde{\varphi}(z) + \Lambda_2$. In other words, the function

$$f_\lambda(z) := \tilde{\varphi}(z + \lambda) - \tilde{\varphi}(z)$$

takes values in Λ_2 , which is discrete. Since f_λ is continuous and $\mathbb{C}$ is connected, f_λ is constant, so $f'_\lambda = 0$, that is, $\tilde{\varphi}'(z + \lambda) = \tilde{\varphi}'(z)$ for every $z \in \mathbb{C}$. So $\tilde{\varphi}'$ is Λ_1 -periodic; it is therefore determined by its values on the compact set $\overline{\mathcal{F}_{\Lambda_1}}$ from the proof of Lemma 1.2.4, hence bounded. By Liouville's theorem, $\tilde{\varphi}' = \alpha$ is constant, so $\tilde{\varphi}(z) = \alpha z + \beta$. Moreover $\alpha\lambda = f_\lambda(z) \in \Lambda_2$ for every $\lambda \in \Lambda_1$, that is, $\alpha\Lambda_1 \subseteq \Lambda_2$.

If $\alpha = 0$, then φ is constant, so not injective. If $\alpha \neq 0$ and $\alpha\Lambda_1 \subsetneq \Lambda_2$, pick $w \in \Lambda_2 - \alpha\Lambda_1$. Then $w/\alpha \notin \Lambda_1$, and $\varphi(w/\alpha + \Lambda_1) = w + \beta + \Lambda_2 = \beta + \Lambda_2 = \varphi(0 + \Lambda_1)$, so φ is not injective. If $\alpha\Lambda_1 = \Lambda_2$, then $\psi(z + \Lambda_2) := \frac{1}{\alpha}(z - \beta) + \Lambda_1$ is a well defined holomorphic inverse of φ . $\square$

COROLLARY 1.3.2. *A map $\varphi: \mathbb{C}/\Lambda_1 \rightarrow \mathbb{C}/\Lambda_2$ is a holomorphic group homomorphism if and only if $\varphi(z + \Lambda_1) = \alpha z + \Lambda_2$ for some $\alpha \in \mathbb{C}$ with $\alpha\Lambda_1 \subseteq \Lambda_2$.*

PROOF. In Proposition 1.3.1, φ is a homomorphism if and only if $\varphi(0 + \Lambda_1) = 0 + \Lambda_2$, that is, $\beta \in \Lambda_2$. $\square$

DEFINITION 1.3.3. An **isogeny** is a non-zero holomorphic homomorphism between complex tori. An **isomorphism** of complex tori is an invertible isogeny.

EXAMPLE 1.3.4. Let Λ be any lattice, and let $m \geq 1$. Then $m\Lambda \subseteq \Lambda$, and the **multiplication by m map**

$$[m]: \mathbb{C}/\Lambda \rightarrow \mathbb{C}/\Lambda, \quad z + \Lambda \mapsto mz + \Lambda$$

is an isogeny. Its kernel is the m -torsion subgroup of $\mathbb{C}/\Lambda$,

$$(\mathbb{C}/\Lambda)[m] := \{z + \Lambda : mz \in \Lambda\} = \frac{1}{m}\Lambda/\Lambda \cong (\mathbb{Z}/m\mathbb{Z})^2.$$

Figure 1.3.1 shows the 4-torsion points in the fundamental parallelogram. This is the analogue, one dimension up, of the m -torsion subgroup of the circle

$$\mathbb{R}/\mathbb{Z} \cong \{z \in \mathbb{C} : |z| = 1\} \cong \mathbb{C}^\times / \mathbb{R}_{>0},$$

which is $\frac{1}{m}\mathbb{Z}/\mathbb{Z} \cong \mathbb{Z}/m\mathbb{Z}$, identified via $t \mapsto e^{2\pi it}$ with the group of complex m -th roots of unity

$$\mu_m(\mathbb{C}) := \{z \in \mathbb{C} : z^m = 1\}.$$

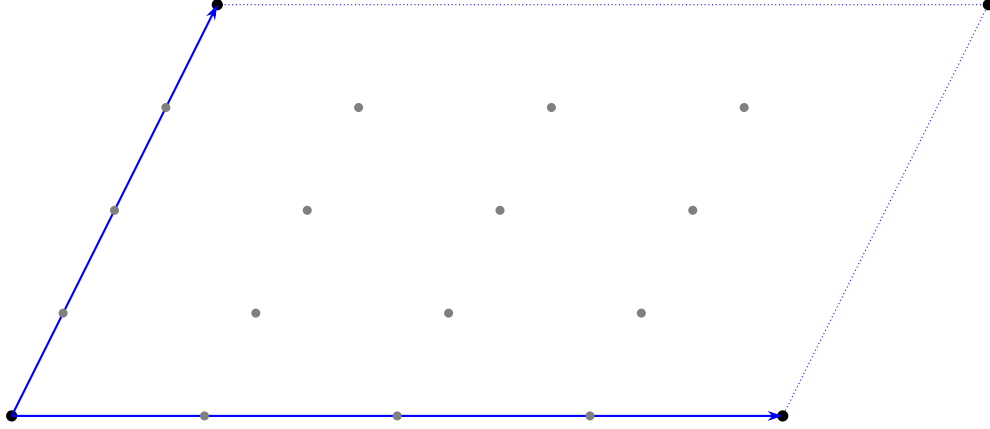


FIGURE 1.3.1. The 4-torsion points of $\mathbb{C}/\Lambda$ (gray) in the fundamental parallelogram, whose vertices are lattice points (black).

EXAMPLE 1.3.5. Let $\Lambda = \mathbb{Z}\omega_1 \oplus \mathbb{Z}\omega_2$ be a lattice, and let $N \geq 1$. The lattice $\Lambda' := \mathbb{Z}\frac{\omega_1}{N} \oplus \mathbb{Z}\omega_2$ contains Λ , and $\Lambda'/\Lambda \cong \mathbb{Z}/N\mathbb{Z}$ is generated by $\frac{\omega_1}{N} + \Lambda$. By Corollary 1.3.2 with $\alpha = 1$, the map

$$\varphi: \mathbb{C}/\Lambda \rightarrow \mathbb{C}/\Lambda', \quad z + \Lambda \mapsto z + \Lambda'$$

is an isogeny. Its kernel Λ'/Λ is a cyclic subgroup of order N of $(\mathbb{C}/\Lambda)[N]$, so φ is a **cyclic isogeny**. Figure 1.3.2 shows the case $N = 4$. Conversely, every cyclic subgroup $C \subseteq \mathbb{C}/\Lambda$ of order N is the kernel of such an isogeny: take $\Lambda' := \pi^{-1}(C)$, where $\pi: \mathbb{C} \rightarrow \mathbb{C}/\Lambda$ is the quotient map.

Since $N\Lambda' = \mathbb{Z}\omega_1 \oplus \mathbb{Z}N\omega_2 \subseteq \Lambda$, multiplication by N also defines an isogeny $\hat{\varphi}: \mathbb{C}/\Lambda' \rightarrow \mathbb{C}/\Lambda$, $z + \Lambda' \mapsto Nz + \Lambda$. Both composites $\hat{\varphi} \circ \varphi$ and $\varphi \circ \hat{\varphi}$ are multiplication by N (Example 1.3.4).

In Example 1.3.5 the isogeny φ has a partner $\hat{\varphi}$ going the other way, and both composites are multiplication by the order of the kernel. Every isogeny has such a partner. For a lattice Λ , write $\text{area}(\Lambda)$ for the area of the fundamental parallelogram $\mathcal{F}_\Lambda$ from the proof of Lemma 1.2.4. It does not depend on the basis, since a change of basis has determinant ± 1 .

LEMMA 1.3.6. *Let $\varphi: \mathbb{C}/\Lambda_1 \rightarrow \mathbb{C}/\Lambda_2$, $z + \Lambda_1 \mapsto \alpha z + \Lambda_2$, be an isogeny. Then $\ker \varphi = \alpha^{-1}\Lambda_2/\Lambda_1$ is finite, and*

$$\#\ker \varphi = [\Lambda_2 : \alpha\Lambda_1] = |\alpha|^2 \frac{\text{area}(\Lambda_1)}{\text{area}(\Lambda_2)}.$$

PROOF. We have $z + \Lambda_1 \in \ker \varphi$ if and only if $\alpha z \in \Lambda_2$, that is, $z \in \alpha^{-1}\Lambda_2$. Since $\varphi \neq 0$, $\alpha \neq 0$, and multiplication by α is a group isomorphism $\alpha^{-1}\Lambda_2/\Lambda_1 \rightarrow \Lambda_2/\alpha\Lambda_1$. Now $\alpha\Lambda_1 \subseteq \Lambda_2$ are both lattices. For lattices $\Lambda' \subseteq \Lambda$, write a basis of Λ' in terms of a basis of Λ by a matrix $M \in M_2(\mathbb{Z})$ with $\det M \neq 0$. By the Smith normal form, $[\Lambda : \Lambda'] = |\det M|$, and the $\mathbb{R}$ -linear map of $\mathbb{C}$ taking the one basis to the other

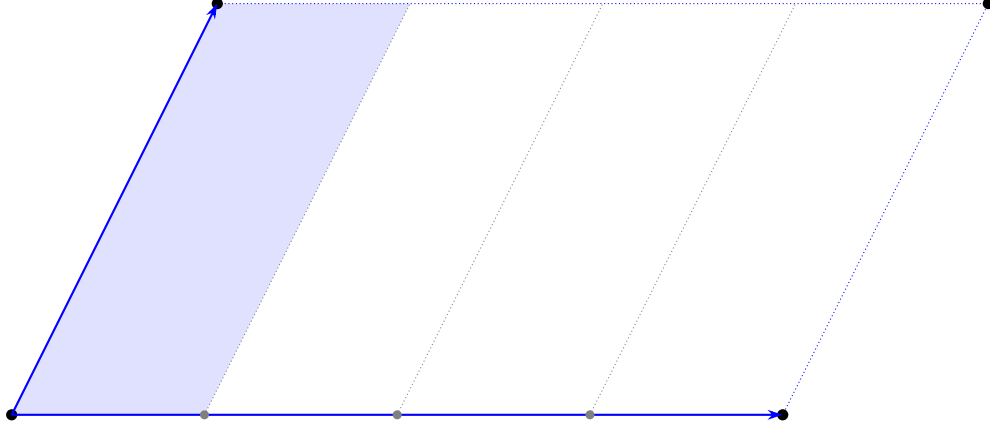


FIGURE 1.3.2. The kernel of the cyclic isogeny $\mathbb{C}/\Lambda \rightarrow \mathbb{C}/\Lambda'$ for $N = 4$ (gray) in the fundamental parallelogram of Λ , whose vertices are lattice points (black). A fundamental parallelogram of Λ' is shaded, and its translates by the kernel are dotted.

scales areas by $|\det M|$; so $[\Lambda : \Lambda'] = \text{area}(\Lambda')/\text{area}(\Lambda)$. Finally, $z \mapsto \alpha z$ is $\mathbb{R}$ -linear with determinant $|\alpha|^2$, so $\text{area}(\alpha\Lambda_1) = |\alpha|^2 \text{area}(\Lambda_1)$. $\square$

DEFINITION 1.3.7. The **degree** of an isogeny φ of complex tori is $\deg \varphi := \# \ker \varphi$.

For instance, $\deg[m] = m^2$ by Example 1.3.4, the cyclic isogeny of Example 1.3.5 has degree N , and by Proposition 1.3.1 the isogenies of degree 1 are exactly the isomorphisms. If $\lambda: \mathbb{C}/\Lambda_2 \rightarrow \mathbb{C}/\Lambda_3$ is another isogeny, given by $z \mapsto \beta z$, then $\lambda \circ \varphi$ is given by $z \mapsto \beta \alpha z$, and the area formula of Lemma 1.3.6 gives $\deg(\lambda \circ \varphi) = \deg \lambda \cdot \deg \varphi$. The scalar α is determined by φ : if $\alpha z - \alpha' z \in \Lambda_2$ for every $z \in \mathbb{C}$, then $\alpha - \alpha' = 0$, because a continuous map from $\mathbb{C}$ to the discrete set Λ_2 is constant.

THEOREM 1.3.8. Let $\varphi: \mathbb{C}/\Lambda_1 \rightarrow \mathbb{C}/\Lambda_2$, $z + \Lambda_1 \mapsto \alpha z + \Lambda_2$, be an isogeny of degree d .

- (1) There is a unique isogeny $\hat{\varphi}: \mathbb{C}/\Lambda_2 \rightarrow \mathbb{C}/\Lambda_1$, the **dual isogeny**, with $\hat{\varphi} \circ \varphi = [d]$. It is $\hat{\varphi}(z + \Lambda_2) = \frac{d}{\alpha} z + \Lambda_1$.
- (2) $\varphi \circ \hat{\varphi} = [d]$ on $\mathbb{C}/\Lambda_2$.
- (3) $\widehat{\lambda \circ \varphi} = \hat{\varphi} \circ \hat{\lambda}$ for an isogeny $\lambda: \mathbb{C}/\Lambda_2 \rightarrow \mathbb{C}/\Lambda_3$.
- (4) $\widehat{\varphi + \psi} = \hat{\varphi} + \hat{\psi}$ for an isogeny $\psi: \mathbb{C}/\Lambda_1 \rightarrow \mathbb{C}/\Lambda_2$ with $\varphi + \psi \neq 0$.
- (5) $\widehat{[m]} = [m]$ for all $m \geq 1$.
- (6) $\deg \hat{\varphi} = \deg \varphi$ and $\hat{\hat{\varphi}} = \varphi$.

PROOF. By Lemma 1.3.6, $\Lambda_2/\alpha\Lambda_1$ is a group of order d , so it is killed by d : $d\Lambda_2 \subseteq \alpha\Lambda_1$, that is, $\frac{d}{\alpha}\Lambda_2 \subseteq \Lambda_1$. By Corollary 1.3.2, $\hat{\varphi}$ is an isogeny, and $\hat{\varphi} \circ \varphi$ is $z \mapsto dz$, which is $[d]$. Composition of isogenies multiplies their scalars, and the scalar determines the isogeny, so an isogeny $z \mapsto \gamma z$ with this property has $\gamma\alpha = d$. This proves part 1, and part 2 follows from $\alpha \cdot \frac{d}{\alpha} = d$.

The rest is arithmetic with scalars. Set $c := \text{area}(\Lambda_1)/\text{area}(\Lambda_2)$. By Lemma 1.3.6, $d = c|\alpha|^2$, so

$$\frac{d}{\alpha} = c\bar{\alpha}.$$

For part 3, let λ be given by $z \mapsto \beta z$, of degree d' . Then $\lambda \circ \varphi$ has scalar $\beta\alpha$ and degree dd' , so its dual has scalar $\frac{dd'}{\beta\alpha} = \frac{d}{\alpha} \cdot \frac{d'}{\beta}$, which is the scalar of $\hat{\varphi} \circ \hat{\lambda}$. For part 4, let ψ be given by $z \mapsto \beta z$. Then $\varphi + \psi$ has scalar $\alpha + \beta$, and since c depends only on Λ_1 and Λ_2 , its dual has scalar $c\overline{(\alpha + \beta)} = c\bar{\alpha} + c\bar{\beta}$. For part 5, $[m]$ has scalar m and degree m^2 , so its dual has scalar $m^2/m = m$. For part 6, Lemma 1.3.6 gives $\deg \hat{\varphi} = |d/\alpha|^2/c = d^2/d = d$, and so the dual of $\hat{\varphi}$ has scalar $d/(d/\alpha) = \alpha$. $\square$

In Example 1.3.5, $\alpha = 1$ and $d = N$, so the isogeny $\hat{\varphi}$ there is the dual of φ . The kernel of the dual is

$$\ker \hat{\varphi} = \frac{\alpha}{d} \Lambda_1 / \Lambda_2 = \varphi((\mathbb{C}/\Lambda_1)[d]),$$

since $(\mathbb{C}/\Lambda_1)[d] = \frac{1}{d} \Lambda_1 / \Lambda_1$. The formula $d/\alpha = c\bar{\alpha}$ says that, up to the positive constant c , taking the dual is complex conjugation of the scalar. In particular, if we allow the zero map and set $\deg 0 := 0$, then $\varphi \mapsto \deg \varphi = c|\alpha|^2$ is a positive definite quadratic form on the group of holomorphic homomorphisms $\mathbb{C}/\Lambda_1 \rightarrow \mathbb{C}/\Lambda_2$. For an endomorphism of $\mathbb{C}/\Lambda$ we have $c = 1$, so the dual is $\bar{\alpha}$ and the degree is $|\alpha|^2$. These statements return for elliptic curves over an arbitrary field in Theorem 4.3.10 and Corollary 4.3.11, where there is no scalar to conjugate and the proofs are harder.

EXAMPLE 1.3.9. Let $\Lambda_\omega = \mathbb{Z} \oplus \omega\mathbb{Z}$ be the lattice of Example 1.2.3. Since $\mathbb{Z}[\omega]$ is a ring, $\alpha\Lambda_\omega \subseteq \Lambda_\omega$ for every $\alpha \in \mathbb{Z}[\omega]$, and $z \mapsto \alpha z$ is an endomorphism of degree $|\alpha|^2$ with dual $z \mapsto \bar{\alpha}z$.

- (1) For $\alpha = \omega$, the degree is 1, so $z \mapsto \omega z$ is an automorphism, and its dual $z \mapsto \bar{\omega}z = \omega^2 z$ is its inverse.
- (2) For $\alpha = 1 - \omega$, the degree is $|1 - \omega|^2 = 3$, and the dual is $z \mapsto (1 - \omega^2)z$; indeed $(1 - \omega)(1 - \omega^2) = 3$. The kernel is cyclic of order 3, generated by $\frac{1}{1-\omega} + \Lambda_\omega = \frac{2+\omega}{3} + \Lambda_\omega$.

Compare part 3 of Proposition 4.8.10, where the same identity appears for the curves C_d . Similarly, on $\mathbb{C}/\Lambda_i$ the endomorphism $z \mapsto (1 + i)z$ has degree 2 and dual $z \mapsto (1 - i)z$.

1.4. Meromorphic functions

DEFINITION 1.4.1. Let X be a Riemann surface. A **meromorphic function** on X is a holomorphic map $X \rightarrow \mathbb{P}^1(\mathbb{C})$ that is not identically ∞ . Denote the set of meromorphic functions on X by $\mathbb{C}(X)$.

Equivalently, a meromorphic function is a function f with values in $\mathbb{C}$, defined and holomorphic away from a discrete set of points of X , at each of which f has a pole; there we set $f = \infty$ [Mir95, Proposition II.3.13].

EXAMPLE 1.4.2. Let $F_0, F_1 \in \mathbb{C}[X_0, X_1]$ be homogeneous polynomials of equal degree with no common zero on $\mathbb{P}^1(\mathbb{C})$, and $F_0 \neq 0$. Then the map

$$F: \mathbb{P}^1(\mathbb{C}) \rightarrow \mathbb{P}^1(\mathbb{C}), \quad (X_0 : X_1) \mapsto (F_0 : F_1)$$

is a meromorphic function. Moreover, every meromorphic function on $\mathbb{P}^1(\mathbb{C})$ is of this form.

EXAMPLE 1.4.3. Let $\Lambda \subseteq \mathbb{C}$ be a lattice. Then there is a bijective correspondence between

- meromorphic functions on $\mathbb{C}/\Lambda$, and
- Λ -periodic meromorphic functions on $\mathbb{C}$,

given by $f \mapsto \tilde{f} := f \circ \pi$.

$$\begin{array}{ccc} \mathbb{C} & \xrightarrow{\tilde{f}} & \mathbb{P}^1(\mathbb{C}) \\ \pi \downarrow & \searrow & \\ \mathbb{C}/\Lambda & \xrightarrow{f} & \mathbb{P}^1(\mathbb{C}) \end{array}$$

Given $f, g \in \mathbb{C}(X)$, let $S \subseteq X$ be the set of poles of f or g . It is discrete [Mir95, Theorem II.1.33], and on $X - S$ the pointwise sum $f + g$ and product fg are holomorphic. At each point of S they have either a removable singularity or a pole [Mir95, Example II.1.17]; filling in the removable singularities and setting the value ∞ at the poles defines $f + g, fg \in \mathbb{C}(X)$. The constant maps $0, 1 \in \mathbb{C}(X)$ serve as additive and multiplicative identities, respectively.

LEMMA 1.4.4. *Let X be a Riemann surface. With the operations defined above, the set $\mathbb{C}(X)$ of meromorphic functions on X is a field.*

PROOF. The ring axioms hold pointwise away from a discrete set, hence everywhere by continuity. If $f \neq 0$, then since X is connected the zeros of f form a discrete set [Mir95, Theorem II.1.33], so $1/f$ is a meromorphic function [Mir95, Example II.1.17] and $f \cdot (1/f) = 1$. $\square$

1.5. Invariant differentials

On an open subset of $\mathbb{C}$ we can write down differentials $f(z) dz$. A Riemann surface has no global coordinate z , so we write a differential in every chart and ask that the expressions agree under the chain rule [Mir95, Section IV.1].

DEFINITION 1.5.1. Let X be a Riemann surface. A **meromorphic differential** ω on X assigns to every chart $\phi: U \rightarrow V$ of X a meromorphic function f_ϕ on V , written $\omega = f_\phi(z) dz$, such that for any two charts ϕ_1 and ϕ_2 , with transition map $T := \phi_2 \circ \phi_1^{-1}$,

$$f_{\phi_1}(z) = f_{\phi_2}(T(z)) T'(z) \quad \text{for } z \in \phi_1(U_1 \cap U_2).$$

If every f_ϕ is holomorphic, then ω is a **holomorphic differential**.

By the chain rule, it is enough to give functions f_ϕ satisfying the transformation rule for the charts of a single complex structure on X ; they extend uniquely to all charts. Since T' does not vanish, whether f_ϕ has a zero or a pole at $\phi(p)$ depends only on p , so a non-zero meromorphic differential has a discrete set of zeros and poles. If $\eta \neq 0$ is a meromorphic differential, then the quotients f_ϕ/h_ϕ , where $\eta = h_\phi(z) dz$, glue to a meromorphic function, because the factors T' cancel. So every meromorphic differential is $g\eta$ for a unique $g \in \mathbb{C}(X)$.

Differentials pull back along holomorphic maps. Let $F: X \rightarrow Y$ be a non-constant holomorphic map and ω a meromorphic differential on Y . For charts $\phi: U \rightarrow V$ of X and $\psi: U' \rightarrow V'$ of Y with $F(U) \subseteq U'$, write $F_{\psi\phi} := \psi \circ F \circ \phi^{-1}$ and $\omega = f_\psi(w) dw$, and set

$$F^*\omega := f_\psi(F_{\psi\phi}(z)) F'_{\psi\phi}(z) dz.$$

Such charts ϕ cover X , and the chain rule shows that $F^*\omega$ is a meromorphic differential on X . If ω is holomorphic, the same formula makes sense for constant F , and gives $F^*\omega = 0$. Pullback satisfies $(G \circ F)^* = F^* \circ G^*$ and $F^*(g\omega) = (g \circ F) F^*\omega$ for $g \in \mathbb{C}(Y)$. For instance, if F is a map between open subsets of $\mathbb{C}$, then $F^*(dw) = F'(z) dz$, which is the familiar substitution $w = F(z)$.

The Riemann surfaces $\mathbb{C}^\times$ and $\mathbb{C}/\Lambda$ are also abelian groups, and for every element a the translation τ_a , namely $z \mapsto az$ on $\mathbb{C}^\times$ and $z + \Lambda \mapsto z + a + \Lambda$ on $\mathbb{C}/\Lambda$, is an isomorphism of Riemann surfaces.

DEFINITION 1.5.2. Let G be $\mathbb{C}^\times$ or a complex torus. A meromorphic differential ω on G is **invariant** if $\tau_a^*\omega = \omega$ for every $a \in G$.

LEMMA 1.5.3. Let G be $\mathbb{C}^\times$ or a complex torus, and let η be a non-zero invariant differential on G . Then η has no zeros or poles, and every invariant differential on G is $c\eta$ for a unique $c \in \mathbb{C}$.

PROOF. Since $\tau_a^*\eta = \eta$, a point p of G is a zero of η if and only if $\tau_a(p)$ is, and likewise for poles. The translations act transitively on G , so if η had a zero or a pole it would have one everywhere, contradicting discreteness. Let ω be invariant, and write $\omega = g\eta$ with $g \in \mathbb{C}(G)$. Then

$$g\eta = \omega = \tau_a^*\omega = (g \circ \tau_a) \tau_a^*\eta = (g \circ \tau_a) \eta,$$

so $g \circ \tau_a = g$ for every $a \in G$. Hence g is constant, and its value $c \neq \infty$, since g is not identically ∞ . $\square$

EXAMPLE 1.5.4. On $\mathbb{C}^\times$, with its single chart id , consider the holomorphic differential

$$\omega := \frac{dz}{z}.$$

For $a \in \mathbb{C}^\times$, the translation $\tau_a(z) = az$ gives

$$\tau_a^*\omega = \frac{a dz}{az} = \frac{dz}{z} = \omega,$$

so ω is invariant, and by Lemma 1.5.3 every invariant differential on $\mathbb{C}^\times$ is a multiple of dz/z . By contrast, dz is also a holomorphic differential on $\mathbb{C}^\times$, but $\tau_a^*dz = a dz$, so it is not invariant. The holomorphic differentials $f(z) dz$, with f holomorphic on $\mathbb{C}^\times$, form an infinite-dimensional space; only a line of them is invariant.

- (1) **Endomorphisms.** For $m \in \mathbb{Z}$, the map $[m]: z \mapsto z^m$ is a holomorphic group homomorphism $\mathbb{C}^\times \rightarrow \mathbb{C}^\times$, with kernel $\mu_m(\mathbb{C})$ when $m \neq 0$ (Example 1.3.4). It pulls ω back to

$$[m]^* \omega = \frac{mz^{m-1} dz}{z^m} = m \frac{dz}{z} = m\omega.$$

- (2) **Uniformization.** The exponential $\exp: \mathbb{C} \rightarrow \mathbb{C}^\times$ is a surjective holomorphic homomorphism with kernel $2\pi i\mathbb{Z}$, so it induces an isomorphism $\mathbb{C}/2\pi i\mathbb{Z} \cong \mathbb{C}^\times$ of Riemann surfaces and of groups. In the coordinate w on $\mathbb{C}$,

$$\exp^* \omega = \frac{e^w dw}{e^w} = dw,$$

so the invariant differential on $\mathbb{C}^\times$ corresponds to the translation-invariant differential dw on $\mathbb{C}$.

- (3) **Periods.** If γ is a loop in $\mathbb{C}^\times$, then $\int_\gamma dz/z$ is $2\pi i$ times the winding number of γ around 0. So the set of $\int_\gamma \omega$, as γ runs over loops, is $2\pi i\mathbb{Z}$, the kernel of $\exp$, and it is generated by the integral over the unit circle, the image of the segment from 0 to $2\pi i$. The inverse of the isomorphism in part 2 is $z \mapsto \int_1^z \omega \bmod 2\pi i\mathbb{Z}$, the logarithm, which is well defined modulo $2\pi i\mathbb{Z}$.
- (4) **At 0 and ∞ .** In the chart ϕ_1 of Example 1.1.4, with coordinate $u = 1/z$, the transformation rule gives $\omega = -du/u$. So ω is a meromorphic differential on $\mathbb{P}^1(\mathbb{C})$ with simple poles at 0 and ∞ . Unlike dz on $\mathbb{C}/\Lambda$ below, it does not extend to a holomorphic differential on this compactification.

Algebraically, $\mathbb{C}^\times$ is the group of complex points of the multiplicative group $\mathbb{G}_m$ (Example 2.1.5), and ω is dx/x ; see Examples 3.4.7 and 3.4.8 for the algebraic version over an arbitrary field.

EXAMPLE 1.5.5. Let $\Lambda \subseteq \mathbb{C}$ be a lattice, and $\pi: \mathbb{C} \rightarrow \mathbb{C}/\Lambda$ the quotient map. Among the charts constructed in the proof of Lemma 1.2.4, take those coming from $\phi = \text{id}$, that is, the inverses of the restrictions of π to small open subsets of $\mathbb{C}$. Their transition maps are locally $u \mapsto u + \lambda$ with $\lambda \in \Lambda$, whose derivative is 1. So setting $f_\phi := 1$ on each of them satisfies the transformation rule, and defines a holomorphic differential

$$dz \text{ on } \mathbb{C}/\Lambda, \quad \text{with } \pi^* dz = dz.$$

In these charts the translation τ_a reads $u \mapsto u + a$, up to adding an element of Λ , so $\tau_a^* dz = dz$. Hence dz is invariant, it has no zeros or poles, and every invariant differential on $\mathbb{C}/\Lambda$ is a multiple of dz , by Lemma 1.5.3.

- (1) **All holomorphic differentials are invariant.** Let ω be a holomorphic differential on $\mathbb{C}/\Lambda$, and write $\omega = g dz$ with $g \in \mathbb{C}(\mathbb{C}/\Lambda)$. Since dz has no zeros, g has no poles. By Example 1.4.3, $g \circ \pi$ is a Λ -periodic entire function; it is bounded on the compact set $\overline{\mathcal{F}_\Lambda}$, hence bounded, hence constant by Liouville's theorem. So the holomorphic differentials on $\mathbb{C}/\Lambda$ are exactly the multiples of dz , unlike on $\mathbb{C}^\times$. More generally, the meromorphic differentials on $\mathbb{C}/\Lambda$ are the $g dz$ with $g \circ \pi$ a Λ -periodic meromorphic function on $\mathbb{C}$.
- (2) **Isogenies.** By Proposition 1.3.1, a holomorphic map $\varphi: \mathbb{C}/\Lambda_1 \rightarrow \mathbb{C}/\Lambda_2$ has the form $z + \Lambda_1 \mapsto \alpha z + \beta + \Lambda_2$. In the charts above it reads $u \mapsto \alpha u + \beta$, up to adding an element of Λ_2 , so

$$\varphi^* dz = \alpha dz.$$

In particular $[m]^* dz = m dz$ for the multiplication by m map of Example 1.3.4, exactly as $[m]^* \omega = m\omega$ on $\mathbb{C}^\times$. An isogeny is determined by its effect on the invariant differential: by Corollary 1.3.2, $\varphi \mapsto \alpha$ identifies the holomorphic homomorphisms $\mathbb{C}/\Lambda_1 \rightarrow \mathbb{C}/\Lambda_2$ with $\{\alpha \in \mathbb{C} : \alpha\Lambda_1 \subseteq \Lambda_2\}$, and it turns composition into multiplication.

- (3) **Periods.** A path γ in $\mathbb{C}/\Lambda$ lifts along π to a path $\tilde{\gamma}$ in $\mathbb{C}$, unique up to translation by Λ , and we set $\int_\gamma dz := \int_{\tilde{\gamma}} dz$, the difference of the endpoints of $\tilde{\gamma}$. If γ is a loop, then $\tilde{\gamma}$ runs from some z to $z + \lambda$ with $\lambda \in \Lambda$, and $\int_\gamma dz = \lambda$. Every λ occurs, as the image of the segment from 0 to λ . So the periods of dz form the lattice Λ itself, the kernel of π , and $p \mapsto \int_0^p dz \bmod \Lambda$ is the identity of $\mathbb{C}/\Lambda$.

For $c \in \mathbb{C}^\times$, the periods of $c dz$ form the lattice $c\Lambda$, and the isomorphism $\mathbb{C}/\Lambda \rightarrow \mathbb{C}/c\Lambda$, $z + \Lambda \mapsto cz + c\Lambda$, of Proposition 1.3.1 pulls dz back to $c dz$. So a complex torus determines its lattice only up to scaling, but a complex torus together with a non-zero holomorphic differential determines the lattice itself. Under the

isomorphism of Theorem 1.7.1, dz becomes $dx/(2y)$ on $E_{A,B}(\mathbb{C})$, as we will see in Remark 4.1.8; this is the invariant differential of an elliptic curve in Chapter 4.

DRAFT

21 Sep 2026

1.6. Projective plane curves as Riemann surfaces

We follow [Mir95, Section I.3]. The [projective plane](#) is

$$\mathbb{P}^2(\mathbb{C}) := (\mathbb{C}^3 - \{(0, 0, 0)\})/\mathbb{C}^\times,$$

with the quotient topology, and we write its points as $(X : Y : Z)$. It is compact, being the image of the unit sphere in $\mathbb{C}^3$, and it is covered by the open sets U_X , U_Y and U_Z where X , Y or Z does not vanish, each homeomorphic to $\mathbb{C}^2$ by dividing by that coordinate. If $F \in \mathbb{C}[X, Y, Z]$ is homogeneous of degree d , then $F(\lambda z) = \lambda^d F(z)$, so the zero locus

$$C_F := \{(X : Y : Z) \in \mathbb{P}^2(\mathbb{C}) : F(X, Y, Z) = 0\}$$

is a well defined closed subset of $\mathbb{P}^2(\mathbb{C})$.

DEFINITION 1.6.1. A homogeneous polynomial $F \in \mathbb{C}[X, Y, Z]$ is [nonsingular](#) if F , $\partial F/\partial X$, $\partial F/\partial Y$ and $\partial F/\partial Z$ have no common zero in $\mathbb{P}^2(\mathbb{C})$. In that case C_F is a [smooth projective plane curve](#).

PROPOSITION 1.6.2. *Let F be a nonsingular homogeneous polynomial. Then C_F is a compact Riemann surface. Near every point of C_F , some ratio of two homogeneous coordinates is a local coordinate.*

PROOF. See [Mir95, Proposition I.3.6]. The idea: $C_F \cap U_Z$ is the affine curve $f(x, y) := F(x, y, 1) = 0$, where $x = X/Z$ and $y = Y/Z$. At a point $(x : y : 1)$ of C_F we have $\partial f/\partial x = \partial F/\partial X$ and $\partial f/\partial y = \partial F/\partial Y$, and Euler's formula

$$X \frac{\partial F}{\partial X} + Y \frac{\partial F}{\partial Y} + Z \frac{\partial F}{\partial Z} = dF$$

gives $\partial F/\partial Z = dF - x \partial F/\partial X - y \partial F/\partial Y$. Hence the singular points of F in U_Z are exactly the common zeros of f , $\partial f/\partial x$ and $\partial f/\partial y$, and likewise in U_X and U_Y . Where $\partial f/\partial y \neq 0$, the implicit function theorem writes C_F locally as a graph $y = g(x)$, so x is a local coordinate; similarly y is a local coordinate where $\partial f/\partial x \neq 0$. The transition maps between these charts are holomorphic, being built from such functions g and ratios of coordinates. As a subspace of $\mathbb{P}^2(\mathbb{C})$, C_F is Hausdorff, and it is compact, being closed.

That C_F is connected is less elementary; here is the idea. First, F is irreducible: if $F = GH$ with G and H nonconstant, then the curves C_G and C_H meet at some point p of $\mathbb{P}^2(\mathbb{C})$ by Bézout's theorem, and every partial derivative of $F = GH$ vanishes at p by the product rule, so F is singular at p . After permuting the coordinates we may assume $Z \nmid F$. Then f is irreducible too, and C_F meets the line $Z = 0$ in finitely many points. Second, the zero locus of an irreducible f is connected [Mir95, Theorem I.2.3]. Away from finitely many points, the projection $(x, y) \mapsto x$ makes it a finite covering of $\mathbb{C}$ minus a finite set. A connected component would pick out a subset of the roots y of $f(x, \cdot) = 0$ that is preserved as x moves. The elementary symmetric functions of those roots are then rational functions of x , and they give a proper factor of f . Finally, C_F is the closure of the connected set $C_F \cap U_Z$, because the finitely many remaining points are not isolated in C_F . Hence C_F is connected. $\square$

EXAMPLE 1.6.3. Let $A, B \in \mathbb{C}$ and

$$F := Y^2Z - X^3 - AXZ^2 - BZ^3,$$

$$E_{A,B}(\mathbb{C}) := C_F = \{(X : Y : Z) \in \mathbb{P}^2(\mathbb{C}) : Y^2Z = X^3 + AXZ^2 + BZ^3\}.$$

Then F is nonsingular if and only if $4A^3 + 27B^2 \neq 0$. Indeed:

- On $Z = 0$, the equation forces $X = 0$, so the only point of C_F there is $(0 : 1 : 0)$, where $\partial F/\partial Z = Y^2 - 2AXZ - 3BZ^2 = 1 \neq 0$.
- On U_Z , with $x = X/Z$, $y = Y/Z$ and $p(x) := x^3 + Ax + B$, we have $f = y^2 - p(x)$. By Euler's formula, a singular point is a common zero of f , $\partial f/\partial x = -p'(x)$ and $\partial f/\partial y = 2y$, that is, $y = 0$ and x a repeated root of p . Such an x exists if and only if the discriminant of p , which is $-(4A^3 + 27B^2)$, vanishes.

So if $4A^3 + 27B^2 \neq 0$, then $E_{A,B}(\mathbb{C})$ is a compact Riemann surface by Proposition 1.6.2.

1.7. The uniformization theorem

Given any lattice $\Lambda \subseteq \mathbb{C}$, the associated [Weierstrass \$\wp\$ -function](#) is the meromorphic function

$$\wp_\Lambda : \mathbb{C} \rightarrow \mathbb{P}^1(\mathbb{C}), \quad z \mapsto \frac{1}{z^2} + \sum_{\substack{\omega \in \Lambda \\ \omega \neq 0}} \left[\frac{1}{(z - \omega)^2} - \frac{1}{\omega^2} \right].$$

Its derivative is

$$\wp'_\Lambda : \mathbb{C} \rightarrow \mathbb{P}^1(\mathbb{C}), \quad z \mapsto -2 \sum_{\omega \in \Lambda} \frac{1}{(z - \omega)^3}.$$

For even $k > 2$, we have the [Eisenstein series](#)

$$G_k(\Lambda) := \sum_{\substack{\omega \in \Lambda \\ \omega \neq 0}} \frac{1}{\omega^k}.$$

We have the following beautiful result.

THEOREM 1.7.1. *Given $A, B \in \mathbb{C}$ with $4A^3 + 27B^2 \neq 0$, there exists a lattice $\Lambda \subseteq \mathbb{C}$ such that the map*

$$\phi : \mathbb{C}/\Lambda \rightarrow E_{A,B}(\mathbb{C}), \quad z + \Lambda \mapsto \begin{cases} (\wp_\Lambda(z) : \frac{1}{2}\wp'_\Lambda(z) : 1) & \text{if } z \notin \Lambda, \\ (0 : 1 : 0) & \text{if } z \in \Lambda, \end{cases}$$

is an isomorphism of Riemann surfaces. Moreover,

$$A = -15G_4(\Lambda), \quad B = -35G_6(\Lambda).$$

QUESTION 1.7.2 (Diamond–Shurman). This is [DS05, Exercise 1.4.2]. Let $\Lambda \subseteq \mathbb{C}$ be a lattice.

- (1) Show that $\wp_\Lambda$ is even and that $\wp'_\Lambda$ is Λ -periodic.
- (2) Let $\{\omega_1, \omega_2\}$ be a $\mathbb{Z}$ -basis for Λ . For $j = 1, 2$, show that $\wp_\Lambda(z + \omega_j) - \wp_\Lambda(z)$ is some constant c_j . Substitute $z = -\omega_j/2$ to conclude that $c_j = 0$, and deduce that $\wp_\Lambda$ is Λ -periodic.

1.8. Isomorphism classes of complex tori

Proposition [1.3.1](#) lets us classify complex tori up to isomorphism. Start with an arbitrary lattice $\Lambda \subseteq \mathbb{C}$, and choose a $\mathbb{Z}$ -basis $\{\omega_1, \omega_2\}$. Since $\omega_1/\omega_2 \notin \mathbb{R}$, after possibly swapping ω_1 and ω_2 we have $\tau := \omega_1/\omega_2 \in \mathbb{H}$. Define $\Lambda_\tau := \mathbb{Z} \oplus \tau\mathbb{Z}$. Since $\frac{1}{\omega_2}\Lambda = \Lambda_\tau$, Proposition [1.3.1](#) implies that the map

$$\varphi_\tau : \mathbb{C}/\Lambda \rightarrow \mathbb{C}/\Lambda_\tau, \quad z + \Lambda \mapsto \frac{z}{\omega_2} + \Lambda_\tau$$

is an isomorphism of complex tori. So every complex torus is isomorphic to one of the form $\mathbb{C}/\Lambda_\tau$ with $\tau \in \mathbb{H}$. The number τ depends on the choice of basis, and the group $\mathrm{SL}_2(\mathbb{Z})$ accounts for this ambiguity. Recall that it acts on $\mathbb{H}$ by

$$\gamma \cdot \tau := \frac{a\tau + b}{c\tau + d}, \quad \gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z}).$$

PROPOSITION 1.8.1. *Let $\tau, \tau' \in \mathbb{H}$. Then $\mathbb{C}/\Lambda_\tau \cong \mathbb{C}/\Lambda_{\tau'}$ if and only if $\tau' = \gamma \cdot \tau$ for some $\gamma \in \mathrm{SL}_2(\mathbb{Z})$. Consequently, $\tau \mapsto \mathbb{C}/\Lambda_\tau$ induces a bijection between $\mathrm{SL}_2(\mathbb{Z}) \backslash \mathbb{H}$ and the set of isomorphism classes of complex tori.*

PROOF. By Proposition [1.3.1](#), $\mathbb{C}/\Lambda_\tau \cong \mathbb{C}/\Lambda_{\tau'}$ if and only if $\alpha\Lambda_\tau = \Lambda_{\tau'}$ for some $\alpha \in \mathbb{C}^\times$.

Suppose $\tau' = \gamma \cdot \tau$, and set $\alpha := 1/(c\tau + d)$, which is non-zero since $\tau \notin \mathbb{R}$. Since γ is invertible over $\mathbb{Z}$, $\{a\tau + b, c\tau + d\}$ is a $\mathbb{Z}$ -basis of Λ_τ , so $\alpha\Lambda_\tau = \mathbb{Z}\tau' \oplus \mathbb{Z} = \Lambda_{\tau'}$.

Conversely, suppose $\alpha\Lambda_\tau = \Lambda_{\tau'}$. Then $\{\alpha\tau, \alpha\}$ is a $\mathbb{Z}$ -basis of $\Lambda_{\tau'} = \mathbb{Z}\tau' \oplus \mathbb{Z}$, so there is $\delta = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{GL}_2(\mathbb{Z})$ with $\alpha\tau = a\tau' + b$ and $\alpha = c\tau' + d$. Dividing, $\tau = (a\tau' + b)/(c\tau' + d)$, and the identity

$$\mathrm{Im}\left(\frac{a\tau' + b}{c\tau' + d}\right) = \frac{\det(\delta) \mathrm{Im}(\tau')}{|c\tau' + d|^2}$$

together with $\tau, \tau' \in \mathbb{H}$ forces $\det(\delta) = 1$. Hence $\tau = \delta \cdot \tau'$, that is, $\tau' = \delta^{-1} \cdot \tau$ with $\delta^{-1} \in \mathrm{SL}_2(\mathbb{Z})$. The last statement follows, since every complex torus is isomorphic to some $\mathbb{C}/\Lambda_\tau$. $\square$

The region in Figure 1.8.1, including its solid boundary and excluding its dotted boundary, contains exactly one point of every $\mathrm{SL}_2(\mathbb{Z})$ -orbit in $\mathbb{H}$. By Proposition 1.8.1, it is in bijection with the set of isomorphism classes of complex tori.

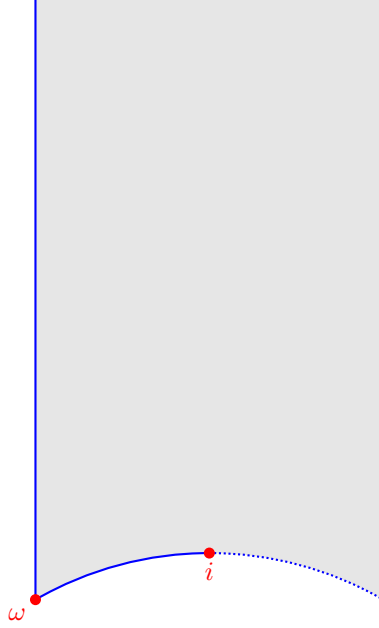


FIGURE 1.8.1. A fundamental domain for the action of $\mathrm{SL}_2(\mathbb{Z})$ on $\mathbb{H}$.

CHAPTER 2

Algebraic varieties

Sources. This chapter follows [Poo06, Chapter 1] and [Sil09, Chapter I]. Unlike these sources, we do not assume that the ground field is perfect.

The goal of this chapter is to describe the categories of affine and projective varieties over a field k . We only state the definitions and results we use, with references; proofs are sketched only where the arguments in [Sil09] rely on the ground field being perfect.

As in the preface, k is an arbitrary field, $k^{\text{sep}} \subseteq \bar{k}$ are a separable and an algebraic closure, and $\text{Gal}_k := \text{Gal}(k^{\text{sep}}/k)$.

2.1. Varieties and their points

DEFINITION 2.1.1. An **affine k -variety** is a scheme over k isomorphic to $\text{Spec } A$ for some k -algebra A of finite type.

Recall that A is of finite type if it admits a surjective homomorphism from a polynomial ring $k[x_1, \dots, x_n] \rightarrow A$. In other words, every such k -algebra is isomorphic to

$$k[x_1, \dots, x_n]/I$$

for some ideal I .

DEFINITION 2.1.2. A **k -variety** is a **separated scheme** of **finite type** over k . A morphism of k -varieties is a morphism of k -schemes.

Whatever the precise definition of these words is, the bottom line is that a k -variety is a topological space X that can be glued from open affine subvarieties: opens $U \subset X$ of the form $U = \text{Spec } A$ for A a finitely generated k -algebra. Likewise, a morphism of k -schemes $f: X \rightarrow Y$ between k -varieties is one that can be glued from morphisms of the form $\text{Spec}(\phi): \text{Spec } B \rightarrow \text{Spec } A$, where $\phi: A \rightarrow B$ is a k -algebra homomorphism.

If you don't know what Spec means, it will be enough for the time being to understand k -varieties through their R -points.

DEFINITION 2.1.3. For a k -algebra R , the set of R -points of a k -variety X is

$$X(R) := \text{Hom}_k(\text{Spec } R, X).$$

The elements of $X(k)$ are the **rational points** of X .

Since Spec is an anti-equivalence between k -algebras and affine k -schemes, for an affine k -variety $X = \text{Spec } A$ there is a bijection, natural in R ,

$$X(R) \cong \text{Hom}_k(A, R).$$

EXAMPLE 2.1.4. Given $n \geq 1$, **affine n -space** is $\mathbb{A}_k^n := \text{Spec } k[x_1, \dots, x_n]$. A k -algebra homomorphism $k[x_1, \dots, x_n] \rightarrow R$ is determined by the images of the variables x_i , so $\mathbb{A}_k^n(R) = R^n$ for every k -algebra R . An ideal $I \subseteq k[x_1, \dots, x_n]$ defines the closed subvariety $V(I) := \text{Spec } k[x_1, \dots, x_n]/I$ of $\mathbb{A}_k^n$, with

$$V(I)(R) = \{a \in R^n : f(a) = 0 \text{ for all } f \in I\}.$$

Up to isomorphism the $V(I)$ are all the affine k -varieties.

DRAFT
21 Sep 2026

EXAMPLE 2.1.5. The **multiplicative group** is $\mathbb{G}_m := \operatorname{Spec} k[x, x^{-1}]$. Since

$$k[x, x^{-1}] \cong k[x, y]/(xy - 1),$$

$\mathbb{G}_m$ can be embedded in the affine plane as $\mathbb{G}_m \cong V(xy - 1) \subseteq \mathbb{A}_k^2$. It is also the open subvariety $\{x \neq 0\}$ of $\mathbb{A}_k^1$. A k -algebra homomorphism $k[x, x^{-1}] \rightarrow R$ is determined by the image of x , which can be any unit of R , so

$$\mathbb{G}_m(R) = R^\times$$

for every k -algebra R . The group structure on $R^\times$ is natural in R ; this is the sense in which $\mathbb{G}_m$ is a group.

In the complex analytic world, $\mathbb{P}^n(\mathbb{C})$ is the set of lines through the origin in $\mathbb{C}^{n+1}$. For $n = 1, 2$ we saw that it is covered by $n + 1$ copies of $\mathbb{C}^n$: see Example 1.1.4 and Section 1.6. In the algebraic world, we want to define $\mathbb{P}_k^n$ in a similar way, as the quotient of $\mathbb{A}_k^{n+1} - \{0\}$ by the scaling action of $\mathbb{G}_m$ (Example 2.1.5). This is what the Proj construction achieves.

EXAMPLE 2.1.6. Projective n -space is $\mathbb{P}_k^n := \operatorname{Proj} k[X_0, \dots, X_n]$. A homogeneous ideal $I \subseteq k[X_0, \dots, X_n]$ defines the closed subvariety $V_+(I) := \operatorname{Proj} k[X_0, \dots, X_n]/I$ of $\mathbb{P}_k^n$.

Let R be a k -algebra. To describe a point of $\mathbb{P}^n(R)$, consider the tuples $(P, s_0, \dots, s_n)$, where P is an **invertible R -module** – a finitely generated projective R -module of constant rank 1 – and $s_0, \dots, s_n$ are elements generating P . Call $(P, s_0, \dots, s_n)$ and $(P', s'_0, \dots, s'_n)$ equivalent if some isomorphism of R -modules $\varphi: P \rightarrow P'$ satisfies $\varphi(s_i) = s'_i$ for every i . A point of $\mathbb{P}^n(R)$ is then an equivalence class $[P, s_0, \dots, s_n]$ [Har77, Theorem II.7.1].

To evaluate a homogeneous polynomial at such a point, note that the tensor power $P^{\otimes d} := P \otimes_R \dots \otimes_R P$ is again invertible. Write $\mathbf{X} := (X_0, \dots, X_n)$ and $\mathbf{s} := (s_0, \dots, s_n)$, and for a multi-index $\mathbf{a} = (a_0, \dots, a_n)$ of non-negative integers write $|\mathbf{a}| := a_0 + \dots + a_n$. A monomial $\mathbf{X}^{\mathbf{a}} = X_0^{a_0} \dots X_n^{a_n}$ with $|\mathbf{a}| = d$ has a value

$$\mathbf{s}^{\mathbf{a}} := \underbrace{s_0 \otimes \dots \otimes s_0}_{a_0} \otimes \dots \otimes \underbrace{s_n \otimes \dots \otimes s_n}_{a_n} \in P^{\otimes d}.$$

So a homogeneous $F = \sum_{|\mathbf{a}|=d} c_{\mathbf{a}} \mathbf{X}^{\mathbf{a}} \in k[X_0, \dots, X_n]$ of degree d has a value

$$F(\mathbf{s}) := \sum_{|\mathbf{a}|=d} c_{\mathbf{a}} \mathbf{s}^{\mathbf{a}} \in P^{\otimes d},$$

and $\varphi^{\otimes d}$ carries $F(\mathbf{s})$ to $F(\mathbf{s}')$, so whether this value vanishes depends only on the R -point. The R -point lies in $V_+(I)(R)$ if and only if $F(\mathbf{s}) = 0$ for every homogeneous $F \in I$.

If every invertible R -module is free, e.g. if R is a field, a local ring or a principal ideal domain, then choosing a basis of P gives

$$\mathbb{P}^n(R) = \{(a_0, \dots, a_n) \in R^{n+1} : a_0 R + \dots + a_n R = R\} / R^\times$$

and

$$V_+(I)(R) = \{(a_0 : \dots : a_n) \in \mathbb{P}^n(R) : F(a) = 0 \text{ for all homogeneous } F \in I\}.$$

When R is a field, the condition $a_0 R + \dots + a_n R = R$ says that $a \neq 0$, so

$$\mathbb{P}^n(R) = (R^{n+1} - \{0\}) / R^\times.$$

For general R there is no such description: an R -point whose P is not free has no homogeneous coordinates. The open subschemes $U_i := \{X_i \neq 0\}$ of $\mathbb{P}_k^n$ are isomorphic to $\mathbb{A}_k^n$, and cover it; the affine coordinates on U_0 are the ratios $x_i = X_i/X_0$. For a closed subvariety $X \subseteq \mathbb{P}_k^n$, the $X \cap U_i$ are the standard affine patches of X ; the equations of $X \cap U_0$ are obtained by setting $X_0 = 1$ (dehomogenization).

Conversely, identify $\mathbb{A}_k^n$ with the patch $U_0 \subseteq \mathbb{P}_k^n$. The **projective closure** of a closed subvariety $X = V(I) \subseteq \mathbb{A}_k^n$ is the Zariski closure $\overline{X}$ of X in $\mathbb{P}_k^n$. It is cut out by the homogenizations of the elements of I . The **homogenization** of $f \in k[x_1, \dots, x_n]$ is the homogeneous polynomial $F \in k[X_0, \dots, X_n]$ of the same degree given by

$$F(X_0, \dots, X_n) := X_0^{\deg f} f(X_1/X_0, \dots, X_n/X_0),$$

and $\overline{X} = V_+(J)$, where J is the homogeneous ideal generated by the homogenizations of all elements of I . For a single non-zero f this says that the projective closure of $V(f)$ is $V_+(F)$. For several equations one must homogenize every element of the ideal, not only a set of generators.

EXAMPLE 2.1.7. The **twisted cubic** is the affine curve $C := V(I) \subseteq \mathbb{A}_k^3$, where

$$I := (x_2 - x_1^2, x_3 - x_1^3),$$

the image of $t \mapsto (t, t^2, t^3)$. Homogenizing the two generators of I gives the quadric surface $V_+(X_0X_2 - X_1^2)$ and the cubic surface $V_+(X_0^2X_3 - X_1^3)$ in $\mathbb{P}_k^3$. On the hyperplane $X_0 = 0$ both equations reduce to $X_1 = 0$, so their intersection contains the whole line $\{X_0 = X_1 = 0\}$. But $\overline{C}$ meets $X_0 = 0$ only in $(0 : 0 : 0 : 1)$; the limit of the points $(1 : t : t^2 : t^3)$ of C for large t ; the projective closure is therefore strictly smaller.

The identity

$$x_1x_3 - x_2^2 = x_1(x_3 - x_1^3) + (x_1^2 - x_2)(x_1^2 + x_2)$$

shows that $x_1x_3 - x_2^2$ lies in I . Its homogenization $X_1X_3 - X_2^2$ is again of degree 2, and the quadric surface $V_+(X_1X_3 - X_2^2)$ it defines cuts the spurious line down to the rational point $(0 : 0 : 0 : 1)$: on $\{X_0 = X_1 = 0\}$ the form $X_1X_3 - X_2^2$ restricts to $-X_2^2$, which forces $X_2 = 0$. So this quadric is needed to cut out $\overline{C}$, although it is not in the ideal generated by the homogenizations of the two generators of I .

DEFINITION 2.1.8. Let X be a k -variety.

- (1) X is **irreducible** if its underlying topological space is non-empty and is not the union of two proper closed subsets.
- (2) X is **reduced** if for every open $U \subseteq X$, the ring $\mathcal{O}_X(U)$ has no non-zero nilpotent elements.
- (3) X is **integral** if for every non-empty open $U \subseteq X$, the ring $\mathcal{O}_X(U)$ is an integral domain.

A k -variety is integral if and only if it is reduced and irreducible [Har77, Proposition II.3.1]. For an affine variety $X = V(I) \subseteq \mathbb{A}_k^n$: X is reduced if and only if I is radical, irreducible if and only if $\sqrt{I}$ is prime, and integral if and only if I is prime.

WARNING 2.1.9. Varieties need not be reduced or irreducible (Definition 2.1.8). The variety $V(x^2) \subseteq \mathbb{A}_k^1$ is irreducible but not reduced, and $V(xy) \subseteq \mathbb{A}_k^2$ is reduced but not irreducible. For every field L , both $V(x^2)(L)$ and $V(x)(L)$ consist of the single L -point 0, but $V(x^2) \neq V(x)$: their points with values in $R = k[\varepsilon]/(\varepsilon^2)$ differ.

The rational points do not determine a variety either.

EXAMPLE 2.1.10. Consider the three closed subvarieties

$$C_1 := V_+(X^2 + Y^2 + Z^2), \quad C_2 := V_+(X^2 + Y^2 - 3Z^2), \quad C_3 := V_+(1)$$

of $\mathbb{P}_{\mathbb{Q}}^2$. None of them has a rational point.

For C_1 there is nothing to prove over $\mathbb{Q}$: already $C_1(\mathbb{R}) = \emptyset$, since a sum of three squares of real numbers vanishes only when all three do. The variety C_3 is empty, so it has no points with values in any $\mathbb{Q}$ -algebra at all.

For C_2 we argue by descent. A point of $C_2(\mathbb{Q})$ can be scaled to a solution $(X_0, Y_0, Z_0) \in \mathbb{Z}^3$ with $\gcd(X_0, Y_0, Z_0) = 1$. Reduce the equation $X_0^2 + Y_0^2 = 3Z_0^2$ modulo 3. Since -1 is not a square modulo 3, the congruence $X_0^2 + Y_0^2 \equiv 0 \pmod{3}$ forces $3 \mid X_0$ and $3 \mid Y_0$. Then $9 \mid 3Z_0^2$, so $3 \mid Z_0$, contradicting $\gcd(X_0, Y_0, Z_0) = 1$.

The three varieties are nevertheless pairwise distinct: their homogeneous ideals differ, and this is already visible on points with values in larger fields. Indeed $C_3(\mathbb{C}) = \emptyset$, while $C_1(\mathbb{C}) \neq \emptyset$; and $C_1(\mathbb{R}) = \emptyset$, while $(\sqrt{3} : 0 : 1) \in C_2(\mathbb{R})$.

EXAMPLE 2.1.11. The cubic $E := V_+(Y^2Z - X^3 - 17Z^3) \subseteq \mathbb{P}_{\mathbb{Q}}^2$ is the projective closure of the affine curve $y^2 = x^3 + 17$, where $x = X/Z$ and $y = Y/Z$. Setting $Z = 0$ leaves $X^3 = 0$, so E has a single point at infinity, $(0 : 1 : 0)$, and

$$E(\mathbb{Q}) = \{(x, y) \in \mathbb{Q}^2 : y^2 = x^3 + 17\} \cup \{(0 : 1 : 0)\},$$

which contains for instance $(-2, 3)$ and $(5234, 378661)$ [Sil09, Examples I.1.3.3 and I.2.8].

The Galois group Gal_k acts on $X(k^{\text{sep}})$ through its action on k^{sep} ; on k^{sep} -points of $\mathbb{A}^n$ or $\mathbb{P}^n$ it acts on coordinates.

PROPOSITION 2.1.12. *Let X be a k -variety. Then $X(k) = X(k^{\text{sep}})^{\text{Gal}_k}$.*

PROOF. Every k^{sep} -point of X lies in $U(k^{\text{sep}})$ for some affine open subvariety $U \subseteq X$, and $U(k^{\text{sep}})$ is stable under Gal_k . So it suffices to treat closed subvarieties of $\mathbb{A}_k^n$, where the claim is $(k^{\text{sep}})^{\text{Gal}_k} = k$. For a closed subvariety of $\mathbb{P}_k^n$ one can also argue in homogeneous coordinates: let $P = (a_0 : \dots : a_n)$ be fixed by Gal_k . Then $\sigma(a_i) = \lambda_\sigma a_i$ for some $\lambda_\sigma \in (k^{\text{sep}})^\times$, and $\sigma \mapsto \lambda_\sigma$ is a 1-cocycle. By Hilbert's Theorem 90 there is $\alpha \in (k^{\text{sep}})^\times$ with $\lambda_\sigma = \alpha/\sigma(\alpha)$, and then all αa_i lie in k [Sil09, Exercise 1.12]. $\square$

WARNING 2.1.13. [Sil09] works with $\bar{k}$ instead of k^{sep} . The fixed field of $\text{Aut}(\bar{k}/k)$ is the perfect closure k^{perf} , so $X(\bar{k})^{\text{Aut}(\bar{k}/k)} = X(k^{\text{perf}})$, which can be larger than $X(k)$ when k is imperfect. For example, if $k = \mathbb{F}_p(t)$ and $X = V(x^p - t) \subseteq \mathbb{A}_k^1$, then $X(k) = \emptyset$, while the unique point $t^{1/p}$ of $X(\bar{k})$ is fixed by $\text{Aut}(\bar{k}/k)$.

REMARK 2.1.14. In [Sil09, Section I.1], whose ground field K is our k , a variety V/K is a subset of $\mathbb{A}^n(\bar{k})$ or $\mathbb{P}^n(\bar{k})$, cut out by an ideal $I(V/K)$ with coefficients in K . It corresponds to the k -variety $X := V(I(V/K))$ or $V_+(I(V/K))$: then $V = X(\bar{k})$ and $V(K) = X(k)$. The standing assumptions of [Sil09] (that $I(V/K)$ generates the ideal of V , and that it is prime over $\bar{k}$) say that X is geometrically integral (Definition 2.2.2).

2.2. Base extension and geometric properties

DEFINITION 2.2.1. Let X be a k -variety and $L \supseteq k$ a field extension. The **base extension** of X to L is the L -variety $X_L := X \times_{\text{Spec } k} \text{Spec } L$.

Concretely, X_L is defined by the same equations as X , with the coefficients viewed in L : if $X = V(f_1, \dots, f_m) \subseteq \mathbb{A}_k^n$, then $X_L = V(f_1, \dots, f_m) \subseteq \mathbb{A}_L^n$, and likewise for projective varieties. We have $X(L) = X_L(L)$ [Poo06, Section 1.3].

DEFINITION 2.2.2. A k -variety X is geometrically reduced, geometrically irreducible or **geometrically integral** if $X_{\bar{k}}$ is reduced, irreducible or integral (Definition 2.1.8), respectively.

DEFINITION 2.2.3. Let X be an integral k -variety. Its **function field** $k(X)$ is the local ring at the generic point of X : for $X = \text{Spec } A$ it is $\text{Frac } A$, and in general it is the function field of any non-empty affine open subvariety. Its elements are the rational functions on X [Poo06, Definitions 1.4.4 and 1.4.8].

PROPOSITION 2.2.4. *Let X be an integral k -variety.*

- (1) *X is geometrically irreducible if and only if k is separably closed in $k(X)$.*
- (2) *X is geometrically reduced if and only if the extension $k(X)/k$ is separable, i.e. separably generated.*
- (3) *If k is perfect, then X is geometrically reduced.*

PROOF. Part 1 is [Poo06, Remarks 1.4.6 and 1.4.7]. Part 3 follows from part 2, since every finitely generated extension of a perfect field is separably generated. $\square$

EXAMPLE 2.2.5. The curve $X := V(x^2 - 2y^2) \subseteq \mathbb{A}_{\mathbb{Q}}^2$ is integral, since $x^2 - 2y^2$ is irreducible in $\mathbb{Q}[x, y]$. Over $\mathbb{Q}(\sqrt{2})$ it becomes the union of the lines $x = \pm\sqrt{2}y$, so X is not geometrically irreducible. Accordingly $x/y \in \mathbb{Q}(X)$ satisfies $(x/y)^2 = 2$ [Poo06, Example 1.4.3].

EXAMPLE 2.2.6. Let $k := \mathbb{F}_p(a)$ and $f := x^p + ay^p$. Then f is irreducible in $k[x, y]$, so $X := V(f) \subseteq \mathbb{A}_k^2$ is integral. Over $\bar{k}$, $f = (x + \alpha y)^p$ with $\alpha^p = a$, so X is not geometrically reduced [Mil20, Example I.1.2]. Accordingly $(x/y)^p = -a$ in $k(X)$, so $k(X)$ contains the inseparable extension $k(a^{1/p})$ of k .

2.3. Morphisms and rational maps

DEFINITION 2.3.1. Let X be an integral k -variety and Y a k -variety. A **rational map** $\varphi: X \dashrightarrow Y$ is an equivalence class of morphisms $U \rightarrow Y$ with $U \subseteq X$ a non-empty open subvariety, two being equivalent if they agree on the intersection of their domains. It is dominant if its image is dense. Integral k -varieties

X and Y are birational if there are dominant rational maps $X \dashrightarrow Y$ and $Y \dashrightarrow X$ inverse to each other [Poo06, Definitions 1.5.2, 1.5.3 and 1.5.5].

Explicitly, a rational map $\varphi: X \dashrightarrow \mathbb{P}_k^n$ is a tuple $(f_0 : \cdots : f_n)$ of elements of $k(X)$, not all zero, modulo multiplication by $k(X)^\times$. It is regular at a point P of X if there is $g \in k(X)^\times$ such that every gf_i lies in $\mathcal{O}_{X,P}$ and some $(gf_i)(P) \in k(P)$ is non-zero; then $\varphi(P) := ((gf_0)(P) : \cdots : (gf_n)(P))$. A rational map to a closed subvariety $Y \subseteq \mathbb{P}_k^n$ is one whose image lies in Y , and it is a morphism if it is regular at every point [Poo06, Definitions 1.5.8 and 1.5.9], [Sil09, Section I.3]. For $X \subseteq \mathbb{P}_k^m$, clearing denominators writes φ as a tuple of homogeneous polynomials of the same degree, not all in the ideal of X [Sil09, Remark I.3.2].

REMARK 2.3.2. A morphism of k -varieties is automatically defined over k . A map "defined over $\bar{k}$ " in the sense of [Sil09, Section I.3] is a morphism $X_{\bar{k}} \rightarrow Y_{\bar{k}}$ [Poo06, Warning 1.5.7]. Varieties X and Y are isomorphic over L if $X_L \cong Y_L$.

THEOREM 2.3.3. *The functor $X \mapsto k(X)$ is an equivalence between the category of integral k -varieties with dominant rational maps and the opposite of the category of finitely generated field extensions of k with k -algebra homomorphisms. In particular, integral k -varieties X and Y are birational if and only if $k(X) \cong k(Y)$ over k .*

PROOF. See [Poo06, equation (1.5.4) and Definition 1.5.6]. $\square$

EXAMPLE 2.3.4. Suppose $\text{char } k \neq 2$, and let $C := V_+(X^2 + Y^2 - Z^2) \subseteq \mathbb{P}_k^2$. The rational map $\varphi := (X + Z : Y) : C \dashrightarrow \mathbb{P}_k^1$ is regular away from the rational point $(1 : 0 : -1)$. Since $(X + Z)(X - Z) = -Y^2$ on C , also $\varphi = (-Y : X - Z)$, which is regular at $(1 : 0 : -1)$ with value $(0 : 1)$. So φ is a morphism. The morphism

$$\psi: \mathbb{P}_k^1 \rightarrow C, \quad (S : T) \mapsto (S^2 - T^2 : 2ST : S^2 + T^2)$$

is inverse to φ , so $C \cong \mathbb{P}_k^1$ [Sil09, Example I.3.5].

EXAMPLE 2.3.5. Suppose $\text{char } k \neq 2$, and let $C := V_+(Y^2Z - X^3 - X^2Z) \subseteq \mathbb{P}_k^2$. The morphism

$$\psi: \mathbb{P}_k^1 \rightarrow C, \quad (S : T) \mapsto ((S^2 - T^2)T : (S^2 - T^2)S : T^3)$$

and the rational map $\varphi := (Y : X) : C \dashrightarrow \mathbb{P}_k^1$ are inverse to each other where defined, so C is birational to $\mathbb{P}_k^1$. But φ is not regular at $(0 : 0 : 1)$, and ψ sends both $(1 : 1)$ and $(1 : -1)$ to $(0 : 0 : 1)$, so $C \not\cong \mathbb{P}_k^1$. The rational point $(0 : 0 : 1)$ is singular (Example 2.4.5) [Sil09, Example I.3.7].

EXAMPLE 2.3.6. Let $C := V_+(X^2 + Y^2 - Z^2)$ and $C_2 := V_+(X^2 + Y^2 - 3Z^2)$ in $\mathbb{P}_{\mathbb{Q}}^2$. By Examples 2.3.4 and 2.1.10, $C(\mathbb{Q})$ is infinite and $C_2(\mathbb{Q})$ is empty, so $C \not\cong C_2$. But $(X : Y : Z) \mapsto (X : Y : \sqrt{3}Z)$ is an isomorphism $(C_2)_{\mathbb{Q}(\sqrt{3})} \cong C_{\mathbb{Q}(\sqrt{3})}$ [Sil09, Example I.3.8].

2.4. Dimension, regularity and smoothness

The dimension of a k -variety is its dimension as a topological space. If X is integral, then $\dim X = \text{trdeg}_k k(X)$, and $\dim X_L = \dim X$ for every field extension $L \supseteq k$ [Poo06, Section 1.6].

DEFINITION 2.4.1. Let X be a k -variety.

- (1) X is **regular** at a point P if the local ring $\mathcal{O}_{X,P}$ is regular, i.e. $\dim_{k(P)} \mathfrak{m}_P / \mathfrak{m}_P^2 = \dim \mathcal{O}_{X,P}$. It is regular if it is regular at every point.
- (2) Suppose $X = V(f_1, \dots, f_m) \subseteq \mathbb{A}_k^n$ has pure dimension d . Then X is **smooth** at $P \in X(\bar{k})$ if the Jacobian matrix $(\partial f_i / \partial x_j(P))$ has rank $n - d$. A projective variety is smooth at P if some (equivalently, every) affine patch containing P is. X is smooth if it is smooth at every $P \in X(\bar{k})$ [Poo06, Definition 1.7.1].
- (3) X is **nice** if it is smooth, projective and geometrically integral [Poo06, Definition 1.7.8].

Smoothness is tested at $\bar{k}$ -points, so it is a geometric property: X is smooth if and only if X_L is, for any field $L \supseteq k$ [Poo06, Remark 1.7.6]. Regularity is not.

PROPOSITION 2.4.2. *Let X be a k -variety.*

- (1) *X is smooth if and only if $X_{\bar{k}}$ is regular. In particular, a smooth variety is regular.*
- (2) *If k is perfect, then X is smooth if and only if it is regular.*
- (3) *Let X be integral, and $P \in X(\bar{k})$. Then X is smooth at P if and only if $\dim_{\bar{k}} \mathfrak{m}_P / \mathfrak{m}_P^2 = \dim X$, where $\mathfrak{m}_P$ is the maximal ideal of $\mathcal{O}_{X_{\bar{k}}, P}$.*

PROOF. Part 3 is [Sil09, Proposition I.1.7], and it gives part 1 at closed points. Over an imperfect field, regular does not imply smooth: see Example 2.4.6. $\square$

REMARK 2.4.3. The "smooth curves" of [Sil09, Chapter II] are the nice k -varieties of dimension 1.

A singular point of the hypersurface $V(f) \subseteq \mathbb{A}_k^n$ is a common zero of f and its partial derivatives [Poo06, Example 1.7.4].

EXAMPLE 2.4.4. Suppose $\text{char } k \neq 2$. On $y^2 = x^3 + x$, a singular point satisfies $2y = 3x^2 + 1 = 0$ and $x(x^2 + 1) = 0$, which is impossible. So this curve is smooth [Sil09, Example I.1.6].

EXAMPLE 2.4.5. Suppose $\text{char } k \neq 2$. On $y^2 = x^3 + x^2$, the conditions $2y = 3x^2 + 2x = 0$ and $x^2(x + 1) = 0$ leave only $(0, 0)$, which is a singular point [Sil09, Example I.1.6].

EXAMPLE 2.4.6. Let $k := \mathbb{F}_3(t)$, let $F := Y^2Z - X^3 + tZ^3$, and let $E := V_+(F) \subseteq \mathbb{P}_k^2$, the curve $y^2 = x^3 - t$. Let $s \in \bar{k}$ with $s^3 = t$. Since $\text{char } k = 3$,

$$\frac{\partial F}{\partial X} = 0, \quad \frac{\partial F}{\partial Y} = 2YZ, \quad \frac{\partial F}{\partial Z} = Y^2,$$

and F vanishes at $(s : 0 : 1)$ too, so this is a singular point and E is not smooth. Over $\bar{k}$ the equation is $y^2 = (x - s)^3$, a cuspidal cubic, which is integral; so E is geometrically integral. Yet E is regular. Every other $\bar{k}$ -point is smooth, and the closed point $P := V(y, x^3 - t)$, with residue field $k(s)$ of degree 3, has maximal ideal $(y, x^3 - t) = (y)$ in $\mathcal{O}_{E, P}$, because $x^3 - t = y^2$ there. So $\mathcal{O}_{E, P}$ is a discrete valuation ring.

Algebraic curves

Sources. This chapter follows [Poo06, Chapter 2] and [Sil09, Chapter II], with material on plane curves from [Mil20, Sections I.1 and I.6]. Unlike these sources, we do not assume that the ground field is perfect.

We keep the notation of the previous chapter: k is an arbitrary field, and $\text{Gal}_k := \text{Gal}(k^{\text{sep}}/k)$.

DEFINITION 3.0.1. A **curve** over k is an integral k -variety of dimension 1. A nice curve is a curve that is nice (Definition 2.4.1).

Standing hypothesis. Unless stated otherwise, C , C_1 and C_2 denote nice curves over k . Remarks point out where a curve that is only regular behaves differently; the basic example is Example 2.4.6.

3.1. Closed points and local rings

DEFINITION 3.1.1. A **closed point** of a k -variety X is a point whose residue field $k(P)$ is a finite extension of k . Its degree is $\deg P := [k(P) : k]$ [Poo06, Definition 2.2.1].

The rational points of X are the closed points of degree 1.

PROPOSITION 3.1.2. *Let X be a k -variety. Sending a $\bar{k}$ -point to its image is a surjection from $X(\bar{k})$ onto the set of closed points of X , whose fibers are the $\text{Aut}(\bar{k}/k)$ -orbits. The fiber over P has $[k(P) : k]_s$ elements, and it meets $X(k^{\text{sep}})$ if and only if $k(P)/k$ is separable.*

PROOF. A $\bar{k}$ -point over P is the same as a k -embedding $k(P) \hookrightarrow \bar{k}$. There are $[k(P) : k]_s$ of these, permuted transitively by $\text{Aut}(\bar{k}/k)$, and they land in k^{sep} exactly when $k(P)/k$ is separable. $\square$

EXAMPLE 3.1.3. On $E: y^2 = x^3 + 17$ over $\mathbb{Q}$, the closed point $P := V(x, y^2 - 17)$ has residue field $\mathbb{Q}(\sqrt{17})$ and degree 2. It corresponds to the orbit $\{(0, \sqrt{17}), (0, -\sqrt{17})\}$ in $E(\overline{\mathbb{Q}})$.

EXAMPLE 3.1.4. Let $k := \mathbb{F}_p(t)$. The closed point $P := V(x^p - t)$ of $\mathbb{A}_k^1$ has degree p and residue field $k(t^{1/p})$. It has a single $\bar{k}$ -point above it, and no k^{sep} -point.

PROPOSITION 3.1.5. *Let X be a curve over k , and P a closed point. The local ring $\mathcal{O}_{X,P}$ is a discrete valuation ring if and only if X is regular at P . In particular this holds at every closed point of a nice curve.*

PROOF. A noetherian local domain of dimension 1 is a discrete valuation ring if and only if its maximal ideal is principal, i.e. it is regular [Sil09, Proposition II.1.1 and Exercise 2.1]. $\square$

DEFINITION 3.1.6. Let P be a closed point of C . The normalized valuation of $\mathcal{O}_{C,P}$ extends to $\text{ord}_P: k(C)^\times \rightarrow \mathbb{Z}$. A **uniformizer** at P is a $t \in k(C)$ with $\text{ord}_P(t) = 1$. If $\text{ord}_P(f) > 0$, then f has a zero at P ; if $\text{ord}_P(f) < 0$, a pole. If $\text{ord}_P(f) \geq 0$, then f is regular at P , with value $f(P) \in k(P)$ [Sil09, Section II.1].

PROPOSITION 3.1.7. *Let $f \in k(C)^\times$. Then f has only finitely many zeros and poles. If f has no poles, then $f \in k$.*

PROOF. See [Sil09, Proposition II.1.2]. For the last statement over k , note that k is algebraically closed in $k(C)$, since C is geometrically integral (Proposition 2.2.4). $\square$

EXAMPLE 3.1.8. Let $f(x, y) = 0$ be an affine patch of C , and $P = (a, b) \in C(k)$. If $\partial f / \partial x(P) \neq 0$, then $y - b$ is a uniformizer at P ; if $\partial f / \partial y(P) \neq 0$, then $x - a$ is [Poo06, Example 1.8.5]. For instance, on $y^2 = x^3 + x$ with $\text{char } k \neq 2$, the function y is a uniformizer at $P = (0, 0)$. Since $x = y^2 / (x^2 + 1)$, we get $\text{ord}_P(x) = 2$, and $\text{ord}_P(2y^2 - x) = \text{ord}_P(x(2x^2 + 1)) = 2$ [Sil09, Example II.1.3].

PROPOSITION 3.1.9. *Let $P \in C(k)$, and let t be a uniformizer at P . Then $k(C)$ is a finite separable extension of $k(t)$.*

PROOF. [Sil09, Proposition II.1.4] proves this for perfect k , by taking p -th roots of coefficients. In general: since $k(P) = k$, the map $\mathfrak{m}_P / \mathfrak{m}_P^2 \rightarrow \Omega_{C/k, P} \otimes k(P)$ is an isomorphism [Har77, Proposition II.8.7]. So dt generates the stalk $\Omega_{C/k, P}$, and in particular $dt \neq 0$ in $\Omega_{k(C)/k}$. Conclude by Proposition 3.4.2. $\square$

The hypothesis $P \in C(k)$ cannot be dropped: see Warning 3.4.5.

3.2. Morphisms of curves

PROPOSITION 3.2.1. *Let X be a curve over k , let P be a closed point at which X is regular, and let $Y \subseteq \mathbb{P}_k^n$ be a projective variety. Then every rational map $\varphi: X \dashrightarrow Y$ is regular at P . In particular, if X is regular, then φ is a morphism.*

PROOF. Write $\varphi = (f_0 : \dots : f_n)$, let t be a uniformizer at P and $m := \min_i \text{ord}_P(f_i)$. Then every $t^{-m} f_i$ is regular at P , and some $(t^{-m} f_i)(P) \in k(P)$ is non-zero [Sil09, Proposition II.2.1]. The argument only uses that $\mathcal{O}_{X, P}$ is a discrete valuation ring, so it applies to closed points of any degree. $\square$

If $f: X \rightarrow Y$ is a holomorphic map between Riemann surfaces and X is compact, then f is either constant or surjective. The algebraic analog is the following.

PROPOSITION 3.2.2. *A morphism from a smooth projective curve to a curve is either constant or surjective.*

PROOF. See [Har77, Proposition II.6.8]. $\square$

THEOREM 3.2.3. *The functor $X \mapsto k(X)$ is an equivalence between the category of regular projective curves over k with non-constant morphisms, and the opposite of the category of finitely generated extensions K/k of transcendence degree 1 with k -embeddings. Under this equivalence, X is geometrically integral if and only if k is algebraically closed in $k(X)$ and $k(X)/k$ is separable.*

PROOF. See [Poo06, Section 2.1.2 and Remark 2.1.3] and [Mil20, Theorem I.6.3] for the equivalence. The last statement follows from Proposition 2.2.4. If $k(X)/k$ is separable, an element of $k(X)$ that is algebraic over k is separable over k . $\square$

REMARK 3.2.4. [Sil09, Remark II.2.5] states the equivalence for smooth curves and function fields in which k is algebraically closed. Over an imperfect field, the regular projective model of such a function field need not be smooth. The function field $\mathbb{F}_3(t)(x, y)$, with $y^2 = x^3 - t$, satisfies both conditions of Theorem 3.2.3, but its regular projective model is the curve of Example 2.4.6, which is not smooth. So no nice curve has this function field.

DEFINITION 3.2.5. Let $\varphi: C_1 \rightarrow C_2$ be a non-constant morphism, and $\varphi^*: k(C_2) \hookrightarrow k(C_1)$ the induced embedding. The **degree** of φ is $\deg \varphi := [k(C_1) : \varphi^* k(C_2)]$; a constant morphism has degree 0. The morphism φ is separable, inseparable or purely inseparable if the extension $k(C_1)/\varphi^* k(C_2)$ is, and $\deg_s \varphi$

and $\deg_i \varphi$ are its separable and inseparable degrees [Sil09, Section II.2], [Poo06, Definition 2.1.4]. The norm gives $\varphi_* := (\varphi^*)^{-1} \circ \text{Nm}_{k(C_1)/\varphi^*k(C_2)} : k(C_1) \rightarrow k(C_2)$.

COROLLARY 3.2.6. *A morphism of degree 1 between regular projective curves is an isomorphism.*

PROOF. This follows from Theorem 3.2.3; compare [Sil09, Corollary II.2.4.1]. $\square$

DEFINITION 3.2.7. Let $\varphi : C_1 \rightarrow C_2$ be non-constant, P a closed point of C_1 , and $Q := \varphi(P)$ with uniformizer t_Q . The **ramification index** of φ at P is $e_\varphi(P) := \text{ord}_P(\varphi^*t_Q) \geq 1$, and the residue degree is $f_\varphi(P) := [k(P) : k(Q)]$. The morphism φ is unramified at P if $e_\varphi(P) = 1$ [Sil09, Section II.2].

PROPOSITION 3.2.8. *Let $\varphi : C_1 \rightarrow C_2$ be non-constant.*

- (1) *For every closed point Q of C_2 , $\sum_{P \mapsto Q} e_\varphi(P) f_\varphi(P) = \deg \varphi$.*
- (2) *For all but finitely many $Q \in C_2(\bar{k})$, the fiber of $\varphi_{\bar{k}}$ over Q has $\deg_s \varphi$ elements.*
- (3) *If $\psi : C_2 \rightarrow C_3$ is non-constant, then $e_{\psi \circ \varphi}(P) = e_\varphi(P) e_\psi(\varphi(P))$.*

PROOF. Over $\bar{k}$ all residue degrees are 1, and this is [Sil09, Proposition II.2.6]. Over k , part 1 is the formula $\sum e_i f_i = n$ for the finite extension of Dedekind domains $\mathcal{O}_{C_2, Q} \subseteq \mathcal{O}_{C_1}$ above Q [Poo06, Remark 2.5.3]. $\square$

EXAMPLE 3.2.9. The morphism $\varphi : \mathbb{P}_k^1 \rightarrow \mathbb{P}_k^1$ given by $\varphi = (X^3(X - Y)^2 : Y^5)$ has degree 5. The fiber over $(0 : 1)$ consists of $(0 : 1)$, with $e_\varphi = 3$, and $(1 : 1)$, with $e_\varphi = 2$; and indeed $3 + 2 = 5$ [Sil09, Example II.2.9].

EXAMPLE 3.2.10. Suppose $\text{char } k \neq 2$. Let $E : y^2 = p(x)$ with p a separable cubic, a smooth plane cubic by Example 3.7.9 below, and let $O = (0 : 1 : 0)$. The morphism $x : E \rightarrow \mathbb{P}_k^1$ has degree 2, since $k(E) = k(x)(y)$ with $y^2 \in k(x)$, and it is separable. Above ∞ lies only O , with $e_x(O) = 2$. Above the closed point of $\mathbb{A}_k^1$ given by an irreducible factor g of p lies only the closed point $V(g, y)$, with ramification index 2. All other closed points are unramified. In characteristic 2 the extension $k(x, y)/k(x)$ defined by $y^2 = p(x)$ is purely inseparable [Mil20, Example I.4.25], but in that case this curve is singular, again by Example 3.7.9; see Example 3.6.4 instead.

DEFINITION 3.2.11. Suppose $\text{char } k = p > 0$, and let $q = p^r$. Let $C \subseteq \mathbb{P}_k^n$ be defined by homogeneous polynomials $G_1, \dots, G_m$, and let $C^{(q)}$ be defined by the polynomials $G_i^{(q)}$ obtained by raising every coefficient to the q -th power. Equivalently, $C^{(q)}$ is the base change of C along $k \rightarrow k, a \mapsto a^q$. The **relative Frobenius morphism** is

$$F : C \rightarrow C^{(q)}, \quad (X_0 : \dots : X_n) \mapsto (X_0^q : \dots : X_n^q).$$

It lands in $C^{(q)}$ because $G^{(q)}(X_0^q, \dots, X_n^q) = G(X_0, \dots, X_n)^q$ [Sil09, Section II.2].

PROPOSITION 3.2.12. *Let $F : C \rightarrow C^{(q)}$ be the relative Frobenius morphism.*

- (1) *$F^*k(C^{(q)}) = k \cdot k(C)^q$, the compositum of k and $k(C)^q$ in $k(C)$.*
- (2) *F is purely inseparable of degree q .*

PROOF. For perfect k , part 1 reads $F^*k(C^{(q)}) = k(C)^q$, and this is [Sil09, Proposition II.2.11]. In general, suppose X_0 does not vanish on C , and let $x_i = X_i/X_0$ be the affine coordinates on the patch U_0 . Then $k(C)$ and $k(C^{(q)})$ are generated over k by the x_i , and $F^*(x_i) = x_i^q$. Since F^* is k -linear, its image is $k(x_i^q)$. This equals $k \cdot k(C)^q$: for $g \in k(C)$, written as a rational function in the x_i with coefficients in k , we have $g^q = g^{(q)}(x_i^q)$.

For part 2, pure inseparability follows from part 1. By induction on r it suffices to treat $q = p$. Since C is geometrically reduced, $k(C)/k$ is separably generated (Proposition 2.2.4): choose $t \in k(C)$ with $k(C)/k(t)$ finite separable. A separable algebraic extension L/K satisfies $L = KL^p$, so $k(C) = (k \cdot k(C)^p)(t)$, and $t^p \in k \cdot k(C)^p$. So the degree is at most p . Finally $t \notin k \cdot k(C)^p$, since d vanishes on $k \cdot k(C)^p$ but $dt \neq 0$ (Proposition 3.4.2). $\square$

WARNING 3.2.13. If k is imperfect, then $k(C)^q \subsetneq k \cdot k(C)^q$, because an element $a \in k - k^q$ is not a q -th power in $k(C)$. Indeed, k is algebraically closed in $k(C)$. So [Sil09, Proposition II.2.11(a)] must be read with $k \cdot k(C)^q$ in place of $k(C)^q$.

COROLLARY 3.2.14. Suppose $\text{char } k = p > 0$. Let $\psi: C_1 \rightarrow C_2$ be non-constant, and $q := \deg_i \psi$. Then $\psi = \lambda \circ F$, with $F: C_1 \rightarrow C_1^{(q)}$ the relative Frobenius morphism and λ separable.

PROOF. Let K be the separable closure of $\psi^*k(C_2)$ in $k(C_1)$. Then $k(C_1)/K$ is purely inseparable of degree q , so $k(C_1)^q \subseteq K$. Since also $k \subseteq K$, we get $k \cdot k(C_1)^q \subseteq K$. Both have index q in $k(C_1)$, by Proposition 3.2.12, so they are equal. Theorem 3.2.3 then gives λ . Compare [Sil09, Corollary II.2.12]. $\square$

EXAMPLE 3.2.15. If $E: Y^2Z = X^3 + AXZ^2 + BZ^3$, then $E^{(q)}: Y^2Z = X^3 + A^qXZ^2 + B^qZ^3$ [Sil09, Example II.2.10]. If $A, B \in \mathbb{F}_q$, then $E^{(q)} = E$, and the relative Frobenius morphism is an endomorphism of E of degree q .

DRAFT
21 Sep 2026

3.3. Divisors

DEFINITION 3.3.1. The group $\text{Div } C$ of **divisors** on C is the free abelian group on the closed points of C . The degree of $D = \sum n_P P$ is $\deg D := \sum n_P \deg P$, and $\text{Div}^0 C$ is the kernel of the degree. D is effective, $D \geq 0$, if all $n_P \geq 0$. For $f \in k(C)^\times$, the **principal divisor** of f is $\text{div}(f) := \sum_P \text{ord}_P(f)P$. Divisors are linearly equivalent if their difference is principal. The **Picard group** is $\text{Pic } C := \text{Div } C / \text{div}(k(C)^\times)$, and $\text{Pic}^0 C$ is the image of $\text{Div}^0 C$ [Poo06, Sections 1.9 and 2.2].

PROPOSITION 3.3.2. Let $f \in k(C)^\times$. Then $\text{div}(f) = 0$ if and only if $f \in k^\times$, and $\deg \text{div}(f) = 0$. So there is an exact sequence

$$1 \rightarrow k^\times \rightarrow k(C)^\times \xrightarrow{\text{div}} \text{Div}^0 C \rightarrow \text{Pic}^0 C \rightarrow 0.$$

PROOF. See [Sil09, Proposition II.3.1 and Remark II.3.4], and [Poo06, Section 2.2.2]; the first statement is Proposition 3.1.7. $\square$

DEFINITION 3.3.3. Let $f \in \bar{k}(C)^\times$ and let $D = \sum n_P P \in \text{Div } C_{\bar{k}}$ be a divisor whose support is disjoint from that of $\text{div}(f)$. The value of f at D is

$$f(D) := \prod_P f(P)^{n_P} \in \bar{k}^\times,$$

the product over the points $P \in C(\bar{k})$ in the support of D [Poo06, Section 1.9.2].

Each factor is a well defined element of $\bar{k}^\times$: the function f has neither a zero nor a pole at P , since P is not in the support of $\text{div}(f)$. Two properties are immediate from the definition. First, $D \mapsto f(D)$ is a homomorphism on the group of divisors supported away from $\text{div}(f)$, and $f \mapsto f(D)$ is a homomorphism on the functions whose divisors are supported away from D . Second, evaluation commutes with the Galois action: for $\sigma \in \text{Aut}(\bar{k}/k)$,

$$\sigma(f(D)) = f^\sigma(\sigma D),$$

because $\sigma(f(P)) = f^\sigma(\sigma P)$ on each factor. In particular, if f is defined over k , then $\sigma(f(D)) = f(\sigma D)$.

THEOREM 3.3.4 (Weil reciprocity). Let $f, g \in \bar{k}(C)^\times$ be functions on a nice curve C such that $\text{div}(f)$ and $\text{div}(g)$ have disjoint supports. Then

$$f(\text{div } g) = g(\text{div } f).$$

PROOF. See [Poo06, Theorem 1.9.5] and [Sil09, Exercise 2.11]. Poonen states [Weil reciprocity](#) over an algebraically closed field; since divisors on $C_{\bar{k}}$ are supported on the points of $C(\bar{k})$, that is exactly the statement above. $\square$

The content of the theorem is already visible on $\mathbb{P}^1$. If f and g are polynomials in $k[x]$ with no common root, then the affine zeros contribute $\prod_{g(\beta)=0} f(\beta)$ to the left side and $\prod_{f(\alpha)=0} g(\alpha)$ to the right, and both products are the resultant $\text{Res}(f, g)$ up to the sign $(-1)^{\deg f \deg g}$ and the leading coefficients; the poles at ∞ supply exactly those factors. So reciprocity is a geometric form of the symmetry of the resultant. We use Theorem 3.3.4 to construct the Weil pairing in Section 4.6.

DEFINITION 3.3.5. Let $\varphi: C_1 \rightarrow C_2$ be non-constant. On closed points, set

$$\varphi^*Q := \sum_{P \mapsto Q} e_\varphi(P) P, \quad \varphi_*P := f_\varphi(P) \varphi(P),$$

and extend linearly to $\varphi^*: \text{Div } C_2 \rightarrow \text{Div } C_1$ and $\varphi_*: \text{Div } C_1 \rightarrow \text{Div } C_2$ [Poo06, Section 2.2.3].

PROPOSITION 3.3.6. *Let $\varphi: C_1 \rightarrow C_2$ be non-constant.*

- (1) $\deg \varphi^*D = (\deg \varphi)(\deg D)$ and $\deg \varphi_*D = \deg D$.
- (2) $\varphi^* \text{div}(f) = \text{div}(\varphi^*f)$ and $\varphi_* \text{div}(f) = \text{div}(\varphi_*f)$.
- (3) $\varphi_* \circ \varphi^*$ is multiplication by $\deg \varphi$ on $\text{Div } C_2$.
- (4) $(\psi \circ \varphi)^* = \varphi^* \circ \psi^*$ and $(\psi \circ \varphi)_* = \psi_* \circ \varphi_*$.

So φ^ and φ_* induce homomorphisms between $\text{Pic } C_1$ and $\text{Pic } C_2$, and between $\text{Pic}^0 C_1$ and $\text{Pic}^0 C_2$.*

PROOF. Part 1 follows from Proposition 3.2.8. For the rest see [Sil09, Proposition II.3.6] and [Poo06, Section 2.2.3]. $\square$

[Sil09, Section II.3] works with divisors on $C_{\bar{k}}$. His $\text{Div}_k(C)$ consists of the $\text{Aut}(\bar{k}/k)$ -invariant divisors on $C_{\bar{k}}$, and $\text{Pic}_k(C) := (\text{Pic } C_{\bar{k}})^{\text{Aut}(\bar{k}/k)}$. The dictionary with our groups is as follows; compare [Poo06, Warning 1.9.6].

PROPOSITION 3.3.7.

- (1) *Pulling back along $C_{k^{\text{sep}}} \rightarrow C$ identifies $\text{Div } C$ with $(\text{Div } C_{k^{\text{sep}}})^{\text{Gal}_k}$, compatibly with degrees and principal divisors. A closed point P goes to the sum, with multiplicity 1, of the closed points of $C_{k^{\text{sep}}}$ above P .*
- (2) *Pulling back along $C_{\bar{k}} \rightarrow C$ sends a closed point P to $[k(P) : k]_i$ times the sum of the points of $C(\bar{k})$ above P . This is injective, compatible with degrees and principal divisors, and lands in $\text{Div}_k(C)$. It is onto $\text{Div}_k(C)$ if k is perfect, but not in general.*
- (3) *The homomorphism $\text{Pic } C \rightarrow (\text{Pic } C_{k^{\text{sep}}})^{\text{Gal}_k}$ is injective, but need not be surjective.*

PROOF. For part 1: $k(P) \otimes_k k^{\text{sep}}$ is reduced, since k^{sep}/k is separable, so every closed point of $C_{k^{\text{sep}}}$ above P has multiplicity 1. They form one Gal_k -orbit. For part 2: $k(P) \otimes_k \bar{k}$ has length $[k(P) : k]_i$ at each of its $[k(P) : k]_s$ $\bar{k}$ -points (Proposition 3.1.2). Part 3 follows from Hilbert's Theorem 90 [Poo06, Section 1.9.3 and Exercise 1.6]. $\square$

WARNING 3.3.8. In general the injection $\text{Pic } C \rightarrow (\text{Pic } C_{k^{\text{sep}}})^{\text{Gal}_k}$ of Proposition 3.3.7 is not surjective: its cokernel embeds in the Brauer group $\text{Br } k$ [Poo17, Corollary 6.7.8]. If C has a rational point, then the obstruction vanishes and $\text{Pic } C \rightarrow (\text{Pic } C_{k^{\text{sep}}})^{\text{Gal}_k}$ is an isomorphism, so the two readings of $\text{Pic}^0 C$ agree. In particular Silverman's $\text{Pic}_k(C)$ need not be the quotient of $\text{Div}_k(C)$ by principal divisors, although it is when C has genus 1 and a rational point [Sil09, Exercise 2.13]. Whenever the distinction matters we say which rational, closed or $\bar{k}$ -points are meant.

EXAMPLE 3.3.9. The degree is an isomorphism $\text{Pic } \mathbb{P}_k^1 \cong \mathbb{Z}$. Indeed, the closed points of $\mathbb{P}_k^1$ other than ∞ correspond to monic irreducible $g_P \in k[x]$, with $\deg P = \deg g_P$. A divisor $D = \sum n_P P + n_\infty \infty$ of degree 0 is $\text{div}(\prod g_P^{n_P})$, since $\text{ord}_\infty(g_P) = -\deg g_P$ [Sil09, Example II.3.2].

EXAMPLE 3.3.10. Suppose $\text{char } k \neq 2$. Let $E: y^2 = (x - e_1)(x - e_2)(x - e_3)$ with distinct $e_i \in k$, let $P_i := (e_i, 0)$ and $O := (0 : 1 : 0)$. Then

$$\text{div}(x - e_i) = 2P_i - 2O, \quad \text{div}(y) = P_1 + P_2 + P_3 - 3O$$

[Sil09, Example II.3.3]. If instead $y^2 = p(x)$ with p irreducible of degree 3, then $\text{div}(y) = P - 3O$, where $P := V(y, p(x))$ is a closed point of degree 3.

DRAFT

21 Sep 2026

3.4. Differentials

DEFINITION 3.4.1. Let X be a curve over k . The space $\Omega_{k(X)/k}$ of **meromorphic differentials** on X is the $k(X)$ -vector space generated by symbols dx , $x \in k(X)$, subject to $d(x + y) = dx + dy$, $d(xy) = x dy + y dx$ and $da = 0$ for $a \in k$. A non-constant morphism $\varphi: X_1 \rightarrow X_2$ induces $\varphi^*: \Omega_{k(X_2)/k} \rightarrow \Omega_{k(X_1)/k}$ [Sil09, Section II.4], [Poo06, Section 2.3.1].

PROPOSITION 3.4.2. *Let X be a curve over k with $k(X)/k$ separable, e.g. a geometrically reduced curve.*

- (1) $\Omega_{k(X)/k}$ has dimension 1 over $k(X)$.
- (2) For $x \in k(X)$, the differential dx is a basis if and only if it is non-zero, if and only if $k(X)/k(x)$ is finite separable.
- (3) A non-constant morphism $\varphi: X_1 \rightarrow X_2$ of such curves is separable if and only if $\varphi^* \neq 0$ on differentials.

PROOF. See [Sil09, Proposition II.4.2] and [Poo06, Proposition 2.3.1]. In characteristic p , the last condition of [Poo06, Proposition 2.3.1], $t \notin k(X)^p$, must be read as $t \notin k \cdot k(X)^p$ when k is imperfect. $\square$

Since C is smooth of dimension 1, the sheaf $\Omega_{C/k}$ is invertible. So for every closed point P the stalk $\Omega_{C/k, P} \subseteq \Omega_{k(C)/k}$ is free of rank 1 over $\mathcal{O}_{C, P}$.

DEFINITION 3.4.3. Let $\omega \in \Omega_{k(C)/k}$ be non-zero, P a closed point, and η a generator of $\Omega_{C/k, P}$. Write $\omega = g\eta$, and set $\text{ord}_P(\omega) := \text{ord}_P(g)$. The divisor of ω is $\text{div}(\omega) := \sum_P \text{ord}_P(\omega)P$. The differential ω is **regular** if $\text{div}(\omega) \geq 0$. A **canonical divisor** is a divisor $K_C := \text{div}(\omega)$. Its class in $\text{Pic } C$ does not depend on ω , since any other non-zero differential is $f\omega$ with $f \in k(C)^\times$ [Sil09, Section II.4], [Poo06, Definitions 2.3.2 to 2.3.4].

PROPOSITION 3.4.4. *Let P be a closed point of C with $k(P)/k$ separable, e.g. $P \in C(k)$, and let t be a uniformizer at P . Then dt generates $\Omega_{C/k, P}$, so $\text{ord}_P(\omega) = \text{ord}_P(\omega/dt)$, as in [Sil09, Proposition II.4.3].*

PROOF. For $P \in C(k)$ see the proof of Proposition 3.1.9. $\square$

WARNING 3.4.5. If $k(P)/k$ is inseparable, a uniformizer need not work. Let $k := \mathbb{F}_p(a)$ and $P := V(x^p - a)$ on $\mathbb{P}_k^1$. Then $t := x^p - a$ is a uniformizer at P , but $dt = px^{p-1}dx - da = 0$. The stalk $\Omega_{\mathbb{P}_k^1/k, P}$ is generated by dx , so $\text{ord}_P(dx) = 0$. This is why Definition 3.4.3 uses a generator of the stalk, rather than dt as in [Poo06, Definition 2.3.2].

EXAMPLE 3.4.6. Let t be the coordinate on $\mathbb{P}_k^1$. On $\mathbb{A}_k^1 = \text{Spec } k[t]$, the module of differentials is free on dt , so $\text{ord}_P(dt) = 0$ for every closed point $P \neq \infty$. At ∞ , the function $s := 1/t$ is a uniformizer at a rational point, and $dt = -s^{-2}ds$, so $\text{div}(dt) = -2\infty$. In particular $\mathbb{P}_k^1$ has no non-zero regular differentials [Sil09, Example II.4.5].

The next two examples start the comparison of elliptic curves with the multiplicative group $\mathbb{G}_m$ of Example 2.1.5, which runs through Chapter 4.

EXAMPLE 3.4.7. Let t be the coordinate on $\mathbb{G}_m = \mathbb{A}_k^1 - \{0\} \subseteq \mathbb{P}_k^1$. Then $k(\mathbb{G}_m) = k(\mathbb{P}_k^1) = k(t)$, so a differential on $\mathbb{G}_m$ is the same thing as a differential on $\mathbb{P}_k^1$. Let $\omega := dt/t$.

- (1) **Invariance.** For $a \in k^\times$, the translation $\tau_a: \mathbb{G}_m \rightarrow \mathbb{G}_m, t \mapsto at$, gives $\tau_a^* \omega = d(at)/(at) = \omega$. Conversely, suppose that $g \in \bar{k}(t)$ and $\tau_a^*(g dt) = g dt$ for every $a \in \bar{k}^\times$. Then $a g(at) = g(t)$, so $g(a) = g(1)/a$ for all but finitely many $a \in \bar{k}^\times$, and hence $g dt = g(1) \omega$. So the invariant differentials on $(\mathbb{G}_m)_{\bar{k}}$ form the line spanned by ω .
- (2) **Divisor.** Since t is a unit of $k[t, t^{-1}]$, ω generates $\Omega_{\mathbb{G}_m/k}$, so ω has no zeros or poles on $\mathbb{G}_m$. At $t = 0$ it has a simple pole. At ∞ , with the uniformizer $s := 1/t$,

$$\omega = \frac{dt}{t} = -\frac{ds}{s^2} \cdot s = -\frac{ds}{s},$$

again a simple pole. So $\text{div}(\omega) = -(0) - (\infty)$, of degree -2 , as Example 3.4.6 requires. The coefficients of dt/t and ds/s , the residues, are 1 and -1 , and they sum to 0.

This is the algebraic form of Example 1.5.4. The invariant differential of $\mathbb{G}_m$ has poles at the two points of $\mathbb{P}_k^1 - \mathbb{G}_m$; by contrast, the invariant differential of an elliptic curve has no poles at all (Example 3.4.9), because an elliptic curve is already projective.

EXAMPLE 3.4.8. Let m be a non-zero integer, and let $[m]: \mathbb{P}_k^1 \rightarrow \mathbb{P}_k^1$ be the morphism extending $t \mapsto t^m$ (Proposition 3.2.1). It maps $\mathbb{G}_m$ to $\mathbb{G}_m$, and on L -points it is the homomorphism $L^\times \rightarrow L^\times, x \mapsto x^m$. The polynomial $T^{|m|} - t^{|m|}$ is Eisenstein over $k[t^{|m|}]$, so $\deg[m] = [k(t) : k(t^m)] = |m|$.

- (1) **Pullback.** $[m]^* \omega = d(t^m)/t^m = m t^{m-1} dt/t^m = m \omega$. By Proposition 3.4.2, $[m]$ is separable if and only if $\text{char } k \nmid m$.
- (2) **Additivity.** By the Leibniz rule, $d(uv)/(uv) = du/u + dv/v$. Since $t^{m+n} = t^m \cdot t^n$, this gives $[m+n]^* \omega = [m]^* \omega + [n]^* \omega$ without computing either side. The same statement for elliptic curves is Theorem 4.4.2, whose proof is much harder.
- (3) **Kernel.** The fiber of $[m]$ over 1 is $\mu_m := \text{Spec } k[t]/(t^m - 1)$ for $m > 0$. If $\text{char } k \nmid m$, then $t^m - 1$ is separable, so $\mu_m(\bar{k})$ has $m = \deg[m]$ elements. If $\text{char } k = p$ and $m = p^r$, then $t^{p^r} - 1 = (t - 1)^{p^r}$, so $\mu_{p^r}(\bar{k}) = \{1\}$, with $e_{[p^r]}(1) = p^r$: all of the degree is inseparable degree. Indeed, $[p]$ is the relative Frobenius morphism of $\mathbb{P}_k^1 = (\mathbb{P}_k^1)^{(p)}$ (Definition 3.2.11).
- (4) **Ramification.** Above 0 and ∞ lie only 0 and ∞ , each with ramification index $|m|$. If $\text{char } k \nmid m$, then $[m]$ is unramified elsewhere, by part 3 and translation, and the Hurwitz formula (Theorem 3.6.2) reads $-2 = |m|(-2) + 2(|m| - 1)$.

For an elliptic curve, too, $[m]^* \omega = m \omega$ (Corollary 4.4.3), but $\deg[m] = m^2$ (Theorem 4.3.10), and $[m]$ is unramified everywhere.

EXAMPLE 3.4.9. Suppose $\text{char } k \neq 2$. Let $E: y^2 = p(x)$ with p a separable cubic, and $O = (0 : 1 : 0)$. On the affine part, $2y dy = p'(x) dx$, and y and $p'(x)$ have no common zero. So $dx/y = 2 dy/p'(x)$ generates $\Omega_{E/k}$ there. At O , $\text{ord}_O(x) = -2$ and $\text{ord}_O(y) = -3$, so $\text{ord}_O(dx) = -3$ and $\text{div}(dx/y) = 0$ [Sil09, Example II.4.6].

3.5. The Riemann–Roch theorem

DEFINITION 3.5.1. For $D \in \text{Div } C$, let

$$\mathcal{L}(D) := \{f \in k(C)^\times : \text{div}(f) + D \geq 0\} \cup \{0\},$$

a finite-dimensional k -vector space, and $\ell(D) := \dim_k \mathcal{L}(D)$. If $\deg D < 0$, then $\mathcal{L}(D) = 0$; if $D \sim D'$, then $\ell(D) = \ell(D')$ [Poo06, Definition 2.4.1], [Sil09, Proposition II.5.2].

PROPOSITION 3.5.2. Let $D \in \text{Div } C$, let $L \supseteq k$ be a field extension, and D_L the pullback of D to C_L . Then the natural map $\mathcal{L}(D) \otimes_k L \rightarrow \mathcal{L}(D_L)$ is an isomorphism, and $\deg D_L = \deg D$. In particular $\ell(D_L) = \ell(D)$.

PROOF. $\mathcal{L}(D)$ is the space of global sections of the invertible sheaf $\mathcal{O}_C(D)$, and $\mathcal{O}_C(D)$ pulls back to $\mathcal{O}_{C_L}(D_L)$. Taking global sections commutes with the flat base change $\text{Spec } L \rightarrow \text{Spec } k$ [Har77, Proposition

III.9.3]. For $L = \bar{k}$ and perfect k , [Sil09, Proposition II.5.8] proves this with Galois descent instead, and [Poo06, Section 2.4.1] states it. $\square$

DEFINITION 3.5.3. The **genus** of C is $g := \ell(K_C)$, the dimension of the k -vector space of regular differentials on C [Poo06, Definition 2.4.5].

THEOREM 3.5.4 (Riemann–Roch). *Let C be a nice curve of genus g , and $D \in \text{Div } C$. Then*

$$\ell(D) - \ell(K_C - D) = \deg D - g + 1.$$

PROOF. See [Sil09, Theorem II.5.4] and [Poo06, Theorem 2.4.7]. To reduce to $k = \bar{k}$, note that $\Omega_{C_{\bar{k}}/\bar{k}}$ is the pullback of $\Omega_{C/k}$, so K_C pulls back to a canonical divisor. Then apply Proposition 3.5.2. $\square$

COROLLARY 3.5.5. *Let C be a nice curve of genus g .*

- (1) $\deg K_C = 2g - 2$.
- (2) *If $\deg D > 2g - 2$, then $\ell(D) = \deg D - g + 1$.*
- (3) *For every field extension $L \supseteq k$, the nice curve C_L has genus g .*

PROOF. Parts 1 and 2 are [Sil09, Corollary II.5.5] and [Poo06, Corollaries 2.4.8 and 2.4.9]. Part 3 follows from Proposition 3.5.2, since K_C pulls back to a canonical divisor. $\square$

EXAMPLE 3.5.6. By Example 3.4.6, $\mathbb{P}_k^1$ has genus 0, and $\ell(D) = \deg D + 1$ whenever $\deg D \geq -1$ [Sil09, Example II.5.6].

EXAMPLE 3.5.7. Let $E: y^2 = p(x)$ be as in Example 3.4.9. We may take $K_E = 0$, so E has genus $\ell(0) = 1$, and $\ell(D) = \deg D$ whenever $\deg D \geq 1$. In particular $\ell(P) = 1$ for $P \in E(k)$, so no function has a single simple pole. For $O = (0 : 1 : 0)$, the functions $1, x$ form a basis of $\mathcal{L}(2O)$, and $1, x, y$ a basis of $\mathcal{L}(3O)$. The seven functions $1, x, y, x^2, xy, x^3, y^2$ lie in the 6-dimensional space $\mathcal{L}(6O)$, and the linear relation between them is the equation of E [Sil09, Example II.5.7].

3.6. The Hurwitz formula

DEFINITION 3.6.1. Let $\varphi: C_1 \rightarrow C_2$ be a non-constant separable morphism, and ω a non-zero differential on C_2 , so that $\varphi^*\omega \neq 0$ (Proposition 3.4.2). The **ramification divisor** of φ is

$$R_\varphi := \text{div}(\varphi^*\omega) - \varphi^*\text{div}(\omega) \in \text{Div } C_1.$$

It does not depend on ω [Poo06, Definition 2.5.1].

THEOREM 3.6.2 (Hurwitz). *Let $\varphi: C_1 \rightarrow C_2$ be a non-constant separable morphism of nice curves of genera g_1 and g_2 . Then*

$$2g_1 - 2 = (\deg \varphi)(2g_2 - 2) + \deg R_\varphi.$$

For every $P \in C_1(\bar{k})$, the base change $\varphi_{\bar{k}}$ satisfies $\text{ord}_P(R_{\varphi_{\bar{k}}}) \geq e(P) - 1$, where $e(P) := e_{\varphi_{\bar{k}}}(P)$, with equality if and only if $\text{char } k \nmid e(P)$. So if no ramification index is divisible by $\text{char } k$, then

$$2g_1 - 2 = (\deg \varphi)(2g_2 - 2) + \sum_{P \in C_1(\bar{k})} (e(P) - 1).$$

PROOF. Take degrees in $\text{div}(\varphi^*\omega) = \varphi^*\text{div}(\omega) + R_\varphi$, and use Proposition 3.3.6 and Corollary 3.5.5 [Poo06, Theorem 2.5.7]. Base change to $\bar{k}$ preserves separability, degrees and genera, and over $\bar{k}$ the rest is [Poo06, Proposition 2.5.4 and Corollary 2.5.8] and [Sil09, Theorem II.5.9]. $\square$

EXAMPLE 3.6.3. Let $x: E \rightarrow \mathbb{P}_k^1$ be the morphism of Example 3.2.10, with $\text{char } k \neq 2$. Over $\bar{k}$ it is ramified exactly at the three roots $(e_i, 0)$ of p and at O , each with index 2. Hurwitz gives $2g - 2 = 2 \cdot (-2) + 4 = 0$, so $g = 1$, in agreement with Example 3.5.7.

EXAMPLE 3.6.4. Suppose $\text{char } k = 2$, and let $C: Y^2Z + YZ^2 = X^3$, the curve $y^2 + y = x^3$. The partial derivatives are X^2 , Z^2 and Y^2 , with no common zero, so C is a smooth cubic, of genus 1 by the degree–genus formula (Corollary 3.7.7 below). The morphism $x: C \rightarrow \mathbb{P}_k^1$ has degree 2, and it is separable, since $y^2 + y - x^3$ has derivative 1 in y . Over each $a \in \bar{k}$ the equation $y^2 + y = a^3$ has two distinct roots. So the only ramification is at $O = (0 : 1 : 0)$, with index 2, which is wild. One computes $R_x = 4O$ [Poo06, Example 2.5.6], and Hurwitz gives $2g - 2 = 2 \cdot (-2) + 4 = 0$. The formula for the tame case would give the impossible value $2 \cdot (-2) + 1$.

DRAFT
21 Sep 2026

3.7. Plane curves

We follow [Mil20, Section I.1], and use coordinates $(X : Y : Z)$ on $\mathbb{P}_k^2$, with $x = X/Z$ and $y = Y/Z$ on the affine patch $Z \neq 0$.

DEFINITION 3.7.1. Let $F \in k[X, Y, Z]$ be a non-constant homogeneous polynomial of degree d with no repeated factor in $\bar{k}[X, Y, Z]$. The **projective plane curve** defined by F is $C_F := V_+(F) \subseteq \mathbb{P}_k^2$, and d is its degree. Affine plane curves $V(f) \subseteq \mathbb{A}_k^2$ are defined in the same way.

The condition on F says that C_F is geometrically reduced [Mil20, Aside I.1.7]. The affine patches of C_F are the affine plane curves $F(x, y, 1) = 0$, $F(x, 1, z) = 0$ and $F(1, y, z) = 0$.

PROPOSITION 3.7.2. *Let $C = V_+(F) \subseteq \mathbb{P}_k^2$ be a projective plane curve of degree d . Its singular points are the common zeros in $\mathbb{P}^2(\bar{k})$ of F , $\partial F/\partial X$, $\partial F/\partial Y$ and $\partial F/\partial Z$. If $\text{char } k \nmid d$, the condition $F = 0$ may be omitted, by the Euler relation*

$$dF = X \frac{\partial F}{\partial X} + Y \frac{\partial F}{\partial Y} + Z \frac{\partial F}{\partial Z}.$$

PROOF. See [Poo06, Example 1.7.5] and [Mil20, Exercise I.1.17]. $\square$

When $\text{char } k \mid d$ the condition $F = 0$ cannot be dropped: for $F = Y^2Z - X^3 + tZ^3$ over $\mathbb{F}_3(t)$ (Example 2.4.6) the three partial derivatives vanish at $(0 : 0 : 1)$, which is not a $\bar{k}$ -point of the curve.

DEFINITION 3.7.3. Let $C = V_+(F)$ and $D = V_+(G)$ be projective plane curves with no common component. Then $C \cap D := V_+(F, G)$ is a finite k -scheme. For a closed point P of $C \cap D$, choose an affine patch containing P , with dehomogenizations f and g of F and G . The **intersection number** of C and D at P is

$$I_P(C, D) := \dim_k \mathcal{O}_{\mathbb{P}_k^2, P} / (f, g).$$

For $P \in C(k) \cap D(k)$ this is the intersection number of [Mil20, Proposition I.1.8], and it does not change under base extension [Mil20, Remark I.1.9].

THEOREM 3.7.4 (Bézout). *Let C and D be projective plane curves over k of degrees m and n , with no common component. Then*

$$\sum_{P \in C \cap D} I_P(C, D) = mn.$$

PROOF. Base change to $\bar{k}$, and apply [Mil20, Theorem I.1.18]. $\square$

COROLLARY 3.7.5. *A smooth projective plane curve is geometrically integral, hence nice.*

PROOF. It is geometrically reduced by definition. Over $\bar{k}$, two irreducible components would meet by Theorem 3.7.4, and a $\bar{k}$ -point on two components is singular [Mil20, Remarks I.1.4 and I.1.21]. $\square$

PROPOSITION 3.7.6. *Let $C = V_+(F) \subseteq \mathbb{P}_k^2$ be a smooth plane curve of degree d , not the line $Z = 0$, and let $f(x, y) := F(x, y, 1)$. Let H be the effective divisor cut out by the line $Z = 0$. Then H has degree d , and*

$$\operatorname{div} \left(\frac{dx}{\partial f / \partial y} \right) = (d - 3)H.$$

Here $dx/(\partial f / \partial y)$ stands for $-dy/(\partial f / \partial x)$ if $\partial f / \partial y$ vanishes identically on C .

PROOF. On the affine part, $\frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy = 0$, and the partial derivatives have no common zero (Proposition 3.7.2). So $\omega := dx/(\partial f / \partial y) = -dy/(\partial f / \partial x)$ generates $\Omega_{C/k}$ there, and $\operatorname{ord}_P(\omega) = 0$ at the closed points P of the affine part.

At a closed point P on the line $Z = 0$, say with $Y \neq 0$, use the coordinates $u := X/Y$ and $v := Z/Y$, so that $x = u/v$ and $y = 1/v$, and let $g(u, v) := F(u, 1, v)$. As before, $\eta := du/(\partial g / \partial v) = -dv/(\partial g / \partial u)$ generates $\Omega_{C/k}$ near P . The Euler relation gives $u \partial g / \partial u + v \partial g / \partial v = -\frac{\partial F}{\partial Y}(u, 1, v)$ on C , and $\partial f / \partial y = v^{1-d} \frac{\partial F}{\partial Y}(u, 1, v)$ because $\partial F / \partial Y$ is homogeneous of degree $d - 1$. Substituting into $dx = (v du - u dv)/v^2$ gives

$$\omega = -v^{d-3} \eta,$$

so $\operatorname{ord}_P(\omega) = (d - 3) \operatorname{ord}_P(v)$, where v is a local equation of the line $Z = 0$. The case $X \neq 0$ is symmetric. Hence $\operatorname{div}(\omega) = (d - 3)H$. The degree of H is d by Theorem 3.7.4: on a smooth curve, $I_P(C, L) = \operatorname{ord}_P(\ell) \deg P$ for a local equation ℓ of the line L . $\square$

For $d = 3$ and $f = y^2 - p(x)$ this recovers Example 3.4.9. For $d = 1$, with $f = y$, it recovers Example 3.4.6.

COROLLARY 3.7.7. *A smooth plane curve of degree d has genus $(d - 1)(d - 2)/2$.*

PROOF. By Proposition 3.7.6, $\deg K_C = d(d - 3)$, and $d(d - 3) = 2g - 2$ by Corollary 3.5.5. See also [Sil09, Exercise 2.7] and [Poo06, Section 2.9.2]. $\square$

REMARK 3.7.8. Smoothness cannot be weakened to regularity here. The curve $E: y^2 = x^3 - t$ over $k = \mathbb{F}_3(t)$ of Example 2.4.6 is regular and geometrically integral, and as a plane cubic it has arithmetic genus 1. Over $k' := k(s)$, where $s^3 = t$, it becomes the cuspidal cubic $y^2 = (x - s)^3$, whose normalization $u \mapsto (s + u^2, u^3)$ is $\mathbb{P}_{k'}^1$. So the genus drops under base extension, and Corollary 3.5.5 fails for E . Also $\Omega_{E/k}$ is not invertible at the non-smooth point, so Definition 3.4.3 does not apply.

EXAMPLE 3.7.9. Let $h \in k[x]$ have degree $n \geq 3$ and leading coefficient c , let $H(X, Z) := Z^n h(X/Z)$ be its homogenization, and let $C := V_+(F) \subseteq \mathbb{P}_k^2$, where

$$F := Y^2 Z^{n-2} - H(X, Z),$$

the projective closure of the affine curve $y^2 = h(x)$. Both F and C have degree n .

- (1) **C is a plane curve.** Since $F \equiv -cX^n \pmod{Z}$, the polynomial Z does not divide F , so a repeated factor of F is one of $y^2 - h(x)$. A factorization $y^2 - h(x) = (y - g)(y + g)$ over $\bar{k}$ has a repeated factor only if $2g = 0$, so F has one exactly when $\operatorname{char} k = 2$ and $h \in k[x^2]$. Exclude that case, so that Definition 3.7.1 applies.
- (2) **Singular points of the affine part.** Here $f = y^2 - h(x)$, so $\partial f / \partial y = 2y$ and $\partial f / \partial x = -h'(x)$, and the singular points of $y^2 = h(x)$ are the pairs $(\alpha, \beta) \in \bar{k}^2$ with

$$\beta^2 = h(\alpha), \quad 2\beta = 0, \quad h'(\alpha) = 0.$$

If $\operatorname{char} k \neq 2$, then $\beta = 0$, and the conditions $h(\alpha) = h'(\alpha) = 0$ say that α is a repeated root of h . So the affine curve is singular exactly at the $\bar{k}$ -points $(e, 0)$ with e a repeated root of h in $\bar{k}$, and it is smooth if and only if h is separable. For $h = x^3 + Ax + B$ this says $4A^3 + 27B^2 \neq 0$ [Mil20, Example I.1.5]. If

$\text{char } k = 2$, the condition $2\beta = 0$ is vacuous. When n is odd, h' has degree $n - 1 \geq 2$, hence a root $\alpha \in \bar{k}$, and any β with $\beta^2 = h(\alpha)$ gives a singular point; for $h = x^3 + Ax + B$ it is the $\bar{k}$ -point with $\alpha^2 = A$ and $\beta^2 = \alpha^3 + A\alpha + B$.

- (3) **The point at infinity.** Since $n \geq 3$, we have $F(X, Y, 0) = -cX^n$, so C meets the line $L := V_+(Z)$ only at $O := (0 : 1 : 0)$, and $I_O(C, L) = n$ by Theorem 3.7.4. Now

$$\frac{\partial F}{\partial X} = -\frac{\partial H}{\partial X}, \quad \frac{\partial F}{\partial Y} = 2YZ^{n-2}, \quad \frac{\partial F}{\partial Z} = (n-2)Y^2Z^{n-3} - \frac{\partial H}{\partial Z}.$$

The partial derivatives of H are homogeneous of degree $n - 1 \geq 2$ in (X, Z) , so they vanish at O , as does $\partial F/\partial Y$. For $n = 3$ the remaining derivative is $\partial F/\partial Z = Y^2 - \partial H/\partial Z$, with value 1 at O ; for $n \geq 4$ it vanishes at O as well. So O is a smooth point of C if $n = 3$, and a singular point if $n \geq 4$, in every characteristic.

- (4) **Conclusion.** If $\text{char } k \neq 2$, then C is smooth if and only if $n = 3$ and h is separable; it is then nice by Corollary 3.7.5 and of genus 1 by Corollary 3.7.7. If $\text{char } k = 2$, then C is singular: by part 2 when n is odd, and by part 3 when n is even, since then $n \geq 4$. One needs the general Weierstrass equation instead [Sil09, Section III.1].

So a curve $y^2 = h(x)$ with $\deg h \geq 4$ has no smooth plane model, and must be studied through some other embedding. The case $n = 3$ is the one we pursue, in Section 4.1.

Elliptic curves as algebraic curves

Sources. This chapter follows [Sil09, Chapter III], with some material from [Mil20, Sections II.1, II.2, II.5 and II.6], and [Voi21, Chapters 2, 3, 5 and 10] for quaternion algebras and their orders. As in the previous two chapters, we do not assume that the ground field is perfect.

In Chapter 1 we met complex tori $\mathbb{C}/\Lambda$: compact Riemann surfaces that are also abelian groups. The uniformization theorem (Theorem 1.7.1) identifies each of them with a smooth plane cubic $E_{A,B}(\mathbb{C})$, and under this identification the isogenies of Section 1.3 and the torsion subgroups $(\mathbb{C}/\Lambda)[m] \cong (\mathbb{Z}/m\mathbb{Z})^2$ of Example 1.3.4 become statements about cubic curves. The goal of this chapter is to recover that whole picture over an arbitrary field k , where there is no lattice to divide by. The tools are the ones developed in Chapters 2 and 3: the Riemann–Roch theorem replaces the Weierstrass $\wp$ -function, the Picard group replaces the quotient $\mathbb{C}/\Lambda$, and the invariant differential replaces dz . Alongside, we compare E with the multiplicative group $\mathbb{G}_m$, whose invariant differential, torsion points and Tate module are elementary (Examples 3.4.7 and 3.4.8); the comparison is summarized at the end of Section 4.5.

We keep the notation of the previous chapters: k is an arbitrary field, $k^{\text{sep}} \subseteq \bar{k}$ are a separable and an algebraic closure, and $\text{Gal}_k := \text{Gal}(k^{\text{sep}}/k)$. For a k -variety X and a field $L \supseteq k$, recall that X_L is the base extension.

4.1. Weierstrass equations

DEFINITION 4.1.1. An **elliptic curve** over k is a pair (E, O) , where E is a nice curve over k of genus 1 and $O \in E(k)$. We usually write E for the pair. A morphism of elliptic curves $(E, O) \rightarrow (E', O')$ is a morphism of curves $E \rightarrow E'$ sending O to O' .

The rational point is part of the data. A nice curve of genus 1 need not have one: by a theorem of Selmer [Sel51], the smooth plane cubic $3X^3 + 4Y^3 + 5Z^3 = 0$ over $\mathbb{Q}$, which has genus 1 by Corollary 3.7.7, has no rational points, although it has points over $\mathbb{R}$ and over every $\mathbb{Q}_p$. Such curves will reappear in Section 4.8 and in Chapter 10.

The model for every elliptic curve is the smooth plane cubic $y^2 = h(x)$ of Example 3.7.9. In characteristic 2 that equation is always singular, by the same example, so we need a more general one.

DEFINITION 4.1.2. A **Weierstrass equation** over k is a projective plane cubic

$$(4.1.1) \quad Y^2Z + a_1XYZ + a_3YZ^2 = X^3 + a_2X^2Z + a_4XZ^2 + a_6Z^3$$

with $a_1, a_2, a_3, a_4, a_6 \in k$, which on the affine patch $Z \neq 0$ reads $y^2 + a_1xy + a_3y = x^3 + a_2x^2 + a_4x + a_6$. Intersecting with **the line at infinity** $Z = 0$ leaves $X^3 = 0$, so the only point at infinity is the flex $O := (0 : 1 : 0)$. We define the **b -invariants** and **c -invariants** of the Weierstrass equation by

$$\begin{aligned} b_2 &:= a_1^2 + 4a_2, & b_4 &:= 2a_4 + a_1a_3, \\ b_6 &:= a_3^2 + 4a_6, & b_8 &:= a_1^2a_6 + 4a_2a_6 - a_1a_3a_4 + a_2a_3^2 - a_4^2, \\ c_4 &:= b_2^2 - 24b_4, & c_6 &:= -b_2^3 + 36b_2b_4 - 216b_6, \end{aligned}$$

the **discriminant** $\Delta := -b_2^2b_8 - 8b_4^3 - 27b_6^2 + 9b_2b_4b_6$, and, when $\Delta \neq 0$, the **j -invariant** $j := c_4^3/\Delta$. The **invariant differential** is

$$\omega := \frac{dx}{2y + a_1x + a_3} = \frac{dy}{3x^2 + 2a_2x + a_4 - a_1y}.$$

These quantities satisfy $4b_8 = b_2b_6 - b_4^2$ and $1728\Delta = c_4^3 - c_6^2$.

A **flex** of a plane cubic is a smooth $\bar{k}$ -point P whose tangent line meets the cubic at P with multiplicity 3; the tangent line at O is $Z = 0$. The two expressions for ω agree because $f := y^2 + a_1xy + a_3y - x^3 - a_2x^2 - a_4x - a_6$ satisfies $\frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy = 0$, and $\partial f / \partial y = 2y + a_1x + a_3$.

EXAMPLE 4.1.3 (Recalling the formulas in Magma). Over the field $\mathbb{Q}(a_1, a_2, a_3, a_4, a_6)$ of rational functions in the coefficients, (4.1.1) is an elliptic curve, and Magma prints the quantities of Definition 4.1.2 for it as polynomials in the a_i [BCP97].

```
K<a1,a2,a3,a4,a6> := FunctionField(Rationals(), 5);
E := EllipticCurve([a1, a2, a3, a4, a6]);
aInvariants(E);           // [a1, a2, a3, a4, a6]
bInvariants(E);           // [b2, b4, b6, b8]
cInvariants(E);           // [c4, c6]
```

Printing `Discriminant(E)` and `jInvariant(E)` gives expressions too long to read, but the identities of Definition 4.1.2 fit on one line each. A `true` here is an identity of rational functions with integer coefficients, so it holds for every Weierstrass equation over every field.

```
b2, b4, b6, b8 := Explode(bInvariants(E));
c4, c6 := Explode(cInvariants(E));
4*b8 eq b2*b6 - b4^2;
1728*Discriminant(E) eq c4^3 - c6^2;
jInvariant(E) eq c4^3/Discriminant(E);
```

LEMMA 4.1.4 (Short Weierstrass form). *If $\text{char } k \neq 2$, completing the square in y , and then, if also $\text{char } k \neq 3$, completing the cube in x , brings the equation to the short form*

$$(4.1.2) \quad y^2 = x^3 + Ax + B,$$

whose discriminant, j -invariant and invariant differential are

$$\Delta = -16(4A^3 + 27B^2), \quad j = 1728 \frac{4A^3}{4A^3 + 27B^2}, \quad \omega = \frac{dx}{2y}.$$

PROOF. Both steps are instances of the change of variables $x = u^2x' + r$, $y = u^3y' + su^2x' + t$ with $u \in k^\times$ and $r, s, t \in k$, which reappears in Theorem 4.1.6. Magma performs one with `Transformation`, in the inverse convention: our (u, r, s, t) is Magma's tuple $[-r/u^2, -s/u, (rs - t)/u^3, 1/u]$. We translate once and for all, and use `subst` again in the proofs below; the blocks of this section run in one session, in order.

```
// Weierstrass change of variables, as above:
// x = u^2*x' + r, y = u^3*y' + s*u^2*x' + t.
// Magma's Transformation inverts it, so translate the tuple.
subst := func< E,u,r,s,t | Transformation(E,
    [BaseRing(E) | -r/u^2, -s/u, (r*s - t)/u^3, 1/u]) >;
```

If $\text{char } k \neq 2$, take $u = 1$, $r = 0$, $s = -a_1/2$ and $t = -a_3/2$. This makes the new a_1 and a_3 vanish, and the equation becomes $y^2 = x^3 + \frac{b_2}{4}x^2 + \frac{b_4}{2}x + \frac{b_6}{4}$.

```
K<a1,a2,a3,a4,a6> := FunctionField(Rationals(), 5);
E := EllipticCurve([a1,a2,a3,a4,a6]);
b2, b4, b6, b8 := Explode(bInvariants(E));
c4, c6 := Explode(cInvariants(E));
E1 := subst(E, 1, 0, -a1/2, -a3/2);
aInvariants(E1) eq [K | 0, b2/4, 0, b4/2, b6/4]; // true
```

If also $\text{char } k \neq 3$, take $u = 1$ and $r = -b_2/12$. This makes the new a_2 vanish and leaves (4.1.2), with $A = -c_4/48$ and $B = -c_6/864$.

```

E2 := subst(E1, 1, -b2/12, 0, 0);
A := -c4/48; B := -c6/864;
aInvariants(E2) eq [K | 0, 0, 0, A, B];           // true
Discriminant(E2) eq -16*(4*A^3 + 27*B^2);         // true
jInvariant(E2) eq 1728*4*A^3/(4*A^3 + 27*B^2);    // true
Discriminant(E2) eq Discriminant(E);              // true

```

The last line records that neither substitution changes Δ , as both have $u = 1$; the same holds for j and for ω . The displayed formulas are those of Definition 4.1.2 read off the short form. $\square$

Weierstrass equations with non-zero discriminant, together with the [point at infinity](#), define elliptic curves over any field k .

PROPOSITION 4.1.5. *Let E be the plane cubic (4.1.1).*

- (1) *E is smooth if and only if $\Delta \neq 0$.*
- (2) *If $\Delta \neq 0$, then (E, O) is an elliptic curve over k , and the invariant differential satisfies $\text{div}(\omega) = 0$.*

PROOF. Write F for the cubic form in (4.1.1). At O we have $\partial F/\partial Z = Y^2 = 1$, so O is always a smooth point. That a singular point of the affine part exists if and only if $\Delta = 0$ is [Sil09, Proposition III.1.4(a)], a computation over $\bar{k}$; when $\text{char } k \neq 2, 3$ it is Example 3.7.9, by (4.1.2).

If E is smooth, it is nice by Corollary 3.7.5, of genus 1 by Corollary 3.7.7, and $O \in E(k)$. Finally, let $\omega = dx/(\partial f/\partial y)$, so $\text{div}(\omega) = (3 - 3)O = 0$ by Proposition 3.7.6. $\square$

The converse is the first application of the Riemann–Roch theorem: every elliptic curve is a Weierstrass cubic. Example 3.5.7 already contains the idea, run backwards. The following is [Sil09, Proposition III.3.1].

THEOREM 4.1.6. *Let (E, O) be an elliptic curve over k .*

- (1) *There are $x, y \in k(E)$ such that the morphism $E \rightarrow \mathbb{P}_k^2$ given by $(x : y : 1)$ is an isomorphism onto a smooth Weierstrass cubic (4.1.1), sending O to $(0 : 1 : 0)$. Moreover $k(E) = k(x, y)$ and $[k(E) : k(x)] = 2$.*
- (2) *Any two such pairs, (x, y) and (x', y') , are related by*

$$(4.1.3) \quad x = u^2 x' + r, \quad y = u^3 y' + s u^2 x' + t,$$

for some $u \in k^\times$ and $r, s, t \in k$.

The functions x and y are [Weierstrass coordinates](#) on E .

PROOF. Since E has genus 1, Corollary 3.5.5 gives $\ell(nO) = n$ for every $n \geq 1$. So $\mathcal{L}(O) = k$, and we may choose $x \in \mathcal{L}(2O)$ and $y \in \mathcal{L}(3O)$ with $\{1, x\}$ and $\{1, x, y\}$ bases; then $\text{ord}_O(x) = -2$ and $\text{ord}_O(y) = -3$. The seven functions $1, x, y, x^2, xy, y^2, x^3$ lie in the 6-dimensional space $\mathcal{L}(6O)$, so they satisfy a linear relation over k . The only functions in the list with a pole of order 6 are y^2 and x^3 , so both appear with non-zero coefficients, and rescaling x and y by elements of $k^\times$ gives a Weierstrass equation. Let C be the cubic it defines. The rational map $\varphi := (x : y : 1) : E \dashrightarrow C$ is a morphism by Proposition 3.2.1, and $\varphi(O) = (x/y : 1 : 1/y)(O) = (0 : 1 : 0)$.

The morphism $x : E \rightarrow \mathbb{P}_k^1$ pulls ∞ back to $2O$, so it has degree 2 by Proposition 3.2.8; likewise y has degree 3. Hence $[k(E) : k(x, y)]$ divides 2 and 3, so $k(E) = k(x, y)$ and φ is birational onto C . If C were singular, then $C_{\bar{k}}$, and so $E_{\bar{k}}$, would be birational to $\mathbb{P}_{\bar{k}}^1$ [Sil09, Proposition III.1.6]; compare Example 2.3.5. By Corollary 3.2.6 this would make the nice curve $E_{\bar{k}}$ isomorphic to $\mathbb{P}_{\bar{k}}^1$, which has genus 0, whereas $E_{\bar{k}}$ has genus 1 by Corollary 3.5.5. Since the genus is a geometric invariant, C is smooth, and φ is an isomorphism by Corollary 3.2.6.

For part 2, $\{1, x\}$ and $\{1, x'\}$ are both bases of $\mathcal{L}(2O)$, and $\{1, x, y\}$ and $\{1, x', y'\}$ of $\mathcal{L}(3O)$. So $x = u_1 x' + r$ and $y = u_2 y' + s_2 x' + t$ with $u_1, u_2 \in k^\times$. Comparing the coefficients of x'^3 and y'^2 in the two equations gives $u_1^3 = u_2^2$, and $u := u_2/u_1$ does the job. $\square$

PROPOSITION 4.1.7. *Under the change of variables (4.1.3), the quantities of the primed equation satisfy*

$$u^4 c'_4 = c_4, \quad u^6 c'_6 = c_6, \quad u^{12} \Delta' = \Delta, \quad j' = j, \quad u^{-1} \omega' = \omega.$$

In particular the j -invariant and the class of Δ in $k^\times/k^{\times 12}$ are invariants of the elliptic curve E . In the short form (4.1.2), the only changes of variables preserving the form are $x = u^2 x'$, $y = u^3 y'$, and they give $A' = u^{-4} A$ and $B' = u^{-6} B$.

PROOF. This is a direct computation, which we can do in Magma: over the function field in the five coefficients and the four parameters of (4.1.3), a `true` is an identity of rational functions, hence holds for every Weierstrass equation over every field.

```
L<a1,a2,a3,a4,a6,u,r,s,t> := FunctionField(Rationals(), 9);
E := EllipticCurve([a1,a2,a3,a4,a6]);
Ep := subst(E, u, r, s, t);
c4, c6 := Explode(cInvariants(E));
c4p, c6p := Explode(cInvariants(Ep));
u^4*c4p eq c4; // true
u^6*c6p eq c6; // true
u^12*Discriminant(Ep) eq Discriminant(E); // true
jInvariant(Ep) eq jInvariant(E); // true
```

For ω , differentiating $x = u^2 x' + r$ gives $dx = u^2 dx'$, while $2y + a_1 x + a_3 = u^3(2y' + a'_1 x' + a'_3)$, so $\omega = u^{-1} \omega'$.

A change of variables preserving the short form (4.1.2) must have $a'_1 = a'_3 = 0$, forcing $s = t = 0$, and then $a'_2 = 0$ forces $r = 0$. What is left rescales A and B .

```
M<A,B,u> := FunctionField(Rationals(), 3);
E := EllipticCurve([M | 0, 0, 0, A, B]);
aInvariants(subst(E, u, 0, 0, 0)); // [0, 0, 0, A/u^4, B/u^6]
```

That is $A' = u^{-4} A$ and $B' = u^{-6} B$. $\square$

REMARK 4.1.8. Chapter 1 uses the short form (4.1.2) (Example 1.6.3), and the discriminant of Lemma 4.1.4 explains the condition $4A^3 + 27B^2 \neq 0$ there. Under the isomorphism of Theorem 1.7.1 we have $x = \wp_\Lambda(z)$ and $y = \frac{1}{2} \wp'_\Lambda(z)$, so $\omega = \wp'_\Lambda(z) dz / \wp'_\Lambda(z) = dz$: the invariant differential is the translation-invariant differential dz on $\mathbb{C}/\Lambda$. With $A = -15G_4(\Lambda)$ and $B = -35G_6(\Lambda)$, the j -invariant becomes a function of the lattice, and by Proposition 4.1.7 a function on $\mathrm{SL}_2(\mathbb{Z}) \backslash \mathbb{H}$ (Proposition 1.8.1).

The j -invariant classifies elliptic curves up to isomorphism over $\bar{k}$, but not over k . The failure is already visible in the short form. Suppose $\mathrm{char} k \neq 2$, let $E: y^2 = x^3 + Ax + B$ and let $d \in k^\times$. The quadratic twist of E by d is

$$E^d: y^2 = x^3 + d^2 Ax + d^3 B,$$

which is $dy^2 = x^3 + Ax + B$ in the coordinates dx and $d^2 y$. So $(x, y) \mapsto (dx, d^{3/2} y)$ is an isomorphism $E \rightarrow E^d$ over $k(\sqrt{d})$, and $j(E^d) = j(E)$. Suppose now that $j(E) \neq 0, 1728$, that is $AB \neq 0$. By Proposition 4.1.7 an isomorphism $E \rightarrow E^d$ over k would be a change of variables $x = u^2 x'$, $y = u^3 y'$ with $u^4 = d^{-2}$ and $u^6 = d^{-3}$, forcing $u^2 = d^{-1}$. So E and E^d are isomorphic over k exactly when $d \in k^{\times 2}$, and the quadratic twists of E are indexed by $d \in k^\times / k^{\times 2}$. They all share one j -invariant. For $j = 0$ and $j = 1728$ the curve has extra automorphisms and there are more twists than these; see Theorem 4.7.11 and Chapter 10.

EXAMPLE 4.1.9. Take $E: y^2 = x^3 + x + 1$ over $\mathbb{Q}$ and $d = -1$, so that $E^{-1}: y^2 = x^3 + x - 1$. Here $d^6 = 1$, so the two curves have the same discriminant -496 as well as the same j -invariant $6912/31$; they are nonetheless

not isomorphic over $\mathbb{Q}$, because $-1 \notin \mathbb{Q}^{\times 2}$. They become isomorphic over $\mathbb{Q}(i)$. So Δ and j together do not determine an elliptic curve up to isomorphism over k .

```
E := EllipticCurve([Rationals() | 0, 0, 0, 1, 1]);
Ed := QuadraticTwist(E, -1);
aInvariants(Ed);           // [0, 0, 0, 1, -1]
jInvariant(Ed) eq jInvariant(E); // true
Discriminant(Ed) eq Discriminant(E); // true
IsIsomorphic(E, Ed);       // false
K<i> := QuadraticField(-1);
IsIsomorphic(ChangeRing(E, K), ChangeRing(Ed, K)); // true
```

The curves $y^2 = x^3 - x$ and $y^2 = x^3 + x$, which also share a j -invariant, are **not** quadratic twists of one another: each is its own twist by -1 . They are quartic twists, the extra twists that $j = 1728$ admits.

Twists in general are the subject of Chapter 10; the cubic twists of Section 4.8 are the other case we meet in this chapter.

PROPOSITION 4.1.10.

- (1) Two elliptic curves over k are isomorphic over $\bar{k}$ if and only if they have the same j -invariant.
- (2) Every $j_0 \in k$ is the j -invariant of an elliptic curve over k , namely the one listed here, where $c := 1/(j_0 - 1728)$.

j_0	Weierstrass equation	Δ	needs
$\neq 0, 1728$	$y^2 + xy = x^3 - 36cx - c$	$j_0^2 c^3$	
0	$y^2 + y = x^3$	-27	$\text{char } k \neq 3$
1728	$y^2 = x^3 + x$	-64	$\text{char } k \neq 2$

In characteristic 2 and 3 we have $0 = 1728$, and exactly one of the last two rows applies, so the single special value is covered.

PROOF. In part 1, an isomorphism over $\bar{k}$ is a change of variables (4.1.3) over $\bar{k}$, so it preserves j by Proposition 4.1.7. For the converse, suppose $\text{char } k \neq 2, 3$ and put both curves in the short form (4.1.2). The changes of variables preserving that form are $x = u^2 x'$, $y = u^3 y'$, giving $A' = u^{-4}A$ and $B' = u^{-6}B$, so we must produce $u \in \bar{k}^\times$ with $u^4 = A/A'$ and $u^6 = B/B'$. Now $j = j'$ means $A^3 B'^2 = A'^3 B^2$. If $A = 0$, then $A' = 0$, and $u^6 = B/B'$ will do. If $B = 0$, then $B' = 0$, and $u^4 = A/A'$ will do. Otherwise take $u^4 = A/A'$; then $(u^6)^2 = (A/A')^3 = (B/B')^2$, and replacing u by $\sqrt{-1}u$ if necessary gives $u^6 = B/B'$. The characteristics 2 and 3 need the normal forms of Proposition 4.1.12, and are finished after it.

For part 2, the first row has $b_2 = 1$, $b_4 = -72c$, $b_6 = -4c$ and $b_8 = -c - 1296c^2$, whence $c_4 = 1 + 1728c$ and $\Delta = j_0^2 c^3$. Since $j_0 = 1728 + 1/c$ we have $c_4 = j_0 c$, so $j = c_4^3 / \Delta = j_0^3 c^3 / (j_0^2 c^3) = j_0$. The second row has $b_6 = 1$ as its only non-zero b , so $\Delta = -27$ and $c_4 = 0$, giving $j = 0$; the third has $b_4 = 2$ and $b_8 = -1$, so $\Delta = -64$ and $c_4 = -48$, giving $j = 1728$. Each row asks only that its own Δ be non-zero, which is the condition listed. $\square$

REMARK 4.1.11. If $\Delta = 0$, the cubic (4.1.1) has exactly one singular point S . It is a node (two distinct tangent directions) if $c_4 \neq 0$, and a cusp (one tangent direction) if $c_4 = 0$ [Sil09, Proposition III.1.4(a)]. In both cases the curve is birational to $\mathbb{P}^1$, as for the nodal cubic of Example 2.3.5 [Sil09, Proposition III.1.6]. These curves matter to us because an elliptic curve over $\mathbb{Q}$ acquires such a singularity when its equation is reduced modulo a prime dividing Δ (Chapter 6); see Proposition 4.2.11.

In characteristic 2 and 3 the short form (4.1.2) is not available, and the general equation (4.1.1) carries too many coefficients to compute with. The substitutions (4.1.3) still cut it down to a short list of normal forms: two in each of these characteristics, according as $j = 0$ or not.

PROPOSITION 4.1.12. *Let E be an elliptic curve over k . Then the Weierstrass coordinates of Theorem 4.1.6 can be chosen so that the model (4.1.1) is the short form (4.1.2) when $\text{char } k \neq 2, 3$, and otherwise is the one listed here.*

char k	case	Weierstrass equation	Δ	j
3	$j(E) \neq 0$	$y^2 = x^3 + a_2x^2 + a_6$	$-a_2^3a_6$	$-a_2^3/a_6$
3	$j(E) = 0$	$y^2 = x^3 + a_4x + a_6$	$-a_4^3$	0
2	$j(E) \neq 0$	$y^2 + xy = x^3 + a_2x^2 + a_6$	a_6	$1/a_6$
2	$j(E) = 0$	$y^2 + a_3y = x^3 + a_4x + a_6$	a_3^4	0

PROOF. The case $\text{char } k \neq 2, 3$ is the completion of the square and of the cube that produced (4.1.2). In the two remaining characteristics we start from a model (4.1.1) and apply a substitution (4.1.3), reading off the new coefficients from the transformation table [Sil09, Table 3.1]. Nothing deep goes into the choice of u, r, s, t : one only asks that certain coefficients become 0 or 1. Each step below can be watched in Magma over the function field in the coefficients, in the characteristic at hand, with **subst** as defined above; the blocks run in one session, in order, and each redefines the a_i it needs.

Suppose $\text{char } k = 3$. Completing the square in y is (4.1.3) with $u = 1, r = 0, s = -a_1/2$ and $t = -a_3/2$, and it gives $y^2 = x^3 + a_2x^2 + a_4x + a_6$.

```
K<a1,a2,a3,a4,a6> := FunctionField(GF(3), 5);
E := EllipticCurve([a1,a2,a3,a4,a6]);
aInvariants(subst(E, 1, 0, -a1/2, -a3/2)); // a1 = a3 = 0
```

For such an equation a direct computation in characteristic 3 gives $c_4 = a_2^2$ and $\Delta = a_2^2a_4^2 - a_2^3a_6 - a_4^3$. So $j = a_2^6/\Delta$, and $j = 0$ if and only if $a_2 = 0$.

```
L<a2,a4,a6> := FunctionField(GF(3), 3);
D := EllipticCurve(L | 0, a2, 0, a4, a6);
cInvariants(D)[1] eq a2^2; // true
Discriminant(D) eq a2^2*a4^2 - a2^3*a6 - a4^3; // true
jInvariant(D) eq a2^6/Discriminant(D); // true
```

If $j = 0$, the equation is already the one listed, with $\Delta = -a_4^3$. If $j \neq 0$, then $a_2 \neq 0$, and $x = x' + a_4/a_2$ turns the coefficient of x' into $2a_2(a_4/a_2) + a_4 = 3a_4 = 0$, leaving the equation listed for $j \neq 0$; its discriminant is $-a_2^3a_6$, so $j = a_2^6/(-a_2^3a_6) = -a_2^3/a_6$.

```
D2 := subst(D, 1, a4/a2, 0, 0);
a6new := aInvariants(D2)[5];
aInvariants(D2); // [0, a2, 0, 0, a6new]
Discriminant(D2) eq -a2^3*a6new; // true
jInvariant(D2) eq -a2^3/a6new; // true
```

Suppose $\text{char } k = 2$. Now $c_4 = a_1^4$, so $j = a_1^{12}/\Delta$, and $j = 0$ if and only if $a_1 = 0$. If $j \neq 0$, the substitution

$$x = a_1^2x' + \frac{a_3}{a_1}, \quad y = a_1^3y' + \frac{a_1^2a_4 + a_3^2}{a_1^3}$$

makes the new a_1 equal to 1 and kills the new a_3 and a_4 , leaving the equation listed for $j \neq 0$; there $b_2 = 1, b_4 = b_6 = 0$ and $b_8 = a_6$, so $\Delta = a_6$ and $j = 1/a_6$.

```
K<a1,a2,a3,a4,a6> := FunctionField(GF(2), 5);
E := EllipticCurve([a1,a2,a3,a4,a6]);
cInvariants(E)[1] eq a1^4; // true
F := subst(E, a1, a3/a1, 0, (a1^2*a4 + a3^2)/a1^3);
a6new := aInvariants(F)[5];
```

```

aInvariants(F);           // [1, a2new, 0, 0, a6new]
Discriminant(F) eq a6new; // true
jInvariant(F) eq 1/a6new; // true

```

If $j = 0$, then $a_1 = 0$, and $x = x' + a_2$, $y = y'$ kills the new a_2 , since $4a_2 = 0$, leaving the last equation of the table.

```

M<a2,a3,a4,a6> := FunctionField(GF(2), 4);
G := EllipticCurve([M | 0, a2, a3, a4, a6]);
G2 := subst(G, 1, a2, 0, 0);
aInvariants(G2);           // [0, 0, a3, a2^2 + a4, a2*a4 + a6]
Discriminant(G2) eq a3^4; // true
jInvariant(G2) eq 0;       // true

```

There $b_2 = b_4 = 0$ and $b_6 = a_3^2$, so $\Delta = -27a_3^4 = a_3^4$, while $c_4 = 0$, so $j = 0$. This is [Sil09, Proposition A.1.1]. $\square$

Since $\Delta \neq 0$, the two characteristic 2 rows force $a_6 \neq 0$ and $a_3 \neq 0$ respectively; in characteristic 2 there is no smooth Weierstrass equation with $a_1 = a_3 = 0$.

The normal forms finish the converse in Proposition 4.1.10, which we left open in characteristic 2 and 3. Let E and E' be elliptic curves over $\bar{k}$ with $j(E) = j(E')$, both put in the form of the same row of the table; we must find a change of variables (4.1.3) over $\bar{k}$ carrying one to the other. In each case the substitutions preserving the row's shape are constrained first, by asking that the coefficients the row sets to 0 or 1 stay that way.

- **Characteristic 3, $j \neq 0$.** In $y^2 = x^3 + a_2x^2 + a_6$, the condition $a'_1 = 0$ forces $s = 0$, then $a'_3 = 0$ forces $t = 0$, and then $a'_4 = 0$ reads $2ra_2 = 0$, so $r = 0$ because $a_2 \neq 0$. What is left is $x = u^2x'$, $y = u^3y'$, giving $a'_2 = a_2/u^2$ and $a'_6 = a_6/u^6$. Now $j = -a_2^3/a_6$, so $j = j'$ says $a_2^3a'_6 = a'^3_2a_6$, and $u^2 := a_2/a'_2$ does the job.
- **Characteristic 3, $j = 0$.** In $y^2 = x^3 + a_4x + a_6$ we again get $s = t = 0$, while $a'_2 = 0$ is automatic since $3r = 0$. So $x = u^2x' + r$, $y = u^3y'$, with $a'_4 = a_4/u^4$ and $a'_6 = (a_6 + ra_4 + r^3)/u^6$. Here $a_4 \neq 0$, so choose u with $u^4 = a_4/a'_4$, and then r as a root of $r^3 + a_4r + a_6 - u^6a'_6 = 0$.
- **Characteristic 2, $j \neq 0$.** In $y^2 + xy = x^3 + a_2x^2 + a_6$, the condition $a'_1 = 1$ forces $u = 1$, then $a'_3 = 0$ forces $r = 0$ and $a'_4 = 0$ forces $t = 0$. So $x = x'$, $y = y' + sx'$, with $a'_6 = a_6$ and $a'_2 = a_2 + s + s^2$. Since $j = 1/a_6$, the hypothesis $j = j'$ already gives $a'_6 = a_6$, and it remains to take s a root of $s^2 + s + a_2 + a'_2 = 0$.
- **Characteristic 2, $j = 0$.** In $y^2 + a_3y = x^3 + a_4x + a_6$ the condition $a'_1 = 0$ is automatic and $a'_2 = 0$ forces $r = s^2$. Then $a'_3 = a_3/u^3$, $a'_4 = (a_4 + sa_3 + s^4)/u^4$ and $a'_6 = (a_6 + s^2a_4 + s^6 + ta_3 + t^2)/u^6$. Here $a_3 \neq 0$, so choose u with $u^3 = a_3/a'_3$, then s a root of $s^4 + a_3s + a_4 - u^4a'_4 = 0$, and then t a root of $t^2 + a_3t + a_6 + s^2a_4 + s^6 - u^6a'_6 = 0$.

Each equation has a root because $\bar{k}$ is algebraically closed, so Proposition 4.1.10 holds in every characteristic. Reading the same four computations with $E = E'$ counts the automorphisms, which is Theorem 4.7.11.

DRAFT
21 Sep 2026

4.2. The group law

On $\mathbb{C}/\Lambda$ the group law comes for free from addition in $\mathbb{C}$. On an elliptic curve over k it comes from the group $\text{Pic}^0 E$ of degree-0 divisors modulo principal divisors (Proposition 3.3.2): for a curve of genus 1, the map $P \mapsto [P - O]$ is a bijection $E(k) \rightarrow \text{Pic}^0 E$.

LEMMA 4.2.1. *Let E be a nice curve of genus 1, and $P, Q \in E(k)$. If $P \sim Q$, then $P = Q$.*

PROOF. Say $\text{div}(f) = P - Q$. Then $f \in \mathcal{L}(Q)$. Since $\deg Q = 1 > 2g - 2 = 0$, Corollary 3.5.5 gives $\ell(Q) = 1$, so $\mathcal{L}(Q) = k$ and f is constant. Hence $P - Q = \text{div}(f) = 0$. $\square$

This is what distinguishes genus 1 from genus 0: on $\mathbb{P}^1_k$ any two rational points are linearly equivalent (Example 3.3.9).

PROPOSITION 4.2.2. *Let (E, O) be an elliptic curve over k .*

- (1) *For every $D \in \text{Div}^0 E$ there is a unique $P \in E(k)$ with $D \sim P - O$. Write $\sigma(D) := P$.*
- (2) *The map $\kappa: E(k) \rightarrow \text{Pic}^0 E$, $P \mapsto [P - O]$, is a bijection, with inverse induced by σ .*

PROOF. Uniqueness in part 1 is Lemma 4.2.1. For existence, $\deg(D + O) = 1$, so $\ell(D + O) = 1$ by Corollary 3.5.5. Let f span $\mathcal{L}(D + O)$. Then $\text{div}(f) + D + O$ is an effective divisor of degree 1, so it is a single closed point P of degree 1, that is, a rational point, and $D \sim P - O$. Part 2 follows [Sil09, Proposition III.3.4]. $\square$

Note that the argument works over k itself, with no Galois descent: an effective divisor of degree 1 is automatically a rational point. Since O is a rational point, $\text{Pic}^0 E$ is also $(\text{Pic}^0 E_{k^{\text{sep}}})^{\text{Gal}_k}$ by Warning 3.3.8, so both readings of Pic^0 give $E(k)$. This fails for a nice curve of genus 1 with no rational point, such as $3X^3 + 4Y^3 + 5Z^3 = 0$ (Remark 4.8.13). We use κ to transport the group structure of $\text{Pic}^0 E$ to $E(k)$. So $P + Q$ is the unique rational point with

$$(P + Q) - O \sim (P - O) + (Q - O),$$

and O is the identity. For every field $L \supseteq k$, the same construction on E_L makes $E(L)$ a group, and $E(k) \subseteq E(L)$ is a subgroup. If L/k is Galois, then $\text{Gal}(L/k)$ acts on $E(L)$ by group automorphisms, with fixed points $E(k)$ (Proposition 2.1.12).

COROLLARY 4.2.3. *Let $D = \sum n_P P$ be a divisor on E supported on points of $E(k)$. Then D is principal if and only if $\sum n_P = 0$ and $\sum [n_P]P = O$ in $E(k)$. More generally, there is an exact sequence*

$$1 \rightarrow k^\times \rightarrow k(E)^\times \xrightarrow{\text{div}} \text{Div}^0 E \xrightarrow{\sigma} E(k) \rightarrow 0.$$

PROOF. Combine Proposition 3.3.2 with Proposition 4.2.2 [Sil09, Corollary III.3.5]. $\square$

For a smooth plane cubic whose base point is a flex, such as a Weierstrass cubic, this group law is the classical chord-and-tangent construction, illustrated in Figure 4.2.1. By Definition 4.1.2, the point at infinity of a Weierstrass cubic is a flex, with tangent line $Z = 0$.

PROPOSITION 4.2.4. *Let $E \subseteq \mathbb{P}_k^2$ be a smooth plane cubic, let $O \in E(k)$ be a flex, and give $E(\bar{k})$ the group law with identity O .*

- (1) *If a line meets E in $P, Q, R \in E(\bar{k})$, counted with intersection multiplicity, then $P + Q + R = O$.*
- (2) *If E is a Weierstrass cubic (4.1.1) and $O = (0 : 1 : 0)$, then $-P = (x_0, -y_0 - a_1 x_0 - a_3)$ for $P = (x_0, y_0)$.*

So $P + Q$ is obtained by drawing the line through P and Q (the tangent line if $P = Q$), taking its third intersection point R with E , and reflecting: $P + Q$ is the third intersection point of the line through R and O .

PROOF. We may assume that $k = \bar{k}$. Let L_O be a linear form defining the tangent line at O , which meets E only at O , with multiplicity 3; for a Weierstrass cubic, $L_O = Z$ (Definition 4.1.2). Let L be a linear form defining the line. On a smooth curve the intersection number at a closed point P is $\text{ord}_P(L/L_O) \deg P$ away from $L_O = 0$ (see the proof of Proposition 3.7.6), so by Theorem 3.7.4

$$\text{div}(L/L_O) = P + Q + R - 3O.$$

So $(P - O) + (Q - O) + (R - O) \sim 0$, which is part 1. For part 2, the vertical line $X - x_0 Z$ passes through O , and its other two intersection points with E are the roots in y of $f(x_0, y) = 0$, whose sum is $-a_1 x_0 - a_3$. Part 1 with $R = O$ gives part 2. $\square$

To compute the third intersection point we expand the cubic form along a line. For vectors $P, Q \in k^3$ we write $\nabla F(P) \cdot Q := \sum_i \frac{\partial F}{\partial X_i}(P) Q_i$.

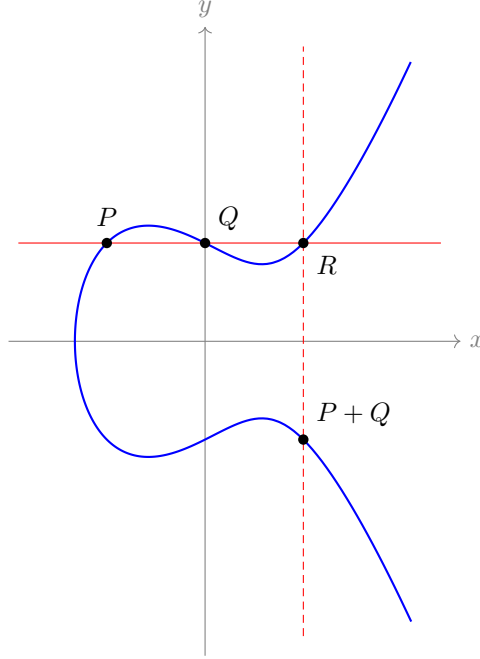


FIGURE 4.2.1. The chord-and-tangent law on $y^2 = x^3 - x + 1$ over $\mathbb{R}$: the line through $P = (-1, 1)$ and $Q = (0, 1)$ meets the curve again at $R = (1, 1)$, and $P + Q = -R = (1, -1)$.

LEMMA 4.2.5 (Polar expansion). *Let $F \in k[X, Y, Z]$ be a cubic form, and $P, Q \in k^3$. Then, for $s, t \in k$,*

$$(4.2.1) \quad F(sP + tQ) = s^3 F(P) + s^2 t \nabla F(P) \cdot Q + s t^2 \nabla F(Q) \cdot P + t^3 F(Q).$$

PROOF. Since F is homogeneous of degree 3, $F(sP + tQ)$ is a form of degree 3 in s and t , say $\sum_{m=0}^3 c_m s^{3-m} t^m$, with $c_0 = F(P)$ and $c_3 = F(Q)$. Setting $s = 1$, c_1 is the coefficient of t in the polynomial $F(P + tQ) \in k[t]$, which is its derivative at $t = 0$; by the chain rule for polynomials, this is $\nabla F(P) \cdot Q$. Exchanging the roles of P and Q gives $c_2 = \nabla F(Q) \cdot P$. $\square$

The tangent line to E at a smooth point P is $\{\nabla F(P) \cdot V = 0\}$, so the coefficient $\nabla F(P) \cdot Q$ vanishes exactly when Q lies on the tangent line at P . This turns the chord-and-tangent construction into formulas in homogeneous coordinates.

PROPOSITION 4.2.6. *Let $E = V_+(F) \subseteq \mathbb{P}_k^2$ be a smooth plane cubic with a flex $O \in E(k)$, and give $E(\bar{k})$ the group law with identity O , as in Proposition 4.2.4. We write $\bar{k}$ -points as vectors in $\bar{k}^3$, and the formulas below are homogeneous.*

- (1) **Chord.** *Let $P_1 \neq P_2$ be points of $E(\bar{k})$, and set $A := \nabla F(P_1) \cdot P_2$ and $B := \nabla F(P_2) \cdot P_1$. Then A and B do not both vanish, the line through P_1 and P_2 meets E in the third intersection point $R := BP_1 - AP_2$, and $P_1 + P_2 = -R$.*
- (2) **Tangent.** *Let $P \in E(\bar{k})$, and let $Q \in \bar{k}^3$ be a vector on the tangent line at P , that is $\nabla F(P) \cdot Q = 0$, which is not a multiple of P . Then $F(Q)$ and $\nabla F(Q) \cdot P$ do not both vanish, the tangent line meets E again in $R := F(Q)P - (\nabla F(Q) \cdot P)Q$, and $[2]P = -R$.*
- (3) **Weierstrass cubics.** *If E is the Weierstrass cubic (4.1.1) with $O = (0 : 1 : 0)$, then $-(X : Y : Z) = (X : -Y - a_1 X - a_3 Z : Z)$, and for $P \neq O$ the vector $Q := (\frac{\partial F}{\partial Y}(P), -\frac{\partial F}{\partial X}(P), 0)$ satisfies the hypotheses of part 2.*

PROOF. For part 1, $F(P_1) = F(P_2) = 0$, so (4.2.1) reads $F(sP_1 + tP_2) = st(As + Bt)$. If $A = B = 0$, this vanishes identically, so the line would be a component of E , which is impossible since E is geometrically integral (Corollary 3.7.5). Hence the line meets E at $(s : t) = (1 : 0)$, $(0 : 1)$ and $(B : -A)$, that is, at P_1 , P_2 and R . If $A = 0$, then P_2 lies on the tangent line at P_1 , and $R = P_1$ is the tangency, counted twice. Now $P_1 + P_2 + R = O$ by Proposition 4.2.4.

For part 2, $F(P) = 0$ and $\nabla F(P) \cdot Q = 0$, so $F(sP + tQ) = t^2(s \nabla F(Q) \cdot P + tF(Q))$. The double root at $t = 0$ is the tangency at P , and the remaining root $(s : t) = (F(Q) : -\nabla F(Q) \cdot P)$ is R ; as before the two coefficients cannot both vanish. Then $P + P + R = O$.

For part 3, negation is part 2 of Proposition 4.2.4, homogenized. The vector Q satisfies $\nabla F(P) \cdot Q = \frac{\partial F}{\partial X} \frac{\partial F}{\partial Y} - \frac{\partial F}{\partial Y} \frac{\partial F}{\partial X} = 0$. It is non-zero: if both partial derivatives vanished at P , the Euler relation $X \frac{\partial F}{\partial X} + Y \frac{\partial F}{\partial Y} + Z \frac{\partial F}{\partial Z} = 3F$ evaluated at P would give $Z_0 \frac{\partial F}{\partial Z}(P) = 0$, so, as $Z_0 \neq 0$ for $P \neq O$, the $\bar{k}$ -point P would be singular. Finally, Q has third coordinate 0 and P does not, so Q is not a multiple of P . $\square$

For $P = O$ there is nothing to compute, since $[2]O = O$. Reading these formulas on the affine patch $Z = 1$, with the line written as $y = \lambda x + \nu$, gives the classical algorithm.

ALGORITHM 4.2.7 (Group law). Let E be a smooth Weierstrass cubic (4.1.1), and $P_i = (x_i, y_i) \in E(k)$ for $i = 1, 2$.

- (1) If $x_1 = x_2$ and $y_1 + y_2 + a_1x_2 + a_3 = 0$, then $P_1 + P_2 = O$.
- (2) Otherwise set

$$\begin{aligned} \lambda &:= \frac{y_2 - y_1}{x_2 - x_1}, & \nu &:= \frac{y_1x_2 - y_2x_1}{x_2 - x_1} & \text{if } x_1 \neq x_2, \\ \lambda &:= \frac{3x_1^2 + 2a_2x_1 + a_4 - a_1y_1}{2y_1 + a_1x_1 + a_3}, & \nu &:= \frac{-x_1^3 + a_4x_1 + 2a_6 - a_3y_1}{2y_1 + a_1x_1 + a_3} & \text{if } x_1 = x_2, \end{aligned}$$

so that $y = \lambda x + \nu$ is the line through P_1 and P_2 , or the tangent line if $P_1 = P_2$. Then $P_1 + P_2 = (x_3, y_3)$ with

$$x_3 = \lambda^2 + a_1\lambda - a_2 - x_1 - x_2, \quad y_3 = -(\lambda + a_1)x_3 - \nu - a_3.$$

In particular, for the short form (4.1.2) and $P = (x, y)$ with $y \neq 0$, the duplication formula is

$$x([2]P) = \frac{x^4 - 2Ax^2 - 8Bx + A^2}{4(x^3 + Ax + B)}.$$

PROOF. Substitute $y = \lambda x + \nu$ into the equation of E ; this is (4.2.1) on the patch $Z = 1$, with the line parametrized by x instead of $(s : t)$. The resulting cubic in x has roots x_1 , x_2 and x_3 , and the coefficient of x^2 gives $x_1 + x_2 + x_3 = \lambda^2 + a_1\lambda - a_2$. Then negate, by part 3 of Proposition 4.2.6 [Sil09, Algorithm III.2.3]. $\square$

EXAMPLE 4.2.8. On $E: y^2 = x^3 + 17$ over $\mathbb{Q}$, let $P_1 := (-2, 3)$ and $P_2 := (-1, 4)$. The line through them has $\lambda = 1$, $\nu = 5$, so $x_3 = 1 + 2 + 1 = 4$, $y_3 = -4 - 5 = -9$, and $P_1 + P_2 = (4, -9)$. Likewise $[2]P_1 = (8, -23)$ and $[2]P_2 = (137/64, -2651/512)$. Denominators appear quickly; the Mordell–Weil theorem (Chapter 8) says that nonetheless $E(\mathbb{Q})$ is finitely generated.

EXAMPLE 4.2.9. A point $P \neq O$ of $E(\bar{k})$ satisfies $[2]P = O$ if and only if $P = -P$, i.e. if and only if $2y + a_1x + a_3 = 0$ at P . Suppose $\text{char } k \neq 2$, and let $E: y^2 = (x - e_1)(x - e_2)(x - e_3)$ with distinct $e_i \in k$, and $P_i := (e_i, 0)$, as in Example 3.3.10. The divisors computed there,

$$\text{div}(x - e_i) = 2P_i - 2O, \quad \text{div}(y) = P_1 + P_2 + P_3 - 3O,$$

say, by Corollary 4.2.3, that $[2]P_i = O$ and $P_1 + P_2 + P_3 = O$. So

$$\{O, P_1, P_2, P_3\} \cong (\mathbb{Z}/2\mathbb{Z})^2,$$

exactly as for $(\mathbb{C}/\Lambda)[2]$ in Example 1.3.4.

The group law is given by morphisms.

THEOREM 4.2.10. *Let E be an elliptic curve over k . The formulas of Algorithm 4.2.7 define morphisms of k -varieties*

$$+ : E \times E \rightarrow E, \quad - : E \rightarrow E.$$

In particular, for $Q \in E(k)$, translation $\tau_Q : E \rightarrow E$, $P \mapsto P + Q$, is an isomorphism of curves.

PROOF. Negation is a rational map $E \dashrightarrow E$, hence a morphism by Proposition 3.2.1. For addition, see [Sil09, Theorem III.3.6]: one shows that τ_Q is a morphism for each Q , and then that the rational map $E \times E \dashrightarrow E$ is regular everywhere by translating. $\square$

So an elliptic curve is a group variety, an abelian variety of dimension 1. We end the section with the singular Weierstrass cubics of Remark 4.1.11.

PROPOSITION 4.2.11. *Let E be a Weierstrass cubic with $\Delta = 0$, singular point S , and $E_{\text{ns}} := E - \{S\}$. The chord-and-tangent construction of Proposition 4.2.4 makes $E_{\text{ns}}(\bar{k})$ an abelian group, and:*

- (1) *if S is a node, with tangent lines $y = \alpha_i x + \beta_i$ for $i = 1, 2$, then $(x, y) \mapsto (y - \alpha_1 x - \beta_1)/(y - \alpha_2 x - \beta_2)$ is an isomorphism $E_{\text{ns}}(\bar{k}) \cong \bar{k}^\times$;*
- (2) *if S is a cusp, with tangent line $y = \alpha x + \beta$, then $(x, y) \mapsto (x - x(S))/(y - \alpha x - \beta)$ is an isomorphism $E_{\text{ns}}(\bar{k}) \cong \bar{k}$.*

PROOF. See [Sil09, Proposition III.2.5]. $\square$

DRAFT
23 Sep 2026

4.3. Isogenies

By Proposition 1.3.1, every holomorphic map of complex tori is a homomorphism followed by a translation, and holomorphic homomorphisms are the maps $z \mapsto \alpha z$ with $\alpha \Lambda_1 \subseteq \Lambda_2$ (Corollary 1.3.2). We now prove the algebraic analogue.

DEFINITION 4.3.1. Let E_1 and E_2 be elliptic curves over k . An **isogeny** is a morphism $\varphi : E_1 \rightarrow E_2$ with $\varphi(O_1) = O_2$. By Proposition 3.2.2, either $\varphi = 0$ is constant, or φ is surjective and has a degree (Definition 3.2.5); we set $\deg 0 := 0$. The curves are isogenous if there is a non-zero isogeny between them. We write $\text{Hom}(E_1, E_2)$ for the set of isogenies, $\text{End}(E) := \text{Hom}(E, E)$ and $\text{Aut}(E)$ for the invertible elements of $\text{End}(E)$. For a field $L \supseteq k$, the isogenies after base extension to L are the elements of $\text{Hom}(E_{1,L}, E_{2,L})$, $\text{End}(E_L)$ and $\text{Aut}(E_L)$.

Unlike Definition 1.3.3, we allow the zero map. By Remark 2.3.2, elements of $\text{Hom}(E_1, E_2)$ are morphisms over k , and $\text{Hom}(E_1, E_2) \subseteq \text{Hom}(E_{1,\bar{k}}, E_{2,\bar{k}})$; Silverman writes $\text{Hom}(E_1, E_2)$ for the latter. By Theorem 4.2.10, $\text{Hom}(E_1, E_2)$ is a group under $(\varphi + \psi)(P) := \varphi(P) + \psi(P)$, and $\text{End}(E)$ is a ring under composition.

EXAMPLE 4.3.2. For $m \in \mathbb{Z}$, the multiplication by m map $[m] : E \rightarrow E$ is an isogeny, by Theorem 4.2.10. Its kernel is the group scheme $E[m]$, and $E[m](\bar{k})$ is the m -torsion subgroup of $E(\bar{k})$.

EXAMPLE 4.3.3. Suppose $\text{char } k = p > 0$ and $q = p^r$. The relative Frobenius morphism $F : E \rightarrow E^{(q)}$ of Definition 3.2.11 sends $(0 : 1 : 0)$ to $(0 : 1 : 0)$, and $E^{(q)}$ is the Weierstrass cubic with coefficients a_i^q , which has discriminant $\Delta^q \neq 0$. So F is an isogeny, purely inseparable of degree q by Proposition 3.2.12. If E is defined over $\mathbb{F}_q$, then $F \in \text{End}(E)$ (Example 3.2.15).

EXAMPLE 4.3.4. If $\text{char } k \neq 2$ and $i \in k$ is a square root of -1 , then $[i](x, y) := (-x, iy)$ is an automorphism of $y^2 = x^3 + x$ with $[i]^2 = [-1]$.

EXAMPLE 4.3.5. For $Q \in E(k)$, $Q \neq O$, the translation τ_Q is a morphism but not an isogeny. Every morphism $\psi : E_1 \rightarrow E_2$ is $\tau_{\psi(O_1)} \circ \varphi$ with $\varphi := \tau_{-\psi(O_1)} \circ \psi$ an isogeny, as in Proposition 1.3.1.

THEOREM 4.3.6. *An isogeny $\varphi : E_1 \rightarrow E_2$ is a group homomorphism $E_1(L) \rightarrow E_2(L)$, for every field $L \supseteq k$.*

PROOF. After base extension we may assume $L = k$, and that $\varphi \neq 0$. By Proposition 3.3.6, $\varphi_*: \text{Pic}^0 E_1 \rightarrow \text{Pic}^0 E_2$ is a homomorphism, and $\varphi_*(P - O_1) = \varphi(P) - O_2$ for $P \in E_1(k)$, because rational points have residue degree 1. So $\varphi = \kappa_2^{-1} \circ \varphi_* \circ \kappa_1$ on $E_1(k)$, a composition of homomorphisms (Proposition 4.2.2) [Sil09, Theorem III.4.8]. $\square$

PROPOSITION 4.3.7.

- (1) If $m \neq 0$, then $[m] \neq 0$.
- (2) $\text{Hom}(E_1, E_2)$ is a torsion-free $\mathbb{Z}$ -module.
- (3) $\text{End}(E)$ is a ring of characteristic 0 without zero divisors.

PROOF. Part 1 is [Sil09, Proposition III.4.2(a)]; when $\text{char } k \nmid m$ it also follows from Corollary 4.4.3 below. Given part 1, $[m] \circ \varphi = 0$ forces $\varphi = 0$, since $\deg$ is multiplicative and non-constant morphisms compose to non-constant morphisms. This gives parts 2 and 3. $\square$

We write $\ker \varphi$ for the kernel of φ on $E_1(\bar{k})$. The kernel of a non-zero isogeny is finite, being a fiber. The next theorem is the analogue of Lemma 1.3.6; it is the Galois theory of function fields of elliptic curves.

THEOREM 4.3.8. Let $\varphi: E_1 \rightarrow E_2$ be a non-zero isogeny.

- (1) For every $Q \in E_2(\bar{k})$, the fiber $\varphi^{-1}(Q) \subseteq E_1(\bar{k})$ has $\deg_s \varphi$ elements, and $e_\varphi(P) = \deg_i \varphi$ for every $P \in E_1(\bar{k})$.
- (2) Suppose φ is separable. Then φ is unramified, $\#\ker \varphi = \deg \varphi$, and $\bar{k}(E_1)/\varphi^* \bar{k}(E_2)$ is a Galois extension, with Galois group $\ker \varphi$ acting by $T \mapsto \tau_T^*$.

PROOF. By Proposition 3.2.8, all but finitely many fibers have $\deg_s \varphi$ elements. But for $Q, Q' \in E_2(\bar{k})$ with $\varphi(P) = Q$ and $\varphi(P') = Q'$, translation by $P' - P$ is a bijection $\varphi^{-1}(Q) \rightarrow \varphi^{-1}(Q')$, by Theorem 4.3.6, and it preserves ramification indices. So all fibers have the same size and the same ramification indices, and part 1 follows from $\sum_{P \mapsto Q} e_\varphi(P) = \deg \varphi$. In part 2, $e_\varphi = 1$, so $\#\ker \varphi = \deg \varphi$. The translations τ_T , $T \in \ker \varphi$, give $\deg \varphi$ distinct automorphisms of $\bar{k}(E_1)$ over $\varphi^* \bar{k}(E_2)$, so the extension is Galois [Sil09, Theorem III.4.10]. $\square$

If φ is separable, the scheme-theoretic fiber $\varphi^{-1}(O_2)$ is étale over k , so $\ker \varphi \subseteq E_1(k^{\text{sep}})$, and it is a Gal_k -stable subgroup. So an isogeny determines a finite Gal_k -module. Conversely, every finite subgroup is a kernel; this replaces the lattice $\Lambda_2 \supseteq \Lambda_1$ of Chapter 1.

PROPOSITION 4.3.9. Let $\Phi \subseteq E(\bar{k})$ be a finite subgroup. There are an elliptic curve E' over $\bar{k}$, unique up to isomorphism, and a separable isogeny $\varphi: E_{\bar{k}} \rightarrow E'$ with $\ker \varphi = \Phi$. We write $E/\Phi := E'$. If $\Phi \subseteq E(k^{\text{sep}})$ is Gal_k -stable, then E' and φ can be defined over k . Moreover, if $\varphi: E_1 \rightarrow E_2$ is separable and $\psi: E_1 \rightarrow E_3$ is an isogeny with $\ker \varphi \subseteq \ker \psi$, then $\psi = \lambda \circ \varphi$ for a unique isogeny $\lambda: E_2 \rightarrow E_3$.

PROOF. Take E' to be the nice curve with function field $\bar{k}(E)^\Phi$, the fixed field of the translations τ_T^* , $T \in \Phi$ (Theorem 3.2.3); see [Sil09, Proposition III.4.12, Remarks III.4.13.1 and III.4.13.2, and Corollary III.4.11]. $\square$

Over $\mathbb{C}$, the structure $(\mathbb{C}/\Lambda)[N] \cong (\mathbb{Z}/N\mathbb{Z})^2$ was immediate from $(\mathbb{C}/\Lambda)[N] = \frac{1}{N}\Lambda/\Lambda$. Over k we know from Theorem 4.3.8 that $\#E[N](\bar{k}) = \deg_s[N]$, so what we need is $\deg[N] = N^2$. The tool is the dual isogeny, the algebraic analogue of Theorem 1.3.8.

THEOREM 4.3.10. Let $\varphi: E_1 \rightarrow E_2$ be an isogeny of degree d .

- (1) There is a unique isogeny $\hat{\varphi}: E_2 \rightarrow E_1$, the *dual isogeny*, with $\hat{\varphi} \circ \varphi = [d]$. If $\varphi \neq 0$, then on rational points $\hat{\varphi} = \sigma \circ \varphi^* \circ \kappa_2$, that is, $Q \mapsto \sigma(\varphi^*(Q - O_2))$, with σ as in Proposition 4.2.2. We set $\hat{0} := 0$.
- (2) $\varphi \circ \hat{\varphi} = [d]$ on E_2 .

- (3) $\widehat{\lambda \circ \varphi} = \hat{\varphi} \circ \hat{\lambda}$ for an isogeny $\lambda: E_2 \rightarrow E_3$.
- (4) $\widehat{\varphi + \psi} = \hat{\varphi} + \hat{\psi}$ for $\psi \in \text{Hom}(E_1, E_2)$.
- (5) $\widehat{[m]} = [m]$ and $\deg[m] = m^2$ for all $m \in \mathbb{Z}$.
- (6) $\deg \hat{\varphi} = \deg \varphi$ and $\hat{\hat{\varphi}} = \varphi$.

PROOF. See [Sil09, Theorems III.6.1 and III.6.2]. We indicate two steps.

For a separable φ and $P \in E_1(\bar{k})$, Theorem 4.3.8 gives $\varphi^*(\varphi(P)) = \sum_{T \in \ker \varphi} (P + T)$ and $\varphi^*(O_2) = \sum_{T \in \ker \varphi} T$. So $\sigma(\varphi^*(\varphi(P) - O_2)) = \sum_T (P + T) - \sum_T T = [d]P$, which is $\hat{\varphi} \circ \varphi = [d]$. The inseparable case reduces to this one and to Frobenius, using Corollary 3.2.14.

The formula in part 1 shows that $\hat{\varphi}$ is defined over k . Apply it to the generic point of E_2 , that is, to the point $\eta \in E_2(k(E_2))$ given by the identity; it gives a point of $E_1(k(E_2))$, which is a rational map $E_2 \dashrightarrow E_1$ over k , hence a morphism by Proposition 3.2.1. Part 5 follows from part 4 by induction on m , and then $[\deg[m]] = \widehat{[m]} \circ [m] = [m^2]$, so $\deg[m] = m^2$ by Proposition 4.3.7. $\square$

COROLLARY 4.3.11. *The degree $\deg: \text{Hom}(E_1, E_2) \rightarrow \mathbb{Z}$ is a positive definite quadratic form: $(\varphi, \psi) \mapsto \deg(\varphi + \psi) - \deg \varphi - \deg \psi$ is bilinear, and $\deg \varphi > 0$ for $\varphi \neq 0$.*

PROOF. Viewing integers in $\text{End}(E_1)$ via $m \mapsto [m]$, the pairing is $\hat{\varphi}\psi + \hat{\psi}\varphi$, which is bilinear by part 4 of Theorem 4.3.10 [Sil09, Corollary III.6.3]. $\square$

In characteristic p , the dual of the Frobenius morphism has a name of its own.

DEFINITION 4.3.12. Suppose $\text{char } k = p > 0$, and let E be an elliptic curve over k . For $r \geq 1$, write

$$F^r: E \rightarrow E^{(p^r)}$$

for the p^r -th power relative Frobenius morphism (Definition 3.2.11 with $q = p^r$), and

$$V^r := \widehat{F^r}: E^{(p^r)} \rightarrow E$$

for its dual isogeny, the [Verschiebung](#).

The exponent is not an abuse: F^r is the composite $E \rightarrow E^{(p)} \rightarrow \cdots \rightarrow E^{(p^r)}$ of p -th power Frobenius morphisms, one for each of the curves $E^{(p^i)}$. By Example 4.3.3 it is purely inseparable of degree p^r , so Theorem 4.3.10 gives

$$V^r \circ F^r = [p^r] \text{ on } E, \quad F^r \circ V^r = [p^r] \text{ on } E^{(p^r)}, \quad \deg V^r = p^r,$$

by parts 1, 2 and 6. If E is defined over $\mathbb{F}_{p^r}$, then $E^{(p^r)} = E$ and both F^r and V^r lie in $\text{End}(E)$. The name is the German word for "shift".

EXAMPLE 4.3.13. Suppose $\text{char } k \neq 2$. Let $a, b \in k$ with $b(a^2 - 4b) \neq 0$, and

$$E_1: y_1^2 = x_1^3 + ax_1^2 + bx_1, \quad E_2: y_2^2 = x_2^3 - 2ax_2^2 + (a^2 - 4b)x_2,$$

with x_i, y_i the affine coordinates of E_i . Both cubics have distinct roots, so both curves are smooth, and $T := (0, 0)$ is a point of order 2 on E_1 . Let

$$\varphi: E_1 \rightarrow E_2, \quad \varphi(x_1, y_1) := \left(\frac{y_1^2}{x_1^2}, \frac{y_1(x_1^2 - b)}{x_1^2} \right)$$

[Sil09, Example III.4.5].

- (1) **Kernel and degree.** The function $\varphi^*x_2 = y_1^2/x_1^2 = x_1 + a + b/x_1$ has poles only at T and O_1 , so $\ker \varphi = \{O_1, T\}$. Since $y_1 = x_1^2 \varphi^*y_2 / (x_1^2 - b)$ and $x_1^2 + (a - \varphi^*x_2)x_1 + b = 0$, the field $k(E_1)$ has degree at most 2 over $\varphi^*k(E_2)$. By Theorem 4.3.8, $\deg \varphi \geq \#\ker \varphi$, so $\deg \varphi = 2$, and φ is separable.
- (2) **The dual.** The same construction applied to E_2 , whose coefficients are $-2a$ and $a^2 - 4b$, gives an isogeny $\psi: E_2 \rightarrow E_3$ with $E_3: y_3^2 = x_3^3 + 4ax_3^2 + 16bx_3$, and $\iota(x_3, y_3) := (x_3/4, y_3/8)$ is an isomorphism $E_3 \rightarrow E_1$. A direct computation with Algorithm 4.2.7 shows that $\iota \circ \psi \circ \varphi = [2]$, so $\hat{\varphi} = \iota \circ \psi$. Its kernel

is $\{O_2, (0,0)\}$, which is $\varphi(E_1[2](\bar{k}))$: the other two points $(e,0)$ of $E_1[2](\bar{k})$ have $e^2 + ae + b = 0$, and $\varphi(e,0) = (0,0)$.

(3) **Example.** For $y_1^2 = x_1^3 - x_1$, that is $a = 0$ and $b = -1$, we get $E_2: y_2^2 = x_2^3 + 4x_2$ and $\varphi(x_1, y_1) = ((x_1^2 - 1)/x_1, y_1(x_1^2 + 1)/x_1^2)$.

The invariant differential detects the sign in part 2 without computing [2] (Example 4.4.7).

REMARK 4.3.14. Compare the multiplicative group. A morphism $(\mathbb{G}_m)_{\bar{k}} \rightarrow (\mathbb{G}_m)_{\bar{k}}$ is $t \mapsto ct^m$, since the units of $\bar{k}[t, t^{-1}]$ are the ct^m , and it is a homomorphism only if $c = 1$. So the endomorphisms of $\mathbb{G}_m$ are the maps $[m]$, and $\deg[m] = |m|$ (Example 3.4.8). There is no dual in the sense of Theorem 4.3.10: for $m \geq 1$, the only $[n]$ with $[n] \circ [m] = [\deg[m]]$ is $[1]$, and the rule $[m] \mapsto [1]$ is not additive. The identity $\deg[m] = m^2$ of part 5 is what makes the dual of an elliptic curve possible. Over $\mathbb{C}$, it reflects the fact that $\mathbb{C}/\Lambda$ is the quotient of $\mathbb{C}$ by a lattice of rank 2, whereas $\mathbb{C}^\times \cong \mathbb{C}/2\pi i\mathbb{Z}$ is the quotient by a subgroup of rank 1 (Example 1.5.4).

4.4. The invariant differential

On $\mathbb{C}/\Lambda$, the differential dz is invariant under translation, and an isogeny $z \mapsto \alpha z$ pulls it back to αdz , so pullback of differentials is additive (Example 1.5.5). By Remark 4.1.8 the invariant differential ω is dz , and all of this survives over k . The consequences in positive characteristic are striking, because pullback detects separability (Proposition 3.4.2).

Throughout, ω is the invariant differential of a Weierstrass model of E . By Proposition 4.1.5, $\text{div}(\omega) = 0$. Any non-zero differential with divisor 0 is $a\omega$ with $a \in k^\times$, by Proposition 3.1.7, so up to a scalar ω does not depend on the model.

PROPOSITION 4.4.1. *For every $Q \in E(k)$, $\tau_Q^*\omega = \omega$.*

PROOF. Since τ_Q is an isomorphism, $\text{div}(\tau_Q^*\omega) = \tau_Q^*\text{div}(\omega) = 0$, so $\tau_Q^*\omega = a_Q\omega$ with $a_Q \in k^\times$. After base extension to $\bar{k}$, the map $E_{\bar{k}} \rightarrow \mathbb{P}_{\bar{k}}^1$, $Q \mapsto a_Q$, is a morphism that omits 0 and ∞ , so it is constant by Proposition 3.2.2, equal to $a_O = 1$ [Sil09, Proposition III.5.1]. $\square$

THEOREM 4.4.2. *Let $\varphi, \psi: E' \rightarrow E$ be isogenies. Then $(\varphi + \psi)^*\omega = \varphi^*\omega + \psi^*\omega$.*

PROOF. See [Sil09, Theorem III.5.2]. The idea: on $E \times E$, the differential $\text{sum}^*\omega$ is a combination of the pullbacks of ω along the two projections, and translation invariance forces both coefficients to be 1. $\square$

The two plus signs have different meanings: the first is addition in $\text{Hom}(E', E)$, the second is addition of differentials.

COROLLARY 4.4.3. *For every $m \in \mathbb{Z}$, $[m]^*\omega = m\omega$. So if $\text{char } k \nmid m$, then $[m]$ is non-zero and separable.*

PROOF. Induction on m , starting from $[0]^*\omega = 0$ and $[1]^*\omega = \omega$, using Theorem 4.4.2. If $m \neq 0$ in k , then $[m]^*\omega \neq 0$, so $[m] \neq 0$, and $[m]$ is separable by Proposition 3.4.2 [Sil09, Corollaries III.5.3 and III.5.4]. $\square$

COROLLARY 4.4.4. *Let E be an elliptic curve over $\mathbb{F}_q$, of characteristic p , and $F \in \text{End}(E)$ its q -th power Frobenius endomorphism (Example 3.2.15). For $m, n \in \mathbb{Z}$, the isogeny $m + nF$ is separable if and only if $p \nmid m$. In particular $1 - F$ is separable.*

PROOF. F is inseparable, so $F^*\omega = 0$ by Proposition 3.4.2. By Theorem 4.4.2 and Corollary 4.4.3, $(m + nF)^*\omega = m\omega$, which vanishes if and only if $p \mid m$ [Sil09, Corollary III.5.5]. $\square$

Since $E(\mathbb{F}_q)$ is the set of fixed points of F on $E(\bar{k})$, it is the kernel of $1 - F$. By Theorem 4.3.8 and Corollary 4.4.4,

$$\#E(\mathbb{F}_q) = \deg(1 - F).$$

Every isogeny acts on invariant differentials by a scalar.

PROPOSITION 4.4.5. *Let ω_1, ω_2 and ω_3 be invariant differentials of Weierstrass models of elliptic curves E_1, E_2 and E_3 .*

- (1) *For every isogeny $\varphi: E_1 \rightarrow E_2$ there is a unique $c_\varphi \in k$ with $\varphi^*\omega_2 = c_\varphi\omega_1$.*
- (2) *$c_{\lambda \circ \varphi} = c_\lambda c_\varphi$ for $\lambda: E_2 \rightarrow E_3$, and $c_{\varphi + \psi} = c_\varphi + c_\psi$ for $\psi \in \text{Hom}(E_1, E_2)$.*
- (3) *Replacing ω_1 by $u\omega_1$, with $u \in k^\times$, replaces c_φ by c_φ/u .*
- (4) *φ is separable if and only if $c_\varphi \neq 0$.*
- (5) *$c_{\hat{\varphi}}c_\varphi = \deg \varphi$, where $c_{\hat{\varphi}}$ is computed with ω_2 and ω_1 .*

PROOF. For part 1 we may assume $\varphi \neq 0$. Over $\bar{k}$, for $P \in E_1(\bar{k})$, we have $\varphi \circ \tau_P = \tau_{\varphi(P)} \circ \varphi$ by Theorem 4.3.6, so by Proposition 4.4.1

$$\tau_P^* \varphi^* \omega_2 = \varphi^* \tau_{\varphi(P)}^* \omega_2 = \varphi^* \omega_2.$$

Write $\varphi^* \omega_2 = g\omega_1$ with $g \in k(E_1)$. Since ω_1 is invariant too, $g \circ \tau_P = g$ for every P . If $g \neq 0$ had a pole, it would have a pole at every point of $E_1(\bar{k})$; so g is constant, and $g \in k$ by Proposition 3.1.7. Part 2 follows from $(\lambda \circ \varphi)^* = \varphi^* \circ \lambda^*$ and Theorem 4.4.2, part 3 is immediate, and part 4 is Proposition 3.4.2. For part 5, by parts 2 and 3 and Corollary 4.4.3, $c_{\hat{\varphi}}c_\varphi = c_{\hat{\varphi} \circ \varphi} = c_{[\deg \varphi]} = \deg \varphi$ [Sil09, Section III.5]. $\square$

COROLLARY 4.4.6. *For $\varphi \in \text{End}(E)$, compute c_φ with $\omega_1 = \omega_2 = \omega$, so that $\varphi^*\omega = c_\varphi\omega$. Then $\varphi \mapsto c_\varphi$ is a ring homomorphism $\text{End}(E) \rightarrow k$ whose kernel consists of 0 and the inseparable endomorphisms. If $\text{char } k = 0$, then $\text{End}(E_{\bar{k}})$ embeds in $\bar{k}$, and so it is commutative.*

PROOF. It is a ring homomorphism by part 2 of Proposition 4.4.5, with the stated kernel by part 4. In characteristic 0 every non-zero endomorphism is separable; apply this to $E_{\bar{k}}$ [Sil09, Corollary III.5.6]. $\square$

Over $\mathbb{C}$, the homomorphism $\varphi \mapsto c_\varphi$ sends $z \mapsto \alpha z$ to α : it identifies $\text{End}(\mathbb{C}/\Lambda)$ with $\{\alpha \in \mathbb{C} : \alpha\Lambda \subseteq \Lambda\}$, as in Corollary 1.3.2.

EXAMPLE 4.4.7. Take the isogenies of Example 4.3.13, with $\omega_i := dx_i/(2y_i)$ on E_i . Since $\varphi^*y_2 = y_1(x_1^2 - b)/x_1^2$ and $\varphi^*x_2 = x_1 + a + b/x_1$, we have $\varphi^*dx_2 = (x_1^2 - b)dx_1/x_1^2$, so

$$\varphi^*\omega_2 = \frac{(x_1^2 - b)dx_1/x_1^2}{2y_1(x_1^2 - b)/x_1^2} = \omega_1, \quad c_\varphi = 1.$$

The same computation gives $c_\psi = 1$, and $\iota^*\omega_1 = (dx_3/4)/(2y_3/8) = 2\omega_3$, so $c_\iota = 2$. Hence $c_{\hat{\varphi}} = 2$ and $c_{\hat{\varphi}}c_\varphi = 2 = \deg \varphi$, as part 5 of Proposition 4.4.5 requires. If we only knew that $\iota \circ \psi \circ \varphi = \pm[2]$, this would decide the sign, since $c_{[-2]} = -2 \neq 2$.

COROLLARY 4.4.8. *Suppose $\text{char } k = p > 0$.*

- (1) *If $p \mid \deg \varphi$, then φ or $\hat{\varphi}$ is inseparable.*
- (2) *$[p]$ is inseparable.*

PROOF. Part 1 is part 5 of Proposition 4.4.5 with $\deg \varphi = 0$ in k . For part 2, $c_{[p]} = p = 0$. $\square$

For $\mathbb{G}_m$ the same two facts are elementary (Examples 3.4.7 and 3.4.8). In both cases $c_{[m]} = m$ is the derivative of $[m]$ at the identity: of $t \mapsto t^m$ at $t = 1$, and of $z \mapsto mz$ on $\mathbb{C}/\Lambda$. The difference is the degree. In characteristic p , $[p]$ on $\mathbb{G}_m$ is inseparable of degree p , so it has no room for a kernel; on E it is inseparable of degree p^2 , and $E[p](\bar{k})$ can have p elements.

WARNING 4.4.9. If k is imperfect, the points of $E[p](\bar{k})$ need not lie in $E(k^{\text{sep}})$. Let $k := \mathbb{F}_5(u)$ and $E: y^2 = x^3 + 3x + u$, with $j = 2/(u^2 + 4)$. Its 5-division polynomial is $(x^5 - c_1)(x^5 - c_2)$, with $c_1 = 4u^2 + 2u + 1$ and $c_2 = u^2 + 2u + 4$. So the four non-zero points of $E[5](\bar{k})$ have x -coordinates $c_1^{1/5}$ and $c_2^{1/5}$, which are purely inseparable over k and not in k . Hence $E[5](\bar{k}) \cong \mathbb{Z}/5\mathbb{Z}$, but $E[5](\bar{k}) \cap E(k^{\text{sep}}) = \{O\}$. This does not contradict the discussion after Theorem 4.3.8, since $[5]$ is inseparable.

DRAFT
23 Sep 2026

4.5. The Tate module

COROLLARY 4.5.1. *Let E be an elliptic curve over k , and $N \geq 1$.*

- (1) *If $\text{char } k \nmid N$, then $E[N](\bar{k}) \subseteq E(k^{\text{sep}})$, and $E[N](\bar{k}) \cong (\mathbb{Z}/N\mathbb{Z})^2$.*
 - (2) *If $\text{char } k = p > 0$, then either $E[p^e](\bar{k}) = 0$ for all $e \geq 1$, or $E[p^e](\bar{k}) \cong \mathbb{Z}/p^e\mathbb{Z}$ for all $e \geq 1$.*
- Here $E[N](\bar{k})$ is the kernel of $[N]$ on $E(\bar{k})$.*

PROOF.

- (1) By Corollary 4.4.3, $[N]$ is separable, so $\#E[N](\bar{k}) = \deg[N] = N^2$ by Theorems 4.3.8 and 4.3.10. The same holds for every divisor of N . A finite abelian group of order N^2 with d^2 elements killed by d , for every $d \mid N$, is $(\mathbb{Z}/N\mathbb{Z})^2$. The points of $E[N](\bar{k})$ lie in $E(k^{\text{sep}})$ by the discussion after Theorem 4.3.8.
- (2) Write $[p] = V \circ F$, with F the p -th power relative Frobenius and V its Verschiebung (Definition 4.3.12); F is purely inseparable of degree p . Then $\#E[p^e](\bar{k}) = \deg_s[p^e] = (\deg_s V)^e$, and $\deg_s V \in \{1, p\}$ [Sil09, Corollary III.6.4].

□

Compare with Example 1.3.4. In case 2, E is ordinary if $E[p](\bar{k}) \cong \mathbb{Z}/p\mathbb{Z}$, and supersingular if $E[p](\bar{k}) = 0$ (Theorem 5.4.1). By part 1, Gal_k acts on $E[N](\bar{k}) \cong (\mathbb{Z}/N\mathbb{Z})^2$ whenever $\text{char } k \nmid N$, giving the mod N Galois representation

$$\bar{\rho}_{E,N}: \text{Gal}_k \rightarrow \text{Aut}(E[N](\bar{k})) \cong \text{GL}_2(\mathbb{Z}/N\mathbb{Z}),$$

the second isomorphism depending on a choice of basis. For instance, for $y^2 = (x - e_1)(x - e_2)(x - e_3)$ as in Example 4.2.9, $E[2](\bar{k})$ consists of rational points and $\bar{\rho}_{E,2}$ is trivial. If $y^2 = p(x)$ with p irreducible, then Gal_k permutes the three roots of p , and $\bar{\rho}_{E,2}$ has image containing an element of order 3 in $\text{GL}_2(\mathbb{F}_2) \cong S_3$.

For $\mathbb{G}_m$ the analogue of Corollary 4.5.1 is elementary (Example 3.4.8). If $\text{char } k \nmid N$, then $\mu_N(\bar{k}) \subseteq k^{\text{sep}}$ is a subgroup of order N of $(k^{\text{sep}})^\times$, hence cyclic: $\mu_N(\bar{k}) \cong \mathbb{Z}/N\mathbb{Z}$, and an isomorphism is the choice of a primitive N -th root of unity. If $\text{char } k = p$, then $\mu_{p^e}(\bar{k}) = \{1\}$. Torsion of rank 1 for $\mathbb{G}_m$, of rank 2 for E , matches $\deg[N] = N$ versus N^2 .

EXAMPLE 4.5.2. Let $k := \mathbb{F}_7$. Since $\mathbb{F}_{7^m}^\times$ is cyclic of order $7^m - 1$, we have $\mu_{3^n}(\bar{k}) \subseteq \mathbb{F}_{7^m}$ if and only if $3^n \mid 7^m - 1$. By the lifting the exponent lemma, $v_3(7^m - 1) = v_3(7 - 1) + v_3(m) = 1 + v_3(m)$. So $\mu_3(\bar{k}) \subseteq \mathbb{F}_7$, and the field generated by $\mu_{3^n}(\bar{k})$ is $\mathbb{F}_{7^{3^{n-1}}}$. Equivalently, the Frobenius $\zeta \mapsto \zeta^7$ generates the action of $\text{Gal}_{\mathbb{F}_7}$ on $\mu_{3^n}(\bar{k}) \cong \mathbb{Z}/3^n\mathbb{Z}$, and 7 has order 3^{n-1} in $(\mathbb{Z}/3^n\mathbb{Z})^\times$.

The groups $E[\ell^n](k^{\text{sep}})$ for varying n fit together. Throughout this section and the next, ℓ is a prime different from $\text{char } k$.

DEFINITION 4.5.3. The $(\ell$ -adic) **Tate module** of E is the inverse limit

$$T_\ell(E) := \varprojlim_n E[\ell^n](k^{\text{sep}}),$$

taken with respect to the maps $[\ell]: E[\ell^{n+1}](k^{\text{sep}}) \rightarrow E[\ell^n](k^{\text{sep}})$. The ℓ -adic representation of E is the continuous homomorphism $\rho_{E,\ell}: \text{Gal}_k \rightarrow \text{Aut}(T_\ell(E))$ obtained from the action of Gal_k on each $E[\ell^n](k^{\text{sep}})$.

Likewise, $T_\ell(\mathbb{G}_m) := \varprojlim_n \mu_{\ell^n}(k^{\text{sep}})$, with the maps $\zeta \mapsto \zeta^\ell$, is also written $\mathbb{Z}_\ell(1)$. Choosing primitive ℓ^n -th roots of unity ζ_{ℓ^n} with $\zeta_{\ell^{n+1}}^\ell = \zeta_{\ell^n}$, which is possible because $\zeta \mapsto \zeta^\ell$ maps $\mu_{\ell^{n+1}}(k^{\text{sep}})$ onto $\mu_{\ell^n}(k^{\text{sep}})$, gives an isomorphism $\mathbb{Z}_\ell \rightarrow T_\ell(\mathbb{G}_m)$, $(a_n)_n \mapsto (\zeta_{\ell^n}^{a_n})_n$. Over $\mathbb{C}$ one can take $\zeta_{\ell^n} = e^{2\pi i/\ell^n}$. The group

Gal_k acts on $T_\ell(\mathbb{G}_m)$ through the ℓ -adic cyclotomic character $\chi_\ell: \text{Gal}_k \rightarrow \mathbb{Z}_\ell^\times$, defined by $\sigma(\zeta) = \zeta^{\chi_\ell(\sigma)}$. In characteristic p , $T_p(\mathbb{G}_m) = 0$.

By Corollary 4.5.1, $T_\ell(E) \cong \mathbb{Z}_\ell^2$ as a $\mathbb{Z}_\ell$ -module, so $\rho_{E,\ell}$ takes values in $\text{GL}_2(\mathbb{Z}_\ell)$ after choosing a basis; and $T_p(E)$ is 0 or $\mathbb{Z}_p$ if $p = \text{char } k > 0$ [Sil09, Proposition III.7.1]. Concretely, $[\ell]: E[\ell^{n+1}](k^{\text{sep}}) \rightarrow E[\ell^n](k^{\text{sep}})$ is surjective, because $[\ell]$ is surjective on $E(\bar{k})$. Choose a basis (P_1, Q_1) of $E[\ell](k^{\text{sep}})$, and inductively P_{n+1}, Q_{n+1} with $[\ell]P_{n+1} = P_n$ and $[\ell]Q_{n+1} = Q_n$. If $[a]P_n + [b]Q_n = O$, then applying $[\ell^{n-1}]$ and induction show that $\ell^n \mid a$ and $\ell^n \mid b$, so (P_n, Q_n) is a basis of $E[\ell^n](k^{\text{sep}}) \cong (\mathbb{Z}/\ell^n\mathbb{Z})^2$, and $(a_n, b_n)_n \mapsto ([a_n]P_n + [b_n]Q_n)_n$ is an isomorphism $\mathbb{Z}_\ell^2 \rightarrow T_\ell(E)$. Over $\mathbb{C}$ this is familiar: since $E[\ell^n](\mathbb{C}) = \ell^{-n}\Lambda/\Lambda \cong \Lambda/\ell^n\Lambda$ compatibly in n ,

$$T_\ell(\mathbb{C}/\Lambda) \cong \Lambda \otimes_{\mathbb{Z}} \mathbb{Z}_\ell, \quad T_\ell(\mathbb{C}^\times) \cong 2\pi i\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}_\ell,$$

and Λ and $2\pi i\mathbb{Z}$ are the first homology groups of $\mathbb{C}/\Lambda$ and $\mathbb{C}^\times$ (Examples 1.5.4 and 1.5.5). So the Tate module is an algebraic substitute for H_1 , and unlike H_1 it carries an action of Gal_k .

EXAMPLE 4.5.4. Let $E: y^2 = x^3 - x$ over $\mathbb{Q}$, and $\ell = 2$. Then $E[2](\bar{\mathbb{Q}}) = \{O, (0,0), (1,0), (-1,0)\}$. By Algorithm 4.2.7, $[2]^*x := x \circ [2] = (x^2 + 1)^2 / (4(x^3 - x))$, and

$$[2]^*x - e = \frac{g_e(x)^2}{4(x^3 - x)}, \quad g_0 = x^2 + 1, \quad g_1 = x^2 - 2x - 1, \quad g_{-1} = x^2 + 2x - 1.$$

So the points of $E(\bar{\mathbb{Q}})$ halving $(0,0)$ have $x = \pm i$, those halving $(1,0)$ have $x = 1 \pm \sqrt{2}$, and those halving $(-1,0)$ have $x = -1 \pm \sqrt{2}$. For instance $P_2 := (i, 1 - i)$ and $Q_2 := (1 + \sqrt{2}, 2 + \sqrt{2})$ lie on E , because $i^3 - i = (1 - i)^2$ and $x^3 - x = 2x^2 = (2 + \sqrt{2})^2$ for $x = 1 + \sqrt{2}$. They satisfy $[2]P_2 = (0,0)$ and $[2]Q_2 = (1,0)$, so, with $P_1 := (0,0)$ and $Q_1 := (1,0)$, (P_2, Q_2) is a basis of $E[4](\bar{\mathbb{Q}})$ as above. Hence $E[4](\bar{\mathbb{Q}}) \subseteq E(\mathbb{Q}(i, \sqrt{2}))$, and $\mathbb{Q}(i, \sqrt{2}) = \mathbb{Q}(\zeta_8)$ is the field generated by $\mu_8(\bar{\mathbb{Q}})$, which it must contain by Corollary 4.6.5. Complex conjugation σ fixes Q_2 and sends P_2 to $(-i, 1 + i) = [3]P_2 + [2]Q_2$, so in this basis

$$\bar{\rho}_{E,4}(\sigma) = \begin{pmatrix} 3 & 0 \\ 2 & 1 \end{pmatrix}, \quad \det \bar{\rho}_{E,4}(\sigma) = 3 \equiv \chi_2(\sigma) \pmod{4},$$

since $\sigma(i) = i^3$. Continuing to halve gives compatible sequences $(P_1, P_2, P_3, \dots)$ and $(Q_1, Q_2, Q_3, \dots)$, a $\mathbb{Z}_2$ -basis of $T_2(E)$, just as $(-1, i, \zeta_8, \dots)$ is a $\mathbb{Z}_2$ -basis of $T_2(\mathbb{G}_m)$.

An isogeny $\varphi: E_1 \rightarrow E_2$ restricts to maps $E_1[\ell^n](k^{\text{sep}}) \rightarrow E_2[\ell^n](k^{\text{sep}})$ compatible with $[\ell]$, and so induces a $\mathbb{Z}_\ell$ -linear map $\varphi_\ell: T_\ell(E_1) \rightarrow T_\ell(E_2)$. Since φ is a morphism of k -varieties, φ_ℓ commutes with Gal_k . The construction respects composition and sums, and $[m]_\ell = m$, so Theorem 4.3.10 gives $\hat{\varphi}_\ell \circ \varphi_\ell = \deg \varphi$. Since multiplication by $\deg \varphi \neq 0$ is injective on the torsion-free module $T_\ell(E_1)$, φ_ℓ is injective if $\varphi \neq 0$, and an isomorphism if $\ell \nmid \deg \varphi$. For $\mathbb{G}_m$, $[m]_\ell$ is multiplication by m on $\mathbb{Z}_\ell(1)$. Note that $c_{\hat{\varphi}}c_\varphi = \deg \varphi$ and $\hat{\varphi}_\ell \circ \varphi_\ell = \deg \varphi$ both come from $\hat{\varphi} \circ \varphi = [\deg \varphi]$: pullback of differentials reverses the order of composition and $\varphi \mapsto \varphi_\ell$ does not, but scalars commute. In characteristic p , the scalar $c_{[p]} = 0$ detects that $[p]$ is inseparable, while T_p is too small to see it.

THEOREM 4.5.5. *The map $\text{Hom}(E_1, E_2) \otimes \mathbb{Z}_\ell \rightarrow \text{Hom}_{\mathbb{Z}_\ell}(T_\ell(E_1), T_\ell(E_2))$, $\varphi \mapsto \varphi_\ell$, is injective. Consequently $\text{Hom}(E_1, E_2)$ is a free $\mathbb{Z}$ -module of rank at most 4.*

PROOF. See [Sil09, Theorem III.7.4 and Corollary III.7.5]. The key input is that $\ker[\ell^n] \subseteq \ker \varphi$ forces φ to factor through $[\ell^n]$ (Proposition 4.3.9). The rank bound follows because the target is free of rank 4 over $\mathbb{Z}_\ell$, and $\text{Hom}(E_1, E_2)$ is torsion-free (Proposition 4.3.7). $\square$

For number fields and finite fields the image of $\text{Hom}(E_1, E_2) \otimes \mathbb{Z}_\ell$ is exactly the Gal_k -equivariant homomorphisms. This is the Tate conjecture, a theorem of Tate over finite fields and of Faltings over number fields [Sil09, Section III.7].

We summarize the comparison between the multiplicative group and an elliptic curve. Here $\ell \neq \text{char } k$, $p = \text{char } k > 0$ in the row about $[p^e]$, and kernels are taken on $\bar{k}$ -points.

	$\mathbb{G}_m$	E
points over $\bar{k}$	$\bar{k}^\times$	$E(\bar{k})$
over $\mathbb{C}$	$\mathbb{C}/2\pi i\mathbb{Z}$	$\mathbb{C}/\Lambda$
invariant differential	dt/t , poles at 0 and ∞	$dx/(2y)$, no zeros or poles
$[m]^*\omega$	$m\omega$	$m\omega$
$\deg[m]$	$ m $	m^2
$\ker[\ell^n]$	$\mathbb{Z}/\ell^n\mathbb{Z}$	$(\mathbb{Z}/\ell^n\mathbb{Z})^2$
$\ker[p^e]$	0	0 or $\mathbb{Z}/p^e\mathbb{Z}$
T_ℓ	$\mathbb{Z}_\ell(1) \cong \mathbb{Z}_\ell$	$\mathbb{Z}_\ell^2$
T_ℓ over $\mathbb{C}$	$2\pi i\mathbb{Z} \otimes \mathbb{Z}_\ell$	$\Lambda \otimes \mathbb{Z}_\ell$

4.6. The Weil pairing

In Example 1.3.4 we compared $(\mathbb{C}/\Lambda)[N]$ with the group $\mu_N(\mathbb{C})$ of roots of unity. There is a deeper link: if $\Lambda = \mathbb{Z}\omega_1 \oplus \mathbb{Z}\omega_2$ with $\omega_1/\omega_2 \in \mathbb{H}$, the pairing

$$\left(\frac{a\omega_1 + b\omega_2}{N}, \frac{c\omega_1 + d\omega_2}{N} \right) \mapsto e^{2\pi i(ad-bc)/N}$$

is a well defined, bilinear, alternating and non-degenerate map $(\mathbb{C}/\Lambda)[N] \times (\mathbb{C}/\Lambda)[N] \rightarrow \mu_N(\mathbb{C})$. Over an arbitrary field there is no $e^{2\pi it}$, but the pairing can be built from functions. Its values lie in $\mu_N(k^{\text{sep}})$, where $\mu_N = \mathbb{G}_m[N]$, so the Weil pairing relates the torsion of E to the torsion of the multiplicative group.

Fix $N \geq 2$ with $\text{char } k \nmid N$, so that $E[N](\bar{k}) = E[N](k^{\text{sep}}) \cong (\mathbb{Z}/N\mathbb{Z})^2$ by Corollary 4.5.1. Let $S, T \in E[N](k^{\text{sep}})$. Choose divisors

$$D_S \sim (S) - (O), \quad D_T \sim (T) - (O)$$

in $\text{Div } E_{k^{\text{sep}}}$ with disjoint supports; this is possible because D_T may be replaced by $(T + R) - (R)$ for any $R \in E(k^{\text{sep}})$ outside the support of D_S , and $E(k^{\text{sep}})$ is infinite. The divisor ND_S is linearly equivalent to $N(S) - N(O)$, which is principal by Corollary 4.2.3, since $[N]S = O$. So there are $f_S, f_T \in k^{\text{sep}}(E)^\times$ with

$$\text{div}(f_S) = ND_S, \quad \text{div}(f_T) = ND_T,$$

and the supports of $\text{div}(f_S)$ and D_T are disjoint, as are those of $\text{div}(f_T)$ and D_S . Both values $f_S(D_T)$ and $f_T(D_S)$ are therefore defined in the sense of Definition 3.3.3.

DEFINITION 4.6.1. The **Weil pairing** is

$$e_N: E[N](k^{\text{sep}}) \times E[N](k^{\text{sep}}) \rightarrow \mu_N(k^{\text{sep}}), \quad e_N(S, T) := \frac{f_S(D_T)}{f_T(D_S)},$$

for any choice of D_S, D_T, f_S and f_T as above.

LEMMA 4.6.2. *The quantity $e_N(S, T)$ lies in $\mu_N(k^{\text{sep}})$ and does not depend on the choices of D_S, D_T, f_S and f_T .*

PROOF. Everything follows from Weil reciprocity (Theorem 3.3.4), applied over $\bar{k}$ to functions that happen to be defined over k^{sep} .

The function f_S is determined by D_S up to a constant $c \in k^{\text{sep}\times}$ by Proposition 3.3.2, and replacing f_S by cf_S multiplies $f_S(D_T)$ by $c^{\deg D_T} = c^0 = 1$. So $e_N(S, T)$ depends only on D_S and D_T .

Now replace D_S by $D_S + \text{div}(h)$, which is the general divisor linearly equivalent to D_S ; moving D_T if necessary, we may assume the supports stay disjoint. Then f_S is replaced by $f_S h^N$, and $e_N(S, T)$ is multiplied by

$$\frac{h(D_T)^N}{f_T(\text{div } h)} = \frac{h(ND_T)}{f_T(\text{div } h)} = \frac{h(\text{div } f_T)}{f_T(\text{div } h)} = 1,$$

the last equality by reciprocity. The same argument applies to D_T .

Finally, $\deg D_S = \deg D_T = 0$ gives

$$e_N(S, T)^N = \frac{f_S(D_T)^N}{f_T(D_S)^N} = \frac{f_S(ND_T)}{f_T(ND_S)} = \frac{f_S(\text{div } f_T)}{f_T(\text{div } f_S)} = 1,$$

again by reciprocity. So $e_N(S, T) \in \mu_N(\bar{k})$, and it lies in $\mu_N(k^{\text{sep}})$ because $\text{char } k \nmid N$. $\square$

The values are N -th roots of unity for the same reason that the pairing is well defined, and neither fact needs $e^{2\pi it}$.

PROPOSITION 4.6.3. *The Weil pairing is*

- (1) *bilinear*;
- (2) *alternating*: $e_N(T, T) = 1$, so $e_N(S, T) = e_N(T, S)^{-1}$;
- (3) *non-degenerate*: if $e_N(S, T) = 1$ for all $S \in E[N](k^{\text{sep}})$, then $T = O$;
- (4) *Galois equivariant*: $\sigma(e_N(S, T)) = e_N(\sigma S, \sigma T)$ for $\sigma \in \text{Gal}_k$;
- (5) *compatible*: $e_{NN'}(S, T) = e_N([N']S, T)$ for $S \in E[NN'](k^{\text{sep}})$ and $T \in E[N](k^{\text{sep}})$;
- (6) *adjoint for isogenies*: $e_N(\varphi S, T) = e_N(S, \hat{\varphi}T)$ for $\varphi: E_1 \rightarrow E_2$, $S \in E_1[N](k^{\text{sep}})$ and $T \in E_2[N](k^{\text{sep}})$.

PROOF. Parts 1 and 4 come straight from the definition. For additivity in the first variable, take D_{S_1} , D_{S_2} and $D_{S_1+S_2}$ supported away from D_T . Their difference is linearly equivalent to $(S_1) + (S_2) - (S_1 + S_2) - (O)$, a divisor of degree 0 that sums to O in $E(k^{\text{sep}})$, hence is $\text{div}(h)$ for some h by Corollary 4.2.3. So $f_{S_1}f_{S_2}/f_{S_1+S_2}$ and h^N have the same divisor, and reciprocity kills the resulting factor exactly as in Lemma 4.6.2. Part 4 is the Galois equivariance of evaluation noted after Definition 3.3.3: applying σ carries a choice of D_S , D_T , f_S , f_T for the pair (S, T) to one for $(\sigma S, \sigma T)$. Swapping S and T inverts the defining ratio, which gives the second half of part 2; the first half, $e_N(T, T) = 1$, is stronger in characteristic 2 and is proved from the construction of Remark 4.6.4. For that, and for parts 3, 5 and 6, see [Sil09, Propositions III.8.1 and III.8.2]. $\square$

REMARK 4.6.4. Silverman [Sil09] defines the pairing differently, by pulling back along $[N]$. Given $T \in E[N](k^{\text{sep}})$, choose f with $\text{div}(f) = N(T) - N(O)$, and choose $T' \in E[N^2](k^{\text{sep}})$ with $[N]T' = T$. The divisor $\sum_{R \in E[N](k^{\text{sep}})} ((T' + R) - (R))$ has degree 0 and sums to $[N^2]T' = O$, so it is $\text{div}(g)$ for some g , and then $f \circ [N]$ and g^N have the same divisor; rescaling f gives $f \circ [N] = g^N$. For $S \in E[N](k^{\text{sep}})$ and $X \in E(k^{\text{sep}})$ we get $g(X + S)^N = f([N]X) = g(X)^N$, so $g(X + S)/g(X)$ is an N -th root of unity, and as a function of X it is a morphism $E \rightarrow \mathbb{P}^1$ that is not surjective, hence constant. That constant is $e_N(S, T)$ [Sil09, Exercise 3.16]. This construction makes the analogy with $\mathbb{G}_m$ visible — the pairing is built from the N -th power map — and it is the form in which part 2 of Proposition 4.6.3 is proved. The definition above is the one to compute with: evaluating f_S at D_T is what Miller's algorithm does.

COROLLARY 4.6.5. *The Weil pairing is surjective onto $\mu_N(k^{\text{sep}})$. In particular, if $E[N](k^{\text{sep}}) \subseteq E(k)$, then $\mu_N(k^{\text{sep}}) \subseteq k$.*

PROOF. The image is a subgroup $\mu_d(k^{\text{sep}})$. Then $e_N([d]S, T) = e_N(S, T)^d = 1$ for all S and T , so $[d]S = O$ for all $S \in E[N](k^{\text{sep}})$ by non-degeneracy, and $d = N$ by Corollary 4.5.1. If $E[N](k^{\text{sep}}) \subseteq E(k)$, then Gal_k fixes every value $e_N(S, T)$ by Galois equivariance [Sil09, Corollary III.8.1.1]. $\square$

EXAMPLE 4.6.6. If E is an elliptic curve over $\mathbb{Q}$ and $N \geq 3$, then $E[N](\bar{\mathbb{Q}}) \not\subseteq E(\mathbb{Q})$, since $\mu_N(\bar{\mathbb{Q}}) \not\subseteq \mathbb{Q}$. So $E(\mathbb{Q})_{\text{tors}}$ never contains a copy of $(\mathbb{Z}/N\mathbb{Z})^2$ with $N \geq 3$, while $(\mathbb{Z}/2\mathbb{Z})^2$ does occur (Example 4.2.9).

By compatibility, the pairings e_{ℓ^n} fit into a pairing $e: T_{\ell}(E) \times T_{\ell}(E) \rightarrow \mathbb{Z}_{\ell}(1) = T_{\ell}(\mathbb{G}_m)$, which is bilinear, alternating, non-degenerate and Galois equivariant [Sil09, Proposition III.8.3]. A bilinear alternating form on a free module of rank 2 transforms under an endomorphism by its determinant. So $e(\varphi_{\ell}x, \varphi_{\ell}y) = e(x, y)^{\det \varphi_{\ell}}$, while by adjointness $e(\varphi_{\ell}x, \varphi_{\ell}y) = e(x, \hat{\varphi}_{\ell}\varphi_{\ell}y) = e(x, y)^{\deg \varphi}$. Similarly, Galois equivariance says that $\det \rho_{E, \ell}$ is the ℓ -adic cyclotomic character.

PROPOSITION 4.6.7. *Let $\varphi \in \text{End}(E)$. Then*

$$\det \varphi_{\ell} = \deg \varphi, \quad \text{tr } \varphi_{\ell} = 1 + \deg \varphi - \deg(1 - \varphi).$$

In particular, the characteristic polynomial of φ_ℓ has integer coefficients independent of ℓ .

PROOF. The determinant was computed above. For the trace, apply $\det(1 - M) = 1 - \text{tr } M + \det M$, valid for 2×2 matrices, to $M = \varphi_\ell$ [Sil09, Proposition III.8.6]. $\square$

For the Frobenius endomorphism of an elliptic curve over $\mathbb{F}_q$, this gives the characteristic polynomial of F_ℓ (Theorem 5.3.8), and positivity of the degree (Corollary 4.3.11) gives Hasse's bound (Theorem 5.2.1).

4.7. Endomorphisms and automorphisms

For complex tori, $\text{End}(\mathbb{C}/\Lambda) = \{\alpha \in \mathbb{C} : \alpha\Lambda \subseteq \Lambda\}$ (Corollary 1.3.2). Such an α satisfies a monic quadratic equation over $\mathbb{Z}$, by the Cayley–Hamilton theorem applied to multiplication by α on $\Lambda \cong \mathbb{Z}^2$. If $\alpha \in \mathbb{R}$, then multiplication by α on $\Lambda \otimes_{\mathbb{Z}} \mathbb{R} = \mathbb{C}$ is the scalar α , so its characteristic polynomial $(T - \alpha)^2$ has integer coefficients, and $\alpha \in \mathbb{Z}$. So if $\alpha \notin \mathbb{Z}$, then $\mathbb{Q}(\alpha)$ is an imaginary quadratic field. Hence $\text{End}(\mathbb{C}/\Lambda)$ is either $\mathbb{Z}$ or an order in an imaginary quadratic field. The Gaussian and Eisenstein lattices of Examples 1.2.2 and 1.2.3 have the extra endomorphisms i and ω . In positive characteristic a third possibility appears, a non-commutative one, and to name it we first collect what we need about quaternion algebras.

4.7.1. Quaternion algebras. We follow [Voi21]. Throughout this subsection, F is a field with $\text{char } F \neq 2$.

DEFINITION 4.7.1. A **quaternion algebra** over F is an F -algebra B with a basis $1, i, j, ij$ such that

$$i^2 = a, \quad j^2 = b, \quad ji = -ij$$

for some $a, b \in F^\times$. These relations determine the multiplication, and for every $a, b \in F^\times$ they define a quaternion algebra, denoted $\left(\frac{a, b}{F}\right)$ or $(a, b \mid F)$ [Voi21, Definition 2.2.1].

EXAMPLE 4.7.2.

- (1) Hamilton's quaternions are $\mathbf{H} := (-1, -1 \mid \mathbb{R})$ [Voi21, Example 2.2.3].
- (2) The matrix algebra $M_2(F)$ is $(1, 1 \mid F)$, via $i \mapsto \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ and $j \mapsto \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ [Voi21, Example 2.2.4].
- (3) For a field extension $K \supseteq F$, $(a, b \mid F) \otimes_F K = (a, b \mid K)$, with the same basis.

Every $\alpha = t + xi + yj + zij \in B$ has a conjugate $\bar{\alpha} := t - xi - yj - zij$. The map $\alpha \mapsto \bar{\alpha}$ is F -linear, fixes 1, and satisfies $\bar{\bar{\alpha}} = \alpha$ and $\alpha\bar{\beta} = \bar{\beta}\bar{\alpha}$; it is the **standard involution** of B . The **reduced trace** and the **reduced norm** are

$$\text{trd}(\alpha) := \alpha + \bar{\alpha} = 2t, \quad \text{nrd}(\alpha) := \alpha\bar{\alpha} = t^2 - ax^2 - by^2 + abz^2,$$

both in F , and every $\alpha \in B$ satisfies

$$\alpha^2 - \text{trd}(\alpha)\alpha + \text{nrd}(\alpha) = 0$$

[Voi21, Sections 3.2 and 3.3]. The reduced norm is multiplicative, and α is a unit if and only if $\text{nrd}(\alpha) \neq 0$, with inverse $\bar{\alpha}/\text{nrd}(\alpha)$. On $M_2(F)$, the reduced trace and norm are the trace and the determinant. More generally, a standard involution on an F -algebra B is an F -linear map $\alpha \mapsto \bar{\alpha}$ with $\bar{1} = 1$, $\bar{\bar{\alpha}} = \alpha$, $\alpha\bar{\beta} = \bar{\beta}\bar{\alpha}$ and $\alpha\bar{\alpha} \in F$ for every α [Voi21, Definitions 3.2.1 and 3.2.4]. The field F , and a quadratic extension of F with its conjugation, have one.

THEOREM 4.7.3. *Let B be a division algebra over F with a standard involution. Then B is F , a quadratic field extension of F , or a division quaternion algebra over F .*

PROOF. See [Voi21, Theorem 3.5.1 and Corollary 3.5.4]. The involution makes every $\alpha \in B$ a root of $T^2 - \text{trd}(\alpha)T + \text{nrd}(\alpha) \in F[T]$, where $\text{trd}(\alpha) := \alpha + \bar{\alpha}$ and $\text{nrd}(\alpha) := \alpha\bar{\alpha}$. The theorem says that a division algebra all of whose elements are quadratic over F has dimension at most 4, with the three shapes listed. $\square$

DEFINITION 4.7.4. A quaternion algebra B over F is **split** if $B \cong M_2(F)$, and a field $K \supseteq F$ is a **splitting field** for B if $B \otimes_F K$ is split [Voi21, Definition 5.4.5]. A quaternion algebra B over $\mathbb{Q}$ is **definite** if $B \otimes_{\mathbb{Q}} \mathbb{R} \cong \mathbf{H}$, that is, if $\mathbb{R}$ is not a splitting field [Voi21, Section 14.1].

PROPOSITION 4.7.5. *Let $B = (a, b \mid F)$. The following are equivalent: B is split; B is not a division ring; the quadratic form nrd on B has a non-trivial zero; b is a norm from $F(\sqrt{a})$. In particular, every quaternion algebra over an algebraically closed field is split, $(a, b \mid \mathbb{R})$ is split if and only if $a > 0$ or $b > 0$, and $(a, b \mid \mathbb{Q})$ is definite if and only if $a < 0$ and $b < 0$.*

PROOF. See [Voi21, Main Theorem 5.4.4 and Section 14.1]. $\square$

The endomorphism ring of an elliptic curve is not a $\mathbb{Q}$ -algebra but a ring of integers inside one.

DEFINITION 4.7.6. Let B be a finite-dimensional $\mathbb{Q}$ -algebra. A **lattice** in B is a finitely generated $\mathbb{Z}$ -submodule $M \subseteq B$ that spans B over $\mathbb{Q}$; it is then free of rank $\dim_{\mathbb{Q}} B$ [Voi21, Definition 9.3.1]. An **order** of B is a lattice that is also a subring of B , and a **maximal order** is an order not properly contained in another order [Voi21, Definitions 10.2.1 and 10.4.1].

Every element α of an order $\mathcal{O}$ is integral over $\mathbb{Z}$: the subring $\mathbb{Z}[\alpha] \subseteq \mathcal{O}$ is a finitely generated $\mathbb{Z}$ -module, so α is a root of a monic polynomial with integer coefficients. In a quaternion algebra this says that $\text{trd}(\alpha)$ and $\text{nrd}(\alpha)$ are integers for every $\alpha \in \mathcal{O}$ [Voi21, Section 10.3].

EXAMPLE 4.7.7.

- (1) In an imaginary quadratic field K , the orders are the rings $\mathbb{Z} + f\mathcal{O}_K$ with $f \geq 1$, and the ring of integers $\mathcal{O}_K$ is the unique maximal order. For instance $\mathbb{Z}[\sqrt{-3}]$ is an order of $\mathbb{Q}(\sqrt{-3})$, properly contained in the maximal order $\mathbb{Z}[\omega]$ of Eisenstein integers (Example 1.2.3).
- (2) In $(-1, -1 \mid \mathbb{Q})$, the Lipschitz order $\mathbb{Z} + \mathbb{Z}i + \mathbb{Z}j + \mathbb{Z}ij$ is not maximal: it is properly contained in the Hurwitz order, generated over it by $\frac{1+i+j+ij}{2}$, which is maximal [Voi21, Section 11.1]. Unlike a number field, a quaternion algebra has in general several maximal orders, none containing the others.

4.7.2. The endomorphism ring.

THEOREM 4.7.8. *Let E be an elliptic curve over k . The ring $\text{End}(E)$ is isomorphic to one of:*

- (1) $\mathbb{Z}$;
- (2) an order in an imaginary quadratic field;
- (3) an order in a definite quaternion algebra over $\mathbb{Q}$.

If $\text{char } k = 0$, case 3 does not occur.

PROOF. The ring $\text{End}(E)$ has no zero divisors (Proposition 4.3.7), it is free of rank at most 4 over $\mathbb{Z}$ (Theorem 4.5.5), and it has the involution $\varphi \mapsto \hat{\varphi}$, with $\hat{\varphi}\varphi = [\deg \varphi]$ a positive integer for $\varphi \neq 0$ (Theorem 4.3.10). These force the three cases, as follows. The degree is a positive definite quadratic form on $\text{End}(E)$ (Corollary 4.3.11), so its associated pairing

$$\langle \varphi, \psi \rangle := \deg(\varphi + \psi) - \deg \varphi - \deg \psi = \hat{\varphi}\psi + \hat{\psi}\varphi$$

is $\mathbb{Z}$ -bilinear. Taking $\psi = 1$ shows that the trace $t(\varphi) := \varphi + \hat{\varphi}$ is the integer $1 + \deg \varphi - \deg(1 - \varphi)$, and multiplying $\varphi + \hat{\varphi} = [t(\varphi)]$ by φ gives the monic integral quadratic relation

$$\varphi^2 - [t(\varphi)]\varphi + [\deg \varphi] = 0.$$

Its discriminant satisfies $(\varphi - \hat{\varphi})^2 = t(\varphi)^2 - 4\deg \varphi = -\deg(\varphi - \hat{\varphi})$, so $t(\varphi)^2 - 4\deg \varphi \leq 0$, with equality only when $\varphi = \hat{\varphi}$. In that case $([2]\varphi - [t(\varphi)])^2 = t(\varphi)^2 - 4\deg \varphi = 0$, so $[2]\varphi = [t(\varphi)]$ because $\text{End}(E)$ has no zero divisors; taking degrees gives $t(\varphi)^2 = 4\deg \varphi$, so $t(\varphi) = 2s$ is even, and $\varphi = [s] \in \mathbb{Z}$ because $\text{End}(E)$ is torsion-free. So every element of $B := \text{End}(E) \otimes_{\mathbb{Z}} \mathbb{Q}$ outside $\mathbb{Q}$ generates an imaginary quadratic field.

Now B is a division algebra: it has no zero divisors and is finite-dimensional over $\mathbb{Q}$, so multiplication by a non-zero element is injective, hence bijective. The map $\varphi \mapsto \hat{\varphi}$ extends $\mathbb{Q}$ -linearly to B , and by Theorem 4.3.10 it is a standard involution, with $\text{trd}(\varphi) = t(\varphi)$ and $\text{nrd}(\varphi) = \deg \varphi$. Theorem 4.7.3 leaves three possibilities for B : the field $\mathbb{Q}$; a quadratic field, which is imaginary because $t(\varphi)^2 - 4\deg \varphi < 0$ for $\varphi \notin \mathbb{Q}$; or a quaternion algebra $(a, b \mid \mathbb{Q})$, in which $a = i^2 = -\text{nrd}(i) = -\deg i < 0$ because $\text{trd}(i) = 0$, and likewise $b < 0$, so that B is definite by Proposition 4.7.5. Finally, $\text{End}(E)$ is a subring of B , and a free $\mathbb{Z}$ -module of

finite rank (Theorem 4.5.5) spanning B ; so it is an order (Definition 4.7.6). Compare [Sil09, Theorem III.9.3 and Corollary III.9.4]. In characteristic 0, $\text{End}(E)$ is commutative by Corollary 4.4.6, which rules out case 3. $\square$

The endomorphism ring of $E_{\bar{k}}$ will be the subject of Chapter 11 in characteristic 0, where E has complex multiplication if $\text{End}(E_{\bar{k}}) \neq \mathbb{Z}$, and of Section 5.4 in characteristic p , where case 3 happens exactly for supersingular curves.

EXAMPLE 4.7.9. Suppose $\text{char } k \neq 2, 3$. On $y^2 = x^3 + x$, the automorphism $[i]$ of Example 4.3.4 satisfies $[i]^2 = [-1]$, so $\mathbb{Z}[i] \subseteq \text{End}(E_{\bar{k}})$.

EXAMPLE 4.7.10. Suppose $\text{char } k \neq 2, 3$. On $y^2 = x^3 + 1$, let $\zeta \in \bar{k}$ be a primitive cube root of 1. Then $[\zeta](x, y) := (\zeta x, y)$ satisfies $[\zeta]^2 + [\zeta] + 1 = 0$, so $\mathbb{Z}[\zeta] \subseteq \text{End}(E_{\bar{k}})$.

These are the algebraic counterparts of Examples 1.2.2 and 1.2.3. Indeed, since $i\Lambda_i = \Lambda_i$, we have $G_6(\Lambda_i) = G_6(i\Lambda_i) = i^{-6}G_6(\Lambda_i)$, so $G_6(\Lambda_i) = 0$, and in Theorem 1.7.1 $B = 0$. Similarly $G_4(\Lambda_\omega) = 0$, so $A = 0$.

Automorphisms are the units of $\text{End}(E_{\bar{k}})$, and so there are very few of them.

THEOREM 4.7.11. *Let E be an elliptic curve over k . Then $\text{Aut}(E_{\bar{k}})$ is finite, of order*

$$\# \text{Aut}(E_{\bar{k}}) = \begin{cases} 2 & \text{if } j(E) \neq 0, 1728, \\ 4 & \text{if } j(E) = 1728 \text{ and } \text{char } k \neq 2, 3, \\ 6 & \text{if } j(E) = 0 \text{ and } \text{char } k \neq 2, 3, \\ 12 & \text{if } j(E) = 0 = 1728 \text{ and } \text{char } k = 3, \\ 24 & \text{if } j(E) = 0 = 1728 \text{ and } \text{char } k = 2. \end{cases}$$

If $\text{char } k \neq 2, 3$, let $n := \# \text{Aut}(E_{\bar{k}})$. Then $\text{Aut}(E_{\bar{k}}) = \text{Aut}(E_{k^{\text{sep}}})$, and $[\zeta](x, y) := (\zeta^2 x, \zeta^3 y)$ defines an isomorphism of Gal_k -modules $\mu_n(k^{\text{sep}}) \rightarrow \text{Aut}(E_{k^{\text{sep}}})$ for a short Weierstrass model.

PROOF. We treat $\text{char } k \neq 2, 3$; for the rest see [Sil09, Theorem III.10.1]. By Theorem 4.1.6 and Proposition 4.1.7, an automorphism of $y^2 = x^3 + Ax + B$ is $(x, y) \mapsto (u^2 x, u^3 y)$ with $u \in \bar{k}^\times$, $u^4 A = A$ and $u^6 B = B$. If $AB \neq 0$, this forces $u = \pm 1$. If $B = 0$, then $j = 1728$ and $u^4 = 1$. If $A = 0$, then $j = 0$ and $u^6 = 1$. In each case the solutions form $\mu_n(\bar{k})$, which lies in k^{sep} since $\text{char } k \nmid n$, and $\sigma(u)$ corresponds to the conjugate automorphism [Sil09, Corollary III.10.2]. $\square$

The last two cases are the non-abelian ones, and they are no accident: in characteristic 2 and 3 the curve with $j = 0$ is supersingular, so $\text{End}(E_{\bar{k}})$ is a quaternion order by case 3 of Theorem 4.7.8, and its unit group need not be cyclic. The description of $\text{Aut}(E_{\bar{k}})$ as $\mu_n(k^{\text{sep}})$ holds exactly in the cases where $\text{End}(E_{\bar{k}})$ is commutative.

REMARK 4.7.12. Case 3 of Theorem 4.7.8 occurs for $E_{\bar{k}}$ exactly when E is supersingular (Theorem 5.4.1), and then $\text{End}(E_{\bar{k}})$ is an order in a quaternion algebra; by a theorem of Deuring it is a maximal order in the definite quaternion algebra over $\mathbb{Q}$ that is split at every prime other than $\text{char } k$ [Voi21, Theorem 42.1.9]. The worksheet on the Weil pairing, endomorphisms and automorphisms works this out for $y^2 = x^3 - x$ over $\mathbb{F}_9$: its endomorphism ring is the maximal order $\mathbb{Z} + \mathbb{Z}i + \mathbb{Z}\frac{1+j}{2} + \mathbb{Z}i\frac{1+j}{2}$ of $(-1, -3 \mid \mathbb{Q})$ [Voi21, Exercise 11.11], whose unit group is the group of order 12 in Theorem 4.7.11.

REMARK 4.7.13. Over $\mathbb{C}$, the values $j = 1728$ and $j = 0$ come from the lattices Λ_i and Λ_ω (Examples 4.7.9 and 4.7.10), that is, from the points i and ω of the fundamental domain in Figure 1.8.1. These are exactly the points of $\mathbb{H}$ whose stabilizers in $\text{SL}_2(\mathbb{Z})$ are larger than $\{\pm 1\}$: the stabilizer of i has order 4 and that of ω has order 6, matching Theorem 4.7.11. Indeed, by the proof of Proposition 1.8.1, an automorphism of $\mathbb{C}/\Lambda_\tau$ is multiplication by some α with $\alpha\Lambda_\tau = \Lambda_\tau$, and these α correspond to the stabilizer of τ .

4.8. Selmer elliptic curves

We end the chapter by running its results on a single family of plane cubics, where everything can be computed by hand. These curves are not given by Weierstrass equations: the base point is a flex, but the line $Z = 0$ is not its tangent, and finding a Weierstrass model is part of the work. Throughout this section $\text{char } k \neq 3$, $d \in k^\times$, and $\zeta \in k^{\text{sep}}$ is a primitive cube root of 1.

DEFINITION 4.8.1. The **Selmer elliptic curve** with parameter d is the plane cubic

$$C_d: X^3 + Y^3 = dZ^3, \quad O := (1 : -1 : 0).$$

We write $F := X^3 + Y^3 - dZ^3$, and $Q_1 := (1 : -\zeta : 0)$, $Q_2 := (1 : -\zeta^2 : 0)$.

PROPOSITION 4.8.2.

- (1) (C_d, O) is an elliptic curve over k .
- (2) The tangent line to C_d at O is $X + Y = 0$, and it meets C_d only at O , with multiplicity 3: the base point O is a flex. The line $Z = 0$ meets C_d in the three distinct k^{sep} -points O , Q_1 and Q_2 .

PROOF. The partial derivatives $3X^2$, $3Y^2$ and $-3dZ^2$ have no common zero in $\mathbb{P}^2(\bar{k})$, because 3 and d are units. So C_d is smooth, hence nice by Corollary 3.7.5 and of genus 1 by Corollary 3.7.7, and $O \in C_d(k)$. This is part 1.

For part 2, $\nabla F(O) = (3, 3, 0)$. Substituting $Y = -X$ into F leaves $-dZ^3$, which vanishes to order 3 at O . On $Z = 0$, F leaves $X^3 + Y^3$, whose roots $X/Y = -1, -\zeta, -\zeta^2$ are distinct since $\text{char } k \neq 3$. $\square$

So O plays the role of the point at infinity of a Weierstrass cubic, with the flex tangent $X + Y = 0$ in place of the line $Z = 0$, and Proposition 4.2.4 applies: if a line meets C_d in $P, Q, R \in C_d(\bar{k})$, counted with multiplicity, then $P + Q + R = O$, and by its proof $\text{div}(L/(X + Y)) = P + Q + R - 3O$ for a linear form L defining the line.

PROPOSITION 4.8.3. Let $w := Z/(X + Y)$ and $v := X/(X + Y)$ in $k(C_d)$.

- (1) $\text{div}(w) = Q_1 + Q_2 - 2O$, and v has a pole of order 3 at O and no other pole. So $\{1, w\}$ is a basis of $\mathcal{L}(2O)$, and $\{1, w, v\}$ is a basis of $\mathcal{L}(3O)$.
- (2) $3v^2 - 3v + 1 = dw^3$.
- (3) The functions $x := 3dw$ and $y := 9dv$ are Weierstrass coordinates: $\varphi := (x : y : 1)$ is an isomorphism from C_d onto

$$(4.8.1) \quad E_d: y^2 - 9dy = x^3 - 27d^2,$$

with inverse $(x : y : 1) \mapsto (y : 9d - y : 3x)$. This equation has $a_3 = -9d$, $a_6 = -27d^2$ and $a_1 = a_2 = a_4 = 0$, and $c_4 = 0$, $\Delta = -3^9d^4$, $j = 0$.

- (4) Negation on C_d is $(X_0 : Y_0 : Z_0) \mapsto (Y_0 : X_0 : Z_0)$.
- (5) Let $\xi := X/Z$ and $\eta := Y/Z$. The invariant differential of (4.8.1) is $\omega = d\xi/(3\eta^2)$.
- (6) If $\text{char } k \neq 2$, then $12dw$ and $36d(X - Y)/(X + Y)$ are Weierstrass coordinates for $y^2 = x^3 - 432d^2$, which has $\Delta = -2^{12}3^9d^4$.

PROOF. For every linear form h , the discussion after Proposition 4.8.2 gives $\text{div}(h/(X + Y)) = (h \cap C_d) - 3O$. For $h = Z$ this is $\text{div}(w)$, by part 2 of that proposition. The line $X = 0$ does not pass through O , so v has a pole of order exactly 3 at O . Since $\ell(nO) = n$ by Corollary 3.5.5, and $1, w, v$ have pole orders 0, 2, 3, this proves part 1. Part 2 is the equation of C_d divided by $(X + Y)^3$, since $X^3 + Y^3 = (X + Y)^3 - 3XY(X + Y)$ and $Y/(X + Y) = 1 - v$.

For part 3, part 2 gives $y^2 - 9dy = 81d^2(v^2 - v) = 27d^2(dw^3 - 1) = x^3 - 27d^2$, and the proof of Theorem 4.1.6 shows that φ is an isomorphism onto (4.8.1). For the inverse, $X : Y : Z = v : 1 - v : w$. The invariants follow from Definition 4.1.2: $b_2 = b_4 = b_8 = 0$ and $b_6 = a_3^2 + 4a_6 = -27d^2$, so $c_4 = 0$ and $\Delta = -27b_6^2 = -3^9d^4$.

For part 4, φ is a homomorphism by Theorem 4.3.6, and by Proposition 4.2.4 negation on E_d is $(x, y) \mapsto (x, 9d - y)$. The inverse of φ turns this into $(y : 9d - y : 3x) \mapsto (9d - y : y : 3x)$.

For part 5, $\omega = dx/(2y-9d)$. On the affine patch $Z \neq 0$ we have $x = 3d/(\xi+\eta)$, $2y-9d = 9d(\xi-\eta)/(\xi+\eta)$ and $\xi^2 d\xi + \eta^2 d\eta = 0$, so

$$dx = -\frac{3d(d\xi + d\eta)}{(\xi + \eta)^2} = \frac{3d(\xi - \eta)}{(\xi + \eta)\eta^2} d\xi, \quad \omega = \frac{d\xi}{3\eta^2}.$$

For part 6, completing the square in (4.8.1) gives $(2y - 9d)^2 = 4x^3 - 27d^2$, and multiplying by 16 gives $(8y - 36d)^2 = (4x)^3 - 432d^2$, with $4x = 12dw$ and $8y - 36d = 36d(X - Y)/(X + Y)$. This is the change of variables (4.1.3) with $u = 1/2$, so the discriminant is $2^{12}\Delta$ by Proposition 4.1.7. $\square$

The differential in part 5 is $d\xi/(\partial f/\partial \eta)$ for $f := \xi^3 + \eta^3 - d$, the differential that Proposition 3.7.6 attaches to the plane cubic C_d , with divisor $(3 - 3)H = 0$. The equation (4.8.1) is valid in characteristic 2, where $\Delta = -3^9 d^4 = d^4$ is still non-zero, but it has $a_3 \neq 0$, and there is no way to complete the square. Part 6 needs $\text{char } k \neq 2$ twice: to define $X - Y$ as a function with a pole of order 3 at O , and to complete the square. This is why the general Weierstrass equation keeps the coefficients a_1 and a_3 . Finally, the divisor in part 1 is written in $\text{Div } C_{d,\bar{k}}$. If $\zeta \notin k$, then Q_1 and Q_2 are conjugate, and in $\text{Div } C_d$ they form a single closed point of degree 2.

The torsion points of C_d are cut out by lines.

PROPOSITION 4.8.4.

- (1) The non-zero points of $C_d[2](\bar{k})$ are the points $(c : c : 1)$ with $2c^3 = d$.
- (2) A point $P \in C_d(\bar{k})$ satisfies $[3]P = O$ if and only if P is a flex.
- (3) Let $\delta \in \bar{k}$ with $\delta^3 = d$. Then $C_d[3](\bar{k}) = (C_d \cap \{XYZ = 0\})(\bar{k})$ consists of the nine $\bar{k}$ -points

$$O, Q_1, Q_2, \quad (0 : \zeta^i \delta : 1), \quad (\zeta^i \delta : 0 : 1), \quad i = 0, 1, 2.$$

PROOF. For part 1, by part 4 of Proposition 4.8.3, $P = -P$ if and only if $(X_0 : Y_0 : Z_0) = (Y_0 : X_0 : Z_0)$. Either $X_0 = Y_0$, or $X_0 = -Y_0$ and $Z_0 = 0$, which is $P = O$. If $X_0 = Y_0$ and $Z_0 = 0$, then $2X_0^3 = 0$, so $\text{char } k = 2$ and again $P = (1 : 1 : 0) = O$.

For part 2, if the tangent line L at P meets C_d in $3P$, then $P + P + P = O$ by Proposition 4.2.4. Conversely, if $[3]P = O$, then $3P - 3O = \text{div}(f)$ for some $f \in \bar{k}(C_d)$ (Corollary 4.2.3). By part 1 of Proposition 4.8.3 and Proposition 3.5.2, f is a $\bar{k}$ -linear combination of $1, w, v$, that is, $f = h/(X + Y)$ for a linear form h . So h cuts out $3P$, and h is the tangent line at P , meeting C_d with multiplicity 3.

For part 3, let $P_0 = (X_0 : Y_0 : Z_0) \in C_d(\bar{k})$ with $X_0 = 0$. The tangent line $Y_0^2 Y = dZ_0^2 Z$ at P_0 is the line $Z_0 Y = Y_0 Z$ through P_0 and $(1 : 0 : 0)$, and on it $F = X^3$, which vanishes to order 3 at P_0 . So P_0 is a flex. The same computation with the roles of the coordinates exchanged applies when $Y_0 = 0$ or $Z_0 = 0$. This gives nine distinct flexes, and $\#C_d[3](\bar{k}) = 9$ by Corollary 4.5.1, so part 2 shows that they are all of $C_d[3](\bar{k})$. $\square$

In characteristic 2, part 1 says that $C_d[2](\bar{k}) = 0$. So by Corollary 4.5.1 every Selmer curve is supersingular in characteristic 2, which fits with $j = 0$.

EXAMPLE 4.8.5. Over $\mathbb{Q}$, C_d has a rational point of order 2 if and only if $d = 2c^3$ with $c \in \mathbb{Q}^\times$, and then it is $(c : c : 1)$. For instance $(1 : 1 : 1) \in C_2(\mathbb{Q})$. Otherwise the three points of order 2 are permuted transitively by $\text{Gal}_{\mathbb{Q}}$, and $\mathbb{Q}(C_d[2](\bar{\mathbb{Q}}))$ is the splitting field $\mathbb{Q}(\zeta, \sqrt[3]{d/2})$ of $2T^3 - d$, of degree 6. So $\bar{\rho}_{C_d,2} : \text{Gal}_{\mathbb{Q}} \rightarrow \text{GL}_2(\mathbb{F}_2) \cong S_3$ is surjective.

EXAMPLE 4.8.6. Over $\mathbb{Q}$, the $\bar{\mathbb{Q}}$ -points Q_1 and Q_2 are never rational, and $(0 : \zeta^i \delta : 1)$ is rational only if $i = 0$ and $d = \delta^3$ is a cube. So $C_d(\mathbb{Q})[3]$ is $\{O, (0 : \delta : 1), (\delta : 0 : 1)\} \cong \mathbb{Z}/3\mathbb{Z}$ if $d = \delta^3$, and trivial otherwise. In either case $\mathbb{Q}(C_d[3](\bar{\mathbb{Q}})) = \mathbb{Q}(\zeta, \sqrt[3]{d})$ contains $\mu_3(\bar{\mathbb{Q}})$, as Corollary 4.6.5 requires, and $C_d[3](\bar{\mathbb{Q}})$ is never contained in $C_d(\mathbb{Q})$ (Example 4.6.6).

EXAMPLE 4.8.7. C_1 is the Fermat cubic $X^3 + Y^3 = Z^3$. Fermat's Last Theorem for exponent 3 says exactly that $C_1(\mathbb{Q}) = \{O, (1 : 0 : 1), (0 : 1 : 1)\}$: every rational point is a flex.

For C_d , the polar expansion of Lemma 4.2.5 takes a symmetric form. For $P = (X_P, Y_P, Z_P)$, Q and R in $\bar{k}^3$, let

$$\langle P, Q, R \rangle := X_P X_Q X_R + Y_P Y_Q Y_R - d Z_P Z_Q Z_R.$$

This form is symmetric and trilinear, with $\langle P, P, P \rangle = F(P)$ and $\nabla F(P) \cdot Q = 3\langle P, P, Q \rangle$, so (4.2.1) reads

$$(4.8.2) \quad F(sP + tQ) = s^3 F(P) + 3s^2 t \langle P, P, Q \rangle + 3st^2 \langle P, Q, Q \rangle + t^3 F(Q).$$

PROPOSITION 4.8.8. *Let $P_1 = (X_1 : Y_1 : Z_1)$ and $P_2 = (X_2 : Y_2 : Z_2)$ be points of $C_d(\bar{k})$.*

(1) *If $P_1 \neq P_2$, let $A := \langle P_1, P_1, P_2 \rangle$ and $B := \langle P_1, P_2, P_2 \rangle$. Then*

$$(4.8.3) \quad P_1 + P_2 = (BY_1 - AY_2 : BX_1 - AX_2 : BZ_1 - AZ_2).$$

(2) *For every $P = (X_0 : Y_0 : Z_0) \in C_d(\bar{k})$,*

$$(4.8.4) \quad [2]P = (-Y_0(2X_0^3 + Y_0^3) : X_0(X_0^3 + 2Y_0^3) : Z_0(X_0^3 - Y_0^3)).$$

PROOF. Proposition 4.2.6 applies to C_d with its flex O . For part 1, the coefficients there are $\nabla F(P_1) \cdot P_2 = 3A$ and $\nabla F(P_2) \cdot P_1 = 3B$, and since $3 \in k^\times$ the third intersection point of the line through P_1 and P_2 is $BP_1 - AP_2$. Its negative is (4.8.3) by part 4 of Proposition 4.8.3.

For part 2, if $Z_0 = 0$, then $P \in \{O, Q_1, Q_2\} \subseteq C_d[3](\bar{k})$, so $[2]P = -P$, and (4.8.4) gives $(Y_0 : X_0 : 0)$ because $X_0^3 = -Y_0^3$. If $Z_0 \neq 0$, let $Q := (Y_0^2, -X_0^2, 0)$, which is not a multiple of P . Then $\nabla F(P) \cdot Q = 3\langle P, P, Q \rangle = 0$, so Q lies on the tangent line at P , and part 2 of Proposition 4.2.6 says that the tangent line meets C_d again at $R := F(Q)P - 3\langle P, Q, Q \rangle Q$. Here $F(Q) = Y_0^6 - X_0^6$ and $\langle P, Q, Q \rangle = X_0 Y_0 (X_0^3 + Y_0^3) = dX_0 Y_0 Z_0^3$. Substituting $dZ_0^3 = X_0^3 + Y_0^3$, which is non-zero, gives

$$R = -(X_0^3 + Y_0^3)(X_0(X_0^3 + 2Y_0^3), -Y_0(2X_0^3 + Y_0^3), Z_0(X_0^3 - Y_0^3)),$$

and $[2]P = -R$ is (4.8.4). $\square$

On the line $X = Y$, (4.8.4) gives $(-3X_0^4 : 3X_0^4 : 0) = O$, recovering part 1 of Proposition 4.8.4. Note that the derivation collapses in characteristic 3, where the middle terms of (4.8.2) vanish, and $F = (X + Y - \delta Z)^3$ over $\bar{k}$ is a triple line.

EXAMPLE 4.8.9. On C_7 , let $P := (2 : -1 : 1)$, since $2^3 + (-1)^3 = 7$. By (4.8.4), $[2]P = (15 : 12 : 9) = (5 : 4 : 3)$, and indeed $5^3 + 4^3 = 189 = 7 \cdot 3^3$. With $P_1 = P$ and $P_2 = [2]P$, (4.8.3) has $A = 3$ and $B = -29$, so

$$[3]P = (-17 : 73 : 38), \quad 73^3 - 17^3 = 7 \cdot 38^3,$$

and doubling $[2]P$ gives $[4]P = (-1256 : 1265 : 183)$. By Examples 4.8.5 and 4.8.6, $C_7(\mathbb{Q})$ has no points of order 2 or 3. The growth of the coordinates suggests that P has infinite order; this will follow from the Nagell–Lutz theorem in Chapter 8. Under φ , P maps to $(21, 126) \in E_7(\mathbb{Q})$.

The cube roots of unity act on C_d by scaling one coordinate, and this is where the endomorphisms beyond $\mathbb{Z}$ come from.

PROPOSITION 4.8.10. *Let $\alpha(X : Y : Z) := (X : Y : \zeta Z)$, a morphism $C_{d,k(\zeta)} \rightarrow C_{d,k(\zeta)}$.*

(1) *α is an automorphism of $C_{d,k(\zeta)}$ with $\alpha^2 + \alpha + 1 = 0$, so $\mathbb{Z}[\alpha] \cong \mathbb{Z}[\zeta]$ is a subring of $\text{End}(C_{d,k(\zeta)})$.*

In the coordinates of (4.8.1), $\alpha(x, y) = (\zeta x, y)$, and $\alpha^ \omega = \zeta \omega$.*

(2) *If $\text{char } k \neq 2$, then $\text{Aut}(C_{d,\bar{k}}) = \{\pm 1, \pm \alpha, \pm \alpha^2\}$. If moreover $\zeta \notin k$, then $\text{Aut}(C_d) = \{\pm 1\}$.*

(3) *The endomorphism $1 - \alpha$ is separable of degree 3, with dual $1 - \alpha^2$ and kernel $\{O, Q_1, Q_2\}$.*

PROOF. For part 1, $F(X, Y, \zeta Z) = F$ and $\alpha(O) = O$, so α is an isogeny with inverse α^2 . Since $\alpha \neq 1$ and $(\alpha - 1)(\alpha^2 + \alpha + 1) = \alpha^3 - 1 = 0$, Proposition 4.3.7 gives $\alpha^2 + \alpha + 1 = 0$. The kernel of $\mathbb{Z}[T]/(T^2 + T + 1) \rightarrow \text{End}(C_d)$ is an ideal of $\mathbb{Z}[\zeta]$, with finite quotient if it is non-zero, which would contradict Proposition 4.3.7. Next, $\alpha^* \omega = \zeta \omega$ and $\alpha^* v = v$, so $\alpha^* x = \zeta x$ and $\alpha^* y = y$, and $\alpha^* \omega = \zeta dx/(2y - 9d) = \zeta \omega$.

For part 2, $j = 0$, so $\# \text{Aut}(C_{d,\bar{k}}) = 6$ by Theorem 4.7.11. The six automorphisms listed pull ω back to $\pm \zeta^i \omega$, six distinct multiples, so they are distinct. An element φ of $\text{Aut}(C_d)$ has $c_\varphi \in k$ by Proposition 4.4.5, which rules out $\pm \alpha$ and $\pm \alpha^2$ if $\zeta \notin k$.

For part 3, α has degree 1, so $\hat{\alpha} = \alpha^{-1} = \alpha^2$ by Theorem 4.3.10, and $\widehat{1 - \alpha} = 1 - \alpha^2$. Then $(1 - \alpha^2)(1 - \alpha) = 2 - \alpha - \alpha^2 = 3$, so $\deg(1 - \alpha) = 3$. It is separable by Proposition 4.4.5, since $c_{1-\alpha} = 1 - \zeta \neq 0$. So $\# \ker(1 - \alpha) = 3$ by Theorem 4.3.8. The kernel is the set of fixed points of α , and $(X : Y : \zeta Z) = (X : Y : Z)$ forces $Z = 0$, since $X = Y = 0$ is not on C_d . $\square$

In characteristic 0, $\text{End}(C_{d,\bar{k}})$ is an order in an imaginary quadratic field by Theorem 4.7.8, and it contains the maximal order $\mathbb{Z}[\zeta]$, so $\text{End}(C_{d,\bar{k}}) = \mathbb{Z}[\zeta]$. The automorphism α is the $[\zeta]$ of Example 4.7.10 in another model, and $1 - \alpha$ is the algebraic counterpart of multiplication by $1 - \zeta$ on the Eisenstein lattice. Note also $(1 - \alpha)^2 = -3\alpha$, so $\ker(1 - \alpha) \subseteq C_d[3](\bar{k})$, in accordance with Proposition 4.8.4.

EXAMPLE 4.8.11. The kernel $\{O, Q_1, Q_2\}$ of $1 - \alpha$ is cut out by the line $Z = 0$, so it is Gal_k -stable, even though $1 - \alpha$ is only an endomorphism of $C_{d,k(\zeta)}$. By Proposition 4.3.9, there is a 3-isogeny over k with this kernel. Suppose $\text{char } k \neq 2$, and use the model $y^2 = x^3 + D$ with $D = -432d^2$ of Proposition 4.8.3, in which Q_1 and Q_2 become the $\bar{k}$ -points $(0, \pm\sqrt{D})$. The isogeny is

$$\psi(x, y) := \left(\frac{x^3 + 4D}{x^2}, \frac{y(x^3 - 8D)}{x^3} \right), \quad \psi: (y^2 = x^3 + D) \rightarrow (y^2 = x^3 - 27D),$$

which lands on the target because $(u+D)(u-8D)^2 = (u+4D)^3 - 27Du^2$ with $u = x^3$. Since $-27D = 3^6 \cdot 16d^2$, the target is isomorphic to $y^2 = x^3 + 16d^2$, by Proposition 4.1.7 with $u = 3$. Over $k(\zeta)$ this curve is isomorphic to C_d , because $-432d^2/(16d^2) = -27 = (\sqrt{-3})^6$ and $\sqrt{-3} = 2\zeta + 1$. Composed with such an isomorphism, ψ has the same kernel as $1 - \alpha$, so by Proposition 4.3.9 it agrees with $1 - \alpha$ up to an automorphism of C_d . If $\zeta \notin k$, the target is not isomorphic over k to any $C_{d'}$: that would need $-27(d'/d)^2 \in k^{\times 6}$, and so $-3 \in k^{\times 2}$.

Finally, the Selmer curves are twists of one another.

PROPOSITION 4.8.12. *Let $d, d' \in k^\times$.*

- (1) *If $d' = c^3 d$ with $c \in k^\times$, then $(X : Y : Z) \mapsto (X : Y : Z/c)$ is an isomorphism of elliptic curves $C_d \rightarrow C_{d'}$ over k .*
- (2) *If $\text{char } k \neq 2$ and $C_d \cong C_{d'}$ over k , then $d'/d \in k^{\times 3}$.*

So the Selmer curves all become isomorphic over $\bar{k}$, and if $\text{char } k \neq 2$ their isomorphism classes over k are indexed by $k^\times/(k^\times)^3$.

PROOF. Part 1 is immediate, since $d'(Z/c)^3 = dZ^3$ and O is fixed. For part 2, use the models $y^2 = x^3 - 432d^2$ and $y^2 = x^3 - 432d'^2$ of Proposition 4.8.3. By Proposition 4.1.7, an isomorphism between them has $d^2 = u^6 d'^2$ for some $u \in k^\times$, so $d/d' = \pm u^3$, and $-1 = (-1)^3$. They become isomorphic over $\bar{k}$ by Proposition 4.1.10, since $j = 0$. $\square$

The C_d are the **cubic twists** of the Fermat cubic C_1 ; twists in general are the subject of Chapter 10.

REMARK 4.8.13. Selmer studied the diagonal cubics $aX^3 + bY^3 + cZ^3 = 0$ [Sel51]. Among them is the curve $3X^3 + 4Y^3 + 5Z^3 = 0$ of Section 4.1, which is a nice curve of genus 1 with no rational point, so not an elliptic curve over $\mathbb{Q}$. It has a Jacobian, an elliptic curve over $\mathbb{Q}$, and in general the Jacobian of $aX^3 + bY^3 + cZ^3 = 0$ is $y^2 = x^3 - 432(abc)^2$, that is, the Selmer curve C_{abc} . So Selmer's counterexample to the Hasse principle is a torsor under C_{60} . We return to this in Chapter 10, where the C_d are the first examples for descent.

Elliptic curves over finite fields

Sources. This chapter follows [Sil09, Chapter V] and [Poo06, Chapter 3], with [Mil22] for the Galois theory of finite fields, [AH17] for the Cartier operator, and [Voi21] for the mass formula.

DRAFT
21 Sep 2026

5.1. The Frobenius automorphism

Let p be a prime, let $q := p^e$ for some $e \geq 1$, and let $k := \mathbb{F}_q$ be the finite field with q elements. Since k is perfect, $k^{\text{sep}} = \bar{k} = \bar{\mathbb{F}}_q$. Moreover,

$$\bar{\mathbb{F}}_q = \varinjlim_r \mathbb{F}_{q^r}.$$

Here $\mathbb{F}_{q^r}$ is the unique subfield of $\bar{\mathbb{F}}_q$ with q^r elements, and $\mathbb{F}_{q^r} \subseteq \mathbb{F}_{q^s}$ if and only if $r \mid s$; the direct limit is the union $\bigcup_{r \geq 1} \mathbb{F}_{q^r}$.

The Galois group $\text{Gal}_{\mathbb{F}_q} = \text{Gal}(\bar{\mathbb{F}}_q/\mathbb{F}_q)$ is topologically generated by the [Frobenius automorphism](#)

$$\sigma: \bar{\mathbb{F}}_q \rightarrow \bar{\mathbb{F}}_q, \quad x \mapsto x^q,$$

that is, the powers of σ are dense in $\text{Gal}_{\mathbb{F}_q} = \varprojlim_r \text{Gal}(\mathbb{F}_{q^r}/\mathbb{F}_q)$.

In particular, for every $r > 0$,

$$\text{Gal}(\mathbb{F}_{q^r}/\mathbb{F}_q) = \langle \sigma|_{\mathbb{F}_{q^r}} \rangle \cong \mathbb{Z}/r\mathbb{Z},$$

the isomorphism given by $(n \bmod r) \mapsto (\sigma|_{\mathbb{F}_{q^r}})^n$. Assembling these maps for all $r \geq 1$ gives

$$(5.1.1) \quad \widehat{\mathbb{Z}} := \varprojlim_r \mathbb{Z}/r\mathbb{Z} \xrightarrow{\sim} \text{Gal}_{\mathbb{F}_q}, \quad (n_r \bmod r)_{r \geq 1} \mapsto ((\sigma|_{\mathbb{F}_{q^r}})^{n_r})_{r \geq 1}.$$

For all of this, see [Mil22].

DRAFT
21 Sep 2026

5.2. Hasse's bound

Let E be an elliptic curve over $\mathbb{F}_q$. We know from Theorem 4.1.6 that E is given by a Weierstrass equation (4.1.1), that is, by $f(x, y) = 0$ with

$$f := y^2 + a_1xy + a_3y - (x^3 + a_2x^2 + a_4x + a_6) \in \mathbb{F}_q[x, y],$$

and that the set $E(\mathbb{F}_q)$ of rational points contains at least one rational point: $O := (0 : 1 : 0)$.

On the other hand,

$$E(\mathbb{F}_q) = \{O\} \cup \{(x, y) \in \mathbb{F}_q^2 : f(x, y) = 0\}.$$

If $x \in \mathbb{F}_q$ is fixed, the polynomial $f(x, y) \in \mathbb{F}_q[y]$ has at most two roots in $\mathbb{F}_q$, so we obtain the trivial upper bound

$$\#E(\mathbb{F}_q) \leq 2q + 1.$$

Let $F \in \text{End}(E)$ be the q -th power Frobenius endomorphism (Definition 3.2.11 and Example 4.3.3). On $\bar{\mathbb{F}}_q$ -points it acts as the Frobenius automorphism σ on coordinates:

$$F: E(\bar{\mathbb{F}}_q) \rightarrow E(\bar{\mathbb{F}}_q), \quad O \mapsto O, \quad (x, y) \mapsto (x^q, y^q).$$

E. Artin conjectured, in his thesis, and H. Hasse proved the optimal upper bound.

THEOREM 5.2.1 (Hasse). *Let E be an elliptic curve over $\mathbb{F}_q$. Then for every $r \geq 1$,*

$$|\#E(\mathbb{F}_{q^r}) - (q^r + 1)| \leq 2q^{r/2}.$$

PROOF. (Sketch; see [Sil09, Theorem V.1.1].) There are three key observations that go into the proof of Hasse's bound.

- (1) $E(\mathbb{F}_{q^r}) = \ker(1 - F^r)(\overline{\mathbb{F}}_q)$. Indeed, since $\mathbb{F}_{q^r} = \overline{\mathbb{F}}_q^{(\sigma^r)}$, we know that $P \in E(\mathbb{F}_{q^r}) \iff F^r(P) = P$.
- (2) The map $\deg: \text{End}(E) \rightarrow \mathbb{Z}$ is a positive definite quadratic form (Corollary 4.3.11).
- (3) For any abelian group A and any positive definite quadratic form $d: A \rightarrow \mathbb{Z}$, the Cauchy–Schwarz inequality gives

$$|d(a_1 - a_2) - d(a_1) - d(a_2)| \leq 2\sqrt{d(a_1)d(a_2)}.$$

To combine them: $1 - F^r$ is separable (Corollary 4.4.4, applied over $\mathbb{F}_{q^r}$), so observation 1 and Theorem 4.3.8 give $\#E(\mathbb{F}_{q^r}) = \deg(1 - F^r)$, and $\deg F^r = q^r$ (Proposition 3.2.12). Now apply observation 3 with $d = \deg$, $a_1 = 1$ and $a_2 = F^r$. $\square$

DRAFT
24 Sep 2026

5.3. The Weil conjectures

In 1949, Weil made a series of remarkable conjectures about the numbers $\#X(\mathbb{F}_{q^n})$, $n \geq 1$, for a variety X over $\mathbb{F}_q$ [Wei49]. Their proof was one of the main motivations for the scheme-theoretic foundations of algebraic geometry and for étale cohomology (Remark 5.3.6). In this section we state the conjectures for curves, check them on $\mathbb{P}^1$, and prove them for elliptic curves. We keep the notation of Section 5.1, and C denotes a nice curve over $\mathbb{F}_q$ (Definition 2.4.1).

5.3.1. The zeta function. Each set $C(\mathbb{F}_{q^n})$ is finite, and we encode all of their sizes in a single generating function.

DEFINITION 5.3.1. The **zeta function** of $C/\mathbb{F}_q$ is the power series

$$(5.3.1) \quad Z(C/\mathbb{F}_q; T) := \exp \left(\sum_{n=1}^{\infty} \#C(\mathbb{F}_{q^n}) \frac{T^n}{n} \right) \in \mathbb{Q}[[T]].$$

Here $\exp F(T) := \sum_{m \geq 0} F(T)^m / m!$ for a power series $F(T)$ with no constant term, so $Z(C/\mathbb{F}_q; 0) = 1$. The zeta function determines the numbers $\#C(\mathbb{F}_{q^n})$: $\#C(\mathbb{F}_{q^n})$ is the coefficient of T^{n-1} in $\frac{d}{dT} \log Z(C/\mathbb{F}_q; T)$. The exponential in (5.3.1) looks odd at first: the more natural series is $\sum_n \#C(\mathbb{F}_{q^n}) T^n$. The choice makes $Z(C/\mathbb{F}_q; T)$ an Euler product over the closed points of C .

PROPOSITION 5.3.2. *For $d \geq 1$, let N_d be the number of closed points of C of degree d .*

(1) *Each N_d is finite, and $\#C(\mathbb{F}_{q^n}) = \sum_{d|n} dN_d$ for every $n \geq 1$.*

(2) *In $\mathbb{Z}[[T]]$,*

$$(5.3.2) \quad Z(C/\mathbb{F}_q; T) = \prod_P (1 - T^{\deg P})^{-1} = \prod_{d \geq 1} (1 - T^d)^{-N_d},$$

where the first product ranges over the closed points P of C . In particular $Z(C/\mathbb{F}_q; T) \in 1 + T\mathbb{Z}[[T]]$.

(3) *For every $d \geq 1$, $N_d = \frac{1}{d} \sum_{e|d} \mu(d/e) \#C(\mathbb{F}_{q^e})$, where μ is the Möbius function.*

PROOF. By Proposition 3.1.2, the points of $C(\overline{\mathbb{F}}_q)$ above a closed point P form a single Galois orbit with $\deg P$ elements, because $\mathbb{F}_q$ is perfect. Such an $\overline{\mathbb{F}}_q$ -point lies in $C(\mathbb{F}_{q^n})$ exactly when the residue field $\mathbb{F}_q(P) \cong \mathbb{F}_{q^{\deg P}}$ embeds in $\mathbb{F}_{q^n}$, that is, when $\deg P \mid n$. So $N_d \leq \#C(\mathbb{F}_{q^d}) < \infty$, and $C(\mathbb{F}_{q^n})$ is the disjoint union of the orbits of size $d \mid n$, which proves part 1.

For part 2, only the finitely many factors with $\deg P < m$ are $\not\equiv 1 \pmod{T^m}$, so the product converges in $\mathbb{Z}[[T]]$. Taking its logarithm and expanding $-\log(1-x) = \sum_{k \geq 1} x^k/k$,

$$\sum_{d \geq 1} N_d \sum_{k \geq 1} \frac{T^{dk}}{k} = \sum_{n \geq 1} \left(\sum_{d|n} d N_d \right) \frac{T^n}{n} = \sum_{n \geq 1} \#C(\mathbb{F}_{q^n}) \frac{T^n}{n},$$

where we set $n = dk$. Part 3 is Möbius inversion of part 1, applied to $n \mapsto \#C(\mathbb{F}_{q^n})$ and $d \mapsto dN_d$. $\square$

So the numbers of rational points over the extensions of $\mathbb{F}_q$ and the numbers of closed points of each degree determine each other. Substituting $T = q^{-s}$, and writing $NP := \#C(\mathbb{F}_{q^n}) = q^{\deg P}$ for the norm of a closed point, gives

$$\zeta_C(s) := Z(C/\mathbb{F}_q; q^{-s}) = \prod_P (1 - NP^{-s})^{-1},$$

which has the shape of the Euler product $\zeta(s) = \prod_p (1 - p^{-s})^{-1}$ of the Riemann zeta function, with closed points in place of primes.

EXAMPLE 5.3.3. Let $C := \mathbb{P}^1$ over $\mathbb{F}_q$. Then $\#\mathbb{P}^1(\mathbb{F}_{q^n}) = q^n + 1$, so

$$\log Z(\mathbb{P}^1/\mathbb{F}_q; T) = \sum_{n \geq 1} \frac{T^n}{n} + \sum_{n \geq 1} \frac{(qT)^n}{n}, \quad Z(\mathbb{P}^1/\mathbb{F}_q; T) = \frac{1}{(1-T)(1-qT)}$$

[Sil09, Example V.2.1]. The Euler product gives the same answer in another way. The closed points of $\mathbb{P}^1$ are the rational point ∞ , of degree 1, and the closed points of $\mathbb{A}^1$, which are the monic irreducible polynomials $f \in \mathbb{F}_q[t]$, with $\deg P = \deg f$. Since $\mathbb{F}_q[t]$ is a unique factorization domain, expanding each factor of (5.3.2) as a geometric series gives

$$(1-T)^{-1} \prod_f (1 - T^{\deg f})^{-1} = \frac{1}{1-T} \sum_{g \text{ monic}} T^{\deg g} = \frac{1}{1-T} \sum_{d \geq 0} q^d T^d,$$

because there are q^d monic polynomials of degree d . Part 3 of Proposition 5.3.2 counts them: since $\sum_{e|d} \mu(d/e) = 0$ for $d > 1$, the $+1$ in $q^e + 1$ drops out, and the number of monic irreducible polynomials of degree d over $\mathbb{F}_q$ is $\frac{1}{d} \sum_{e|d} \mu(d/e) q^e$ for every $d \geq 1$. Over $\mathbb{F}_2$, in degree 2, this is $\frac{1}{2}(4-2) = 1$: the polynomial $t^2 + t + 1$, whose roots form the Galois orbit $\{\omega, \omega^2\}$ of a primitive cube root of unity in $\mathbb{F}_4$.

The zeta function of $\mathbb{P}^1$ is a rational function, and this is no accident. The next lemma turns a formula for $\#C(\mathbb{F}_{q^n})$ as a trace of n -th powers into a formula for $Z(C/\mathbb{F}_q; T)$.

LEMMA 5.3.4. Let A be a square matrix over a field K of characteristic 0. Then

$$(5.3.3) \quad \exp \left(\sum_{n \geq 1} \operatorname{tr}(A^n) \frac{T^n}{n} \right) = \frac{1}{\det(1 - AT)} \quad \text{in } K[[T]].$$

PROOF. (See [Sil09, Exercise 5.3].) Over an algebraic closure $\bar{K}$, triangularize A , with its eigenvalues $\lambda_1, \dots, \lambda_m$ on the diagonal. Then $\operatorname{tr}(A^n) = \sum_i \lambda_i^n$ and $\det(1 - AT) = \prod_i (1 - \lambda_i T)$. Since $\exp(F + G) = \exp F \exp G$, both sides of (5.3.3) are the products over i of the corresponding sides for the 1×1 matrix (λ_i) , and for these

$$\sum_{n \geq 1} \lambda^n \frac{T^n}{n} = -\log(1 - \lambda T), \quad \exp(-\log(1 - \lambda T)) = (1 - \lambda T)^{-1}$$

in $\bar{K}[[T]]$, which needs characteristic 0. Both sides of (5.3.3) have coefficients in K . $\square$

For example, if there are $\alpha_1, \dots, \alpha_r \in \mathbb{C}$ and signs $\varepsilon_i \in \{\pm 1\}$ with $\#C(\mathbb{F}_{q^n}) = \sum_i \varepsilon_i \alpha_i^n$ for all $n \geq 1$, then Lemma 5.3.4, applied to the 1×1 matrices (α_i) , gives $Z(C/\mathbb{F}_q; T) = \prod_{i=1}^r (1 - \alpha_i T)^{-\varepsilon_i}$.

5.3.2. The statement.

THEOREM 5.3.5 (Weil conjectures for curves). *Let C be a nice curve of genus g over $\mathbb{F}_q$ (Definition 3.5.3).*

(1) (Rationality) *There is a polynomial $P \in 1 + T\mathbb{Z}[T]$ of degree $2g$ such that*

$$Z(C/\mathbb{F}_q; T) = \frac{P(T)}{(1-T)(1-qT)}.$$

(2) (Functional equation)

$$Z\left(C/\mathbb{F}_q; \frac{1}{qT}\right) = q^{1-g} T^{2-2g} Z(C/\mathbb{F}_q; T).$$

(3) (Riemann hypothesis) *Over $\mathbb{C}$,*

$$P(T) = \prod_{i=1}^{2g} (1 - \alpha_i T), \quad |\alpha_i| = \sqrt{q} \quad \text{for every } i.$$

PROOF. Rationality and the functional equation are due to F. K. Schmidt, from the Riemann–Roch theorem (Theorem 3.5.4); the Riemann hypothesis is due to Weil [Wei49]. See Remark 5.3.6. We prove the theorem for elliptic curves in Theorem 5.3.9. $\square$

Taking the logarithm of the formula for $Z(C/\mathbb{F}_q; T)$, as in Example 5.3.3, the α_i determine all the numbers $\#C(\mathbb{F}_{q^n})$:

$$\#C(\mathbb{F}_{q^n}) = q^n + 1 - \sum_{i=1}^{2g} \alpha_i^n, \quad \text{so} \quad |\#C(\mathbb{F}_{q^n}) - (q^n + 1)| \leq 2g q^{n/2}.$$

For $C = \mathbb{P}^1$, of genus 0, Example 5.3.3 gives the theorem with $P = 1$.

REMARK 5.3.6 (A brief and incomplete history). F. K. Schmidt proved rationality and the functional equation for curves. Hasse proved the Riemann hypothesis for elliptic curves in the 1930s, and Weil proved it for all curves, and for abelian varieties, using intersection theory on surfaces [Wei49].

Weil’s conjectures were guided by the Lefschetz fixed-point theorem. If $f: M \rightarrow M$ is a reasonable map of a compact manifold of dimension d , then the number of fixed points of f , counted with multiplicity, is

$$\sum_{i=0}^d (-1)^i \operatorname{tr}(f^* | H^i(M, \mathbb{Q})).$$

The set $C(\mathbb{F}_{q^n})$ is exactly the fixed-point set of the n -th power of the Frobenius morphism on $C(\overline{\mathbb{F}}_q)$. If C comes from a curve over $\mathbb{C}$, then H^0 and H^2 of the Riemann surface have rank 1 and H^1 has rank $2g$. So one expects cohomology groups that behave like these but exist in characteristic p , with F acting by 1 on H^0 , by q on H^2 , and with eigenvalues $\alpha_1, \dots, \alpha_{2g}$ on H^1 . This gives the shape of Theorem 5.3.5.

Weil stated his conjectures for varieties of any dimension. Dwork proved rationality in 1960 using p -adic analysis [Dwo60]. The cohomology theory Weil was looking for is étale cohomology, developed by M. Artin, Grothendieck and others: its Grothendieck–Lefschetz trace formula proves every part of the conjectures except the Riemann hypothesis, which Deligne proved in 1973 [Del74]. See [Oor14].

5.3.3. Elliptic curves. We now prove Theorem 5.3.5 for elliptic curves; the result is Theorem 5.3.9. Let E be an elliptic curve over $\mathbb{F}_q$, let $F \in \operatorname{End}(E)$ be its q -th power Frobenius endomorphism (Section 5.2), let $\ell \neq p$ be a prime, and let F_ℓ be the map induced by F on $T_\ell(E)$ (Section 4.5). Then $\deg F = q$ by Proposition 3.2.12, and F^n is the q^n -th power Frobenius endomorphism. So the discussion after Corollary 4.4.4, applied to E over $\mathbb{F}_{q^n}$, gives

$$(5.3.4) \quad \#E(\mathbb{F}_{q^n}) = \deg(1 - F^n).$$

DEFINITION 5.3.7. The **trace of Frobenius** of $E/\mathbb{F}_q$ is $a := q + 1 - \#E(\mathbb{F}_q)$.

The name comes from Proposition 4.6.7 and (5.3.4): for every $\ell \neq p$,

$$\begin{aligned}\mathrm{tr} \mathbf{F}_\ell &= 1 + \deg \mathbf{F} - \deg(1 - \mathbf{F}) = 1 + q - \#E(\mathbb{F}_q) = a, \\ \det \mathbf{F}_\ell &= \deg \mathbf{F} = q\end{aligned}$$

[Sil09, Remark V.2.6].

THEOREM 5.3.8. *Let $E/\mathbb{F}_q$ be an elliptic curve with trace of Frobenius a .*

(1) *The characteristic polynomial of $\mathbf{F}_\ell$ is $T^2 - aT + q$. Its complex roots α and β are complex conjugates with $|\alpha| = |\beta| = \sqrt{q}$, and for every $n \geq 1$,*

$$(5.3.5) \quad \#E(\mathbb{F}_{q^n}) = q^n + 1 - \alpha^n - \beta^n.$$

(2) $\mathbf{F}^2 - a\mathbf{F} + q = 0$ in $\mathrm{End}(E)$.

PROOF. By the computation above, $\det(T - \mathbf{F}_\ell) = T^2 - \mathrm{tr}(\mathbf{F}_\ell)T + \det \mathbf{F}_\ell = T^2 - aT + q$.

For part 1, let $m/n \in \mathbb{Q}$. By Proposition 4.6.7 and Corollary 4.3.11,

$$\det\left(\frac{m}{n} - \mathbf{F}_\ell\right) = \frac{\det(m - n\mathbf{F}_\ell)}{n^2} = \frac{\deg(m - n\mathbf{F})}{n^2} \geq 0.$$

So the real quadratic polynomial $T^2 - aT + q$ is non-negative on $\mathbb{Q}$, hence on $\mathbb{R}$, and its discriminant satisfies $a^2 - 4q \leq 0$. Its roots are therefore complex conjugates or a repeated real root, so in either case $|\alpha|^2 = \alpha\beta = q$. (The inequality $a^2 \leq 4q$ is Hasse's bound, Theorem 5.2.1 with $r = 1$.)

Viewing α and β in an algebraic closure of $\mathbb{Q}_\ell$, we can triangularize $\mathbf{F}_\ell$ with α and β on the diagonal. Then $(\mathbf{F}^n)_\ell = \mathbf{F}_\ell^n$ is triangular with α^n and β^n on the diagonal. By (5.3.4) and Proposition 4.6.7,

$$\#E(\mathbb{F}_{q^n}) = \deg(1 - \mathbf{F}^n) = \det(1 - \mathbf{F}_\ell^n) = (1 - \alpha^n)(1 - \beta^n) = 1 - \alpha^n - \beta^n + q^n.$$

Since $\alpha^n + \beta^n$ is a polynomial in $a = \alpha + \beta$ and $q = \alpha\beta$ with integer coefficients, it does not matter whether α and β are read in $\mathbb{C}$ or in $\overline{\mathbb{Q}_\ell}$.

For part 2, let $\psi := \mathbf{F}^2 - a\mathbf{F} + q \in \mathrm{End}(E)$. By the Cayley–Hamilton theorem, $\mathbf{F}_\ell$ satisfies its characteristic polynomial, so $\psi_\ell = \mathbf{F}_\ell^2 - a\mathbf{F}_\ell + q = 0$. By Theorem 4.5.5, $\psi = 0$ [Sil09, Theorem V.2.3.1]. $\square$

Theorem 5.3.8 recovers Hasse's bound over every extension at once: by (5.3.5),

$$|\#E(\mathbb{F}_{q^n}) - (q^n + 1)| = |\alpha^n + \beta^n| \leq |\alpha|^n + |\beta|^n = 2q^{n/2},$$

which is Theorem 5.2.1 with $r = n$.

THEOREM 5.3.9 (Weil conjectures for elliptic curves). *Let $E/\mathbb{F}_q$ be an elliptic curve with trace of Frobenius a , and let α and β be as in Theorem 5.3.8. Then*

$$(5.3.6) \quad \begin{aligned} Z(E/\mathbb{F}_q; T) &= \frac{\det(1 - \mathbf{F}_\ell T)}{(1 - T)(1 - qT)} = \frac{1 - aT + qT^2}{(1 - T)(1 - qT)}, \\ 1 - aT + qT^2 &= (1 - \alpha T)(1 - \beta T), \quad |\alpha| = |\beta| = \sqrt{q}, \end{aligned}$$

and

$$Z\left(E/\mathbb{F}_q; \frac{1}{qT}\right) = Z(E/\mathbb{F}_q; T).$$

PROOF. For a 2×2 matrix M , $\det(1 - M) = 1 - \mathrm{tr} M + \det M$. By (5.3.4) and Proposition 4.6.7, and since $\det \mathbf{F}_\ell^n = q^n$,

$$\#E(\mathbb{F}_{q^n}) = \deg(1 - \mathbf{F}^n) = \det(1 - \mathbf{F}_\ell^n) = 1 - \mathrm{tr}(\mathbf{F}_\ell^n) + q^n.$$

So, working in $\mathbb{Q}_\ell[[T]]$ and applying Lemma 5.3.4 to 1, q and $\mathbf{F}_\ell$,

$$Z(E/\mathbb{F}_q; T) = \exp\left(\sum_{n \geq 1} (1 + q^n) \frac{T^n}{n}\right) \exp\left(-\sum_{n \geq 1} \mathrm{tr}(\mathbf{F}_\ell^n) \frac{T^n}{n}\right) = \frac{\det(1 - \mathbf{F}_\ell T)}{(1 - T)(1 - qT)}.$$

By Theorem 5.3.8, $\det(1 - F_\ell T) = T^2 \det(T^{-1} - F_\ell) = 1 - aT + qT^2 = (1 - \alpha T)(1 - \beta T)$, and $|\alpha| = |\beta| = \sqrt{q}$. For the functional equation, substitute $T \mapsto 1/(qT)$ and multiply numerator and denominator by qT^2 :

$$\frac{1 - a/(qT) + 1/(qT^2)}{(1 - 1/(qT))(1 - 1/T)} = \frac{qT^2 - aT + 1}{(qT - 1)(T - 1)} = Z(E/\mathbb{F}_q; T)$$

[Sil09, Theorem V.2.4]. □

So E satisfies Theorem 5.3.5 with $g = 1$ and $P(T) = 1 - aT + qT^2 = \det(1 - F_\ell T)$.

COROLLARY 5.3.10. *Let $E/\mathbb{F}_q$ be an elliptic curve with trace of Frobenius a . Put $a_n := q^n + 1 - \#E(\mathbb{F}_{q^n})$ for $n \geq 1$, and $a_0 := 2$. Then $a_1 = a$ and*

$$a_{n+2} = a a_{n+1} - q a_n \quad \text{for all } n \geq 0.$$

In particular, the single number $\#E(\mathbb{F}_q)$ determines $\#E(\mathbb{F}_{q^n})$ for every $n \geq 1$.

PROOF. By (5.3.5), $a_n = \alpha^n + \beta^n$ for $n \geq 1$, and also for $n = 0$. Since α is a root of $T^2 - aT + q$, $\alpha^{n+2} = a\alpha^{n+1} - q\alpha^n$, and likewise for β ; add the two [Sil09, Exercise 5.13]. □

For instance $a_2 = a^2 - 2q$, so $\#E(\mathbb{F}_{q^2}) = q^2 + 1 - a^2 + 2q$.

REMARK 5.3.11. To see why part 3 of Theorem 5.3.5 is called the Riemann hypothesis, set $T = q^{-s}$ as in Section 5.3.1:

$$\zeta_E(s) := Z(E/\mathbb{F}_q; q^{-s}) = \frac{1 - aq^{-s} + q^{1-2s}}{(1 - q^{-s})(1 - q^{1-s})}.$$

The functional equation becomes $\zeta_E(s) = \zeta_E(1 - s)$, and if $\zeta_E(s) = 0$, then $q^s \in \{\alpha, \beta\}$, so $|q^s| = \sqrt{q}$, that is, $\operatorname{Re} s = 1/2$ [Sil09, Remark V.2.5].

EXAMPLE 5.3.12. Let $E: y^2 + y = x^3 - x$ over $\mathbb{F}_2$, the reduction modulo 2 of the curve with LMFDB label 37.a1, which has good reduction away from 37 [LMFDB]. For $x = 0$ and $x = 1$ the equation reads $y^2 + y = 0$, which has the two solutions $y = 0, 1$. With the point at infinity, $\#E(\mathbb{F}_2) = 5$, so $a = 2 + 1 - 5 = -2$ and

$$Z(E/\mathbb{F}_2; T) = \frac{1 + 2T + 2T^2}{(1 - T)(1 - 2T)}.$$

The roots of $T^2 + 2T + 2$ are $\alpha, \beta = -1 \pm i$, of absolute value $\sqrt{2}$. Since $\alpha^2 = -2i$, $\alpha^3 = 2 + 2i$ and $\alpha^4 = -4$, (5.3.5) predicts

$$\#E(\mathbb{F}_4) = 5, \quad \#E(\mathbb{F}_8) = 9 - 4 = 5, \quad \#E(\mathbb{F}_{16}) = 17 + 8 = 25.$$

Magma agrees [BCP97]. It writes the denominator of the zeta function expanded.

```
E := EllipticCurve([GF(2) | 0, 0, 1, -1, 0]);
TraceOfFrobenius(E); // -2
[#BaseChange(E, GF(2^n)) : n in [1..4]]; // [ 5, 5, 5, 25 ]
Z := ZetaFunction(E); K<T> := Parent(Z); Z;
// (2*T^2 + 2*T + 1)/(2*T^2 - 3*T + 1)
```

EXAMPLE 5.3.13. Let $E: y^2 = x^3 + x$ over $\mathbb{F}_3$. The values of $x^3 + x$ at $x = 0, 1, 2$ are 0, 2, 1, and 2 is not a square in $\mathbb{F}_3$, so $E(\mathbb{F}_3) = \{O, (0, 0), (2, 1), (2, 2)\}$. Hence $a = 0$,

$$Z(E/\mathbb{F}_3; T) = \frac{1 + 3T^2}{(1 - T)(1 - 3T)},$$

$\alpha, \beta = \pm i\sqrt{3}$, and $\#E(\mathbb{F}_{3^n}) = 3^n + 1$ for n odd and $3^n + 1 - 2(-3)^{n/2}$ for n even. Part 2 of Theorem 5.3.8 says that $F^2 = [-3]$: applying the 9-th power map to the coordinates of a point of $E(\overline{\mathbb{F}}_3)$ is the same as multiplying it by -3 . We check this on a random point of $E(\mathbb{F}_{3^5})$.

```
E := EllipticCurve([GF(3) | 1, 0]);
```

```

TraceOfFrobenius(E);           // 0
[#BaseChange(E, GF(3^n)) : n in [1..4]]; // [ 4, 16, 28, 64 ]
E5 := BaseChange(E, GF(3^5));
P := Random(E5);
E5![P[1]^9, P[2]^9, P[3]^9] eq -3*P; // true

```

By Corollary 5.5.6, an elliptic curve over $\mathbb{F}_q$ is supersingular if and only if $a \equiv 0 \pmod{p}$; so this curve is supersingular.

EXAMPLE 5.3.14. Let E be 37.a1 again, now reduced modulo the prime $q = 1013$: $E: y^2 + y = x^3 + 1012x$ over $\mathbb{F}_{1013}$. Magma computes $\#E(\mathbb{F}_{1013}) = 978$, so $a = 36$ and

$$Z(E/\mathbb{F}_{1013}; T) = \frac{1 - 36T + 1013T^2}{(1 - T)(1 - 1013T)}.$$

The roots of $T^2 - 36T + 1013$ are $18 \pm \sqrt{689}i \approx 18 \pm 26.2488i$, of absolute value $\sqrt{1013} \approx 31.8277$. Then Corollary 5.3.10 predicts $\#E(\mathbb{F}_{1013^2}) = q^2 + 1 - (a^2 - 2q) = 1026900$.

```

q := 1013;
E := EllipticCurve([GF(q) | 0, 0, 1, -1, 0]);
#E, TraceOfFrobenius(E);           // 978 36
#BaseChange(E, GF(q^2));           // 1026900
Z := ZetaFunction(E); K<T> := Parent(Z); Z;
// (1013*T^2 - 36*T + 1)/(1013*T^2 - 1014*T + 1)

```

PROPOSITION 5.3.15. *Let E_1 and E_2 be elliptic curves over $\mathbb{F}_q$. The following are equivalent:*

- (1) E_1 and E_2 are isogenous over $\mathbb{F}_q$;
- (2) $\#E_1(\mathbb{F}_q) = \#E_2(\mathbb{F}_q)$;
- (3) $Z(E_1/\mathbb{F}_q; T) = Z(E_2/\mathbb{F}_q; T)$;
- (4) E_1 and E_2 have the same number of closed points of each degree.

PROOF. (See [Sil09, Exercise 5.4].) Let F_1 and F_2 be the q -th power Frobenius endomorphisms of E_1 and E_2 . Parts 2, 3 and 4 of Proposition 5.3.15 are equivalent by Theorem 5.3.9 (the zeta function depends only on q and a) and by parts 1 and 3 of Proposition 5.3.2.

Suppose that $\varphi: E_1 \rightarrow E_2$ is a non-zero isogeny over $\mathbb{F}_q$. Its coordinate functions have coefficients in $\mathbb{F}_q$, which the q -th power map fixes, so $\varphi \circ F_1 = F_2 \circ \varphi$, and hence $\varphi \circ (1 - F_1) = (1 - F_2) \circ \varphi$. Taking degrees and cancelling $\deg \varphi \neq 0$, (5.3.4) gives $\#E_1(\mathbb{F}_q) = \#E_2(\mathbb{F}_q)$.

Conversely, suppose that $\#E_1(\mathbb{F}_q) = \#E_2(\mathbb{F}_q)$, so that $F_{1,\ell}$ and $F_{2,\ell}$ have the same characteristic polynomial $f(T) = T^2 - aT + q$. By Tate's theorem, mentioned after Theorem 4.5.5, the map $\text{Hom}(E_1, E_2) \otimes \mathbb{Z}_\ell \rightarrow \text{Hom}_{\text{Gal}_{\mathbb{F}_q}}(T_\ell(E_1), T_\ell(E_2))$ is an isomorphism [Sil09, Theorem III.7.7(a)]. Since σ acts on $E_i(\overline{\mathbb{F}_q})$ as F_i and topologically generates $\text{Gal}_{\mathbb{F}_q}$, a $\mathbb{Z}_\ell$ -linear map is $\text{Gal}_{\mathbb{F}_q}$ -equivariant if and only if it intertwines $F_{1,\ell}$ and $F_{2,\ell}$. It suffices to find a non-zero such map, since then $\text{Hom}(E_1, E_2) \otimes \mathbb{Z}_\ell \neq 0$, and $\text{Hom}(E_1, E_2)$ is a free $\mathbb{Z}$ -module (Theorem 4.5.5), so it is non-zero.

- If $a^2 \neq 4q$, then $F_{i,\ell}$ is not a scalar, so $V_i := T_\ell(E_i) \otimes \mathbb{Q}_\ell$ has a basis $v, F_{i,\ell}v$ and $V_i \cong \mathbb{Q}_\ell[T]/(f)$, with T acting by $F_{i,\ell}$. An isomorphism $V_1 \cong V_2$ of $\mathbb{Q}_\ell[T]$ -modules, multiplied by a large power of ℓ , maps $T_\ell(E_1)$ into $T_\ell(E_2)$.
- If $a^2 = 4q$, then $f(T) = (T - a/2)^2$ with $a/2 \in \mathbb{Z}$, and by Proposition 4.6.7, $\deg(F_i - [a/2]) = \det(F_{i,\ell} - a/2) = f(a/2) = 0$. So $F_i = [a/2]$, and every $\mathbb{Z}_\ell$ -linear map intertwines the two.

□

5.4. The endomorphism ring

In this section and the next, k is a field of characteristic $p > 0$, not necessarily finite, and E is an elliptic curve over k . We use the Frobenius isogeny $F^r: E \rightarrow E^{(p^r)}$ and its Verschiebung $V^r = \widehat{F^r}$ of Definition 4.3.12.

Over $\mathbb{F}_q$ with $q = p^e$ the two readings agree: $E^{(p^e)} = E$, and the q -th power Frobenius endomorphism is F^e , with dual V^e .

Corollary 4.5.1 left two possibilities for $E[p^r](\bar{k})$, and Theorem 4.7.8 three possibilities for $\text{End}(E_{\bar{k}})$. The following theorem of Deuring says that these are not independent: the p -torsion decides the endomorphism ring.

THEOREM 5.4.1 (Supersingular and ordinary elliptic curves). *Let k be a field of characteristic p , and let E be an elliptic curve over k . The following conditions are equivalent.*

- (1) $E[p^r](\bar{k}) = 0$ for some (all) $r \geq 1$.
- (2) V^r is purely inseparable for some (all) $r \geq 1$.
- (3) $[p]: E \rightarrow E$ is purely inseparable and $j(E) \in \mathbb{F}_{p^2}$.
- (4) $\text{End}(E_{\bar{k}})$ is an order in a quaternion algebra.
- (5) The formal group $\hat{E}$ has height 2.

If they hold, E is **supersingular**. Otherwise E is **ordinary**, and then

$$E[p^r](\bar{k}) \cong \mathbb{Z}/p^r\mathbb{Z} \quad \text{for all } r \geq 1;$$

if moreover $j(E) \in \mathbb{F}_{p^2}$, then $\text{End}(E_{\bar{k}})$ is an order in an imaginary quadratic field.

PROOF. This is [Sil09, Theorem V.3.1], due to Deuring [Deu41]. Two of the conditions are already familiar. For conditions 1 and 2, the proof of Corollary 4.5.1 gives $\#E[p^r](\bar{k}) = \deg_s[p^r] = \deg_s V^r$, because F^r is purely inseparable and $[p^r] = V^r \circ F^r$; so the p^r -torsion vanishes exactly when V^r has separable degree 1. Condition 4 is case 3 of Theorem 4.7.8, and the ordinary case is case 2; compare Remark 4.7.12. $\square$

By condition 3 of Theorem 5.4.1 there are only finitely many supersingular elliptic curves over $\mathbb{F}_p$ up to isomorphism, since each has j -invariant in the finite field $\mathbb{F}_{p^2}$. Counting them is the subject of Theorem 5.5.9.

5.5. The Cartier operator and the Hasse invariant

None of the conditions of Theorem 5.4.1 is easy to check on a given Weierstrass equation. The Cartier operator turns the dichotomy into the vanishing of a single coefficient.

Throughout this section k is a **perfect** field of characteristic $p > 0$, so that $a \mapsto a^p$ is an automorphism of k and we may write $a^{1/p}$ for its inverse. Perfectness is also what makes the definition below work at every closed point: it guarantees $dt \neq 0$ for a uniformizer t , which fails over an imperfect field (Warning 3.4.5).

DEFINITION 5.5.1. Let C be a nice curve of genus $g \geq 1$ over a perfect field k of characteristic p , let P be a closed point of C and let t be a uniformizer at P . The functions $1, t, \dots, t^{p-1}$ form a p -basis of $\mathcal{O}_{C,P}$: every differential regular at P is uniquely

$$\omega = \left(\sum_{j=0}^{p-1} f_j^p t^j \right) dt, \quad f_j \in \mathcal{O}_{C,P}.$$

The **Cartier operator** is

$$\mathcal{C}: \Omega_{k(C)/k} \rightarrow \Omega_{k(C)/k}, \quad \mathcal{C}(\omega) := f_{p-1} dt.$$

The value $\mathcal{C}(\omega)$ does not depend on the uniformizer, the local definitions agree on overlaps, and the result extends from differentials regular at P to all of $\Omega_{k(C)/k}$; see [AH17, Section 2.2], which follows [Ser58, Section 10]. The operator is due to Cartier [Car57]. There is a second description, the one used in computations: if

$x \in k(C)$ is a separating element, so that dx is a basis of $\Omega_{k(C)/k}$ (Proposition 3.4.2), then every differential is uniquely

$$(5.5.1) \quad \omega = \left(\sum_{j=0}^{p-1} g_j^p x^j \right) dx, \quad g_j \in k(C), \quad \text{and} \quad \mathcal{C}(\omega) = g_{p-1} dx.$$

PROPOSITION 5.5.2. *Let C be a nice curve of genus $g \geq 1$ over a perfect field k of characteristic p .*

- (1) $\mathcal{C}(\omega_1 + \omega_2) = \mathcal{C}(\omega_1) + \mathcal{C}(\omega_2)$.
- (2) $\mathcal{C}(f^p \omega) = f \mathcal{C}(\omega)$ for $f \in k(C)$. In particular $\mathcal{C}(c\omega) = c^{1/p} \mathcal{C}(\omega)$ for $c \in k$, so $\mathcal{C}$ is semilinear for the inverse of the p -th power map on k .
- (3) $\mathcal{C}(df) = 0$ for $f \in k(C)$.
- (4) If ω is regular, then so is $\mathcal{C}(\omega)$. So $\mathcal{C}$ restricts to an operator on the space of regular differentials of C , which has dimension g .

PROOF. Parts 1 to 3 are immediate from (5.5.1), and part 4 is the statement that $\mathcal{C}$ acts on $H^0(C, \Omega_{C/k})$; see [AH17, Section 2.2] and [Ser58, Section 10]. Dividing by a fixed non-zero regular differential identifies the space of regular differentials with $\mathcal{L}(K_C)$, whose dimension is the genus (Definition 3.5.3). $\square$

Note that the semilinearity in part 2 is for the inverse of the p -th power map, not for the Frobenius automorphism $\sigma: x \mapsto x^q$ of Section 5.1; the two agree only when $q = p$.

For an elliptic curve the space of regular differentials is one-dimensional, spanned by the invariant differential ω , so $\mathcal{C}$ acts on it by a single scalar. That scalar is the Hasse invariant.

DEFINITION 5.5.3. Let q be a power of $p \geq 3$ and let E be an elliptic curve over $\mathbb{F}_q$, given by $y^2 = f(x)$ with $f \in \mathbb{F}_q[x]$ a cubic with distinct roots in $\overline{\mathbb{F}}_q$ (Lemma 4.1.4), with invariant differential $\omega = dx/(2y)$. Write

$$f(x)^{(p-1)/2} = \sum_{m \geq 0} c_m x^m.$$

The [Hasse invariant](#) of the equation is $A := c_{p-1}$, the coefficient of x^{p-1} in $f(x)^{(p-1)/2}$.

The Hasse invariant is exactly the scalar by which the Cartier operator acts. Indeed, using $y^2 = f(x)$,

$$\frac{dx}{y} = \frac{(y^2)^{(p-1)/2}}{y^p} dx = y^{-p} f(x)^{(p-1)/2} dx = y^{-p} \left(\sum_{m \geq 0} c_m x^m \right) dx,$$

so part 2 of Proposition 5.5.2 and (5.5.1) give

$$(5.5.2) \quad \mathcal{C}\left(\frac{dx}{y}\right) = \frac{1}{y} \mathcal{C}\left(\sum_{m \geq 0} c_m x^m dx\right) = A^{1/p} \frac{dx}{y}.$$

Only the terms with $m \equiv p-1 \pmod{p}$ survive, and $f^{(p-1)/2}$ has degree $3(p-1)/2 < 2p-1$, so $m = p-1$ is the only one. Since $2 \in \mathbb{F}_p$ is fixed by $c \mapsto c^{1/p}$, the same scalar appears for $\omega = dx/(2y)$: $\mathcal{C}(\omega) = A^{1/p} \omega$. This is the genus-1 case of the computation in [AH17, Section 3].

REMARK 5.5.4. Three warnings. First, A depends on the equation and not only on E : the change of coordinates $(x, y) \mapsto (u^2 x, u^3 y)$ of (4.1.3) replaces A by $u^{-(p-1)} A$. What is intrinsic is whether A vanishes, equivalently whether $\mathcal{C} = 0$ on the regular differentials of E . Second, the definition asks for $p \geq 3$, although the Cartier operator itself does not; over $\overline{\mathbb{F}}_2$ the only supersingular elliptic curve is $y^2 + y = x^3$ [Sil09, Section V.4]. Third, Silverman [Sil09] uses "Hasse invariant" for the binary invariant of Theorem 5.4.1, which is 0 for a supersingular curve and 1 for an ordinary one; here it is the scalar A attached to an equation.

The proof that A detects supersingularity counts points with the quadratic character.

LEMMA 5.5.5. Let q be a power of $p \geq 3$, let $\chi: \mathbb{F}_q^\times \rightarrow \{\pm 1\}$ be the unique character of order 2, extended by $\chi(0) := 0$, and let $E/\mathbb{F}_q$ be given by $y^2 = f(x)$ as in Definition 5.5.3. Then

$$(5.5.3) \quad \#E(\mathbb{F}_q) = 1 + q + \sum_{x \in \mathbb{F}_q} \chi(f(x)).$$

PROOF. For a fixed $x \in \mathbb{F}_q$, the number of $y \in \mathbb{F}_q$ with $y^2 = f(x)$ is 2 if $f(x)$ is a non-zero square, 1 if $f(x) = 0$ and 0 otherwise, that is, $1 + \chi(f(x))$. Summing over the q values of x and adding the point at infinity gives (5.5.3) [Sil09, Application V.1.3]. $\square$

COROLLARY 5.5.6. Let E be an elliptic curve over $\mathbb{F}_q$, $q = p^e$, with trace of Frobenius a (Definition 5.3.7). Then E is supersingular if and only if $a \equiv 0 \pmod{p}$.

PROOF. Write $F \in \text{End}(E)$ for the q -th power Frobenius endomorphism. It is the e -fold composite of the p -th power Frobenius, so its dual is V^e in the notation of Definition 4.3.12. By part 2 of Theorem 5.3.8 and $\widehat{F} \circ F = [q]$,

$$(F - [a] + \widehat{F}) \circ F = F^2 - aF + q = 0,$$

and $\text{End}(E)$ has no zero divisors (Proposition 4.3.7) with $F \neq 0$, so $V^e = \widehat{F} = [a] - F$. By Corollary 4.4.4, $[a] - F$ is separable if and only if $p \nmid a$. Finally $\deg_s V^e = \#E[p^e](\overline{k}) \in \{1, p^e\}$ by part 2 of Corollary 4.5.1, so V^e is inseparable if and only if it is purely inseparable, and part 2 of Theorem 5.4.1, with $r = e$, finishes the proof [Sil09, Exercise 5.10]. $\square$

THEOREM 5.5.7. Let q be a power of $p \geq 3$ and let E be an elliptic curve over $\mathbb{F}_q$, given by $y^2 = f(x)$ as in Definition 5.5.3, with Hasse invariant A . For $r \geq 1$, let A_{p^r} be the coefficient of x^{p^r-1} in $f(x)^{(p^r-1)/2}$, so that $A_p = A$. The following are equivalent.

- (1) E is supersingular.
- (2) $A_q = 0$.
- (3) $A = 0$.
- (4) $\mathcal{C} = 0$ on the regular differentials of E .

PROOF. This is [Sil09, Theorem V.4.1(a)]; the argument is Manin's [Man61].

Since $\mathbb{F}_q^\times$ is cyclic of order $q - 1$, we have $\chi(z) = z^{(q-1)/2}$ in $\mathbb{F}_q$ for every $z \in \mathbb{F}_q$, including $z = 0$. So Lemma 5.5.5 reads

$$\#E(\mathbb{F}_q) = 1 + \sum_{x \in \mathbb{F}_q} f(x)^{(q-1)/2} \quad \text{in } \mathbb{F}_q,$$

using $q = 0$ in $\mathbb{F}_q$. Again because $\mathbb{F}_q^\times$ is cyclic, $\sum_{x \in \mathbb{F}_q} x^n$ is -1 when $n > 0$ and $q - 1 \mid n$, and 0 otherwise. The polynomial $f(x)^{(q-1)/2}$ has degree $3(q-1)/2 < 2(q-1)$, so the only surviving term is the one with $n = q - 1$, and $\#E(\mathbb{F}_q) = 1 - A_q$ in $\mathbb{F}_q$. Comparing with $\#E(\mathbb{F}_q) = q + 1 - a$ gives $a \equiv A_q \pmod{p}$, so conditions 1 and 2 are equivalent by Corollary 5.5.6.

For conditions 2 and 3, write

$$f(x)^{(p^{r+1}-1)/2} = \left(f(x)^{(p^r-1)/2}\right)^p f(x)^{(p-1)/2}.$$

A term of the right-hand side contributing to $x^{p^{r+1}-1}$ is x^{pn} times x^m with $pn + m = p^{r+1} - 1$, so $m \equiv p - 1 \pmod{p}$; since $m \leq 3(p-1)/2 < 2p - 1$, the only possibility is $m = p - 1$, and then $n = p^r - 1$. Hence $A_{p^{r+1}} = A_{p^r}^p A$, and by induction $A_q = 0$ if and only if $A = 0$.

For conditions 3 and 4, the regular differentials of E are the $\mathbb{F}_q$ -multiples of ω , and $\mathcal{C}(\omega) = A^{1/p}\omega$ by (5.5.2). $\square$

EXAMPLE 5.5.8. Let $E: y^2 = x^3 - x$, and write $m := (p-1)/2$. Then $f^{(p-1)/2} = x^m(x^2-1)^m$, whose exponents are all congruent to m modulo 2. So the coefficient A of $x^{p-1} = x^{2m}$ vanishes unless m is even, that is,

unless $p \equiv 1 \pmod{4}$; and when $p \equiv 1 \pmod{4}$ the one surviving term gives $A = (-1)^{m/2} \binom{m}{m/2}$, which is non-zero in $\mathbb{F}_p$ because $\binom{m}{m/2}$ is a quotient of factorials of integers smaller than p . Hence E is supersingular exactly when $p \equiv 3 \pmod{4}$. For $p = 11$ we get $A = 0$, and for $p = 13$, $A = (-1)^3 \binom{6}{3} = -20 = 6$ in $\mathbb{F}_{13}$. Magma agrees [BCP97], and lists the supersingular j -invariants of each characteristic; note $1728 = 1$ in $\mathbb{F}_{11}$.

```
for p in [11, 13] do
  P<x> := PolynomialRing(GF(p));
  f := x^3 - x;
  E := EllipticCurve([GF(p) | -1, 0]);
  A := Coefficient(f^((p-1) div 2), p-1);
  print p, A, IsSupersingular(E);
  print [ j : j in GF(p^2) | IsSupersingular(EllipticCurveFromjInvariant(j)) ];
end for;
// 11 0 true
// [ 0, 1 ]
// 13 6 false
// [ 5 ]
```

Both lists are as short as part 3 of Theorem 5.4.1 promises, and they satisfy the mass formula of Theorem 5.5.9: in characteristic 11 the two supersingular curves have $j = 0$ and $j = 1728$, so $\# \text{Aut}(E)$ is 6 and 4 by Theorem 4.7.11, and $1/6 + 1/4 = 5/12 = (11 - 1)/24$; in characteristic 13 the only one has $j = 5$, with $\# \text{Aut}(E) = 2$ and $1/2 = (13 - 1)/24$.

We close with a count of the supersingular curves, weighted by their automorphisms.

THEOREM 5.5.9 (Mass formula). *Let p be a prime. Then*

$$(5.5.4) \quad \sum_E \frac{1}{\# \text{Aut}(E)} = \frac{p-1}{24},$$

where E runs over the isomorphism classes of supersingular elliptic curves over $\overline{\mathbb{F}}_p$.

PROOF. Due to Deuring [Deu41]; see [Voi21, (42.3.9)] and [Sil09, Exercise 5.9]. Remark 5.5.10 explains what a proof needs. $\square$

REMARK 5.5.10. The mass formula does not follow from Theorems 5.4.1 and 5.5.7 alone: those say which curves are supersingular, not how many there are. Two routes to (5.5.4) are worth recording.

The first is analytic, and is the one that explains the denominator. By part condition 4 of Theorem 5.4.1 the endomorphism ring of a supersingular curve is an order in a quaternion algebra, and Deuring's correspondence [Voi21, Corollary 42.3.7] makes this precise: the supersingular elliptic curves over $\overline{\mathbb{F}}_p$, up to isomorphism, are in bijection with the class set of a maximal order $\mathcal{O}$ in the definite quaternion algebra over $\mathbb{Q}$ ramified exactly at p and ∞ , and under the bijection $\text{Aut}(E)$ is the unit group of the corresponding order. The count on the quaternionic side is the Eichler mass formula [Voi21, Theorem 25.1.1]: for a maximal order in a definite quaternion algebra over $\mathbb{Q}$ of discriminant D ,

$$\sum_{[J]} \frac{1}{w_J} = \frac{\varphi(D)}{12}, \quad w_J := \#(\mathcal{O}_L(J)^\times / \{\pm 1\}),$$

which for $D = p$ gives $(p-1)/12$; the quotient by $\{\pm 1\}$ accounts for the factor 2 between that and (5.5.4). The Eichler mass formula is proved by analytic means, from the analytic class number formula and the zeta function of the order [Voi21, Main Theorem 26.1.5 and Section 26.2]; over $\mathbb{Q}$ it goes back to Hey and to Brandt.

The second route is a count. The supersingular j -invariants in characteristic p are the roots of an explicit polynomial, and there are $\lfloor p/12 \rfloor + \varepsilon$ of them, with $\varepsilon \in \{0, 1, 2\}$ depending on $p \pmod{12}$ [Sil09, Theorem V.4.1]; this is due to Igusa [Igu58]. Weighting each by $\# \text{Aut}(E)$, which is 2, 4 or 6 according as $j(E) \neq 0, 1728$,

$j(E) = 1728$ or $j(E) = 0$ when $p \geq 5$ (Theorem 4.7.11), and checking $p = 2, 3$ by hand, gives (5.5.4) case by case.

CHAPTER 6

Elliptic curves over local fields

CHAPTER 7

Galois cohomology

CHAPTER 8

The Mordell–Weil theorem

CHAPTER 9

Integral points

CHAPTER 10

Twisting, Selmer, and Tate–Shafarevich

CHAPTER 11

Complex multiplication

CHAPTER 12

The Tate curve

CHAPTER 13

The L -function and BSD

CHAPTER 14

The Sato–Tate conjecture

CHAPTER 15

Introduction to modular curves

Glossary

- affine k -variety:** A scheme over k isomorphic to $\operatorname{Spec} A$ for some k -algebra A of finite type. p. 25
- affine n -space:** The k -variety $\mathbb{A}_k^n = \operatorname{Spec} k[x_1, \dots, x_n]$, with $\mathbb{A}^n(R) = R^n$ for every k -algebra R . p. 25
- b -invariants:** The polynomials b_2, b_4, b_6, b_8 in the coefficients of a Weierstrass equation, from which Δ is built. p. 43
- base extension:** For a k -variety X and a field $L \supseteq k$, the L -variety $X_L = X \times_{\operatorname{Spec} k} \operatorname{Spec} L$, defined by the same equations as X . p. 28
- c -invariants:** The polynomials $c_4 = b_2^2 - 24b_4$ and $c_6 = -b_2^3 + 36b_2b_4 - 216b_6$ of a Weierstrass equation; they satisfy $1728\Delta = c_4^3 - c_6^2$. p. 43
- canonical divisor:** The divisor of a non-zero differential on a nice curve. p. 36
- Cartier operator:** The additive operator $\mathcal{C}$ on the differentials of a curve over a perfect field of characteristic p with $\mathcal{C}(f^p\omega) = f\mathcal{C}(\omega)$, given locally by $\mathcal{C}(\sum_j f_j^p t^j dt) = f_{p-1} dt$. p. 76
- chart:** A homeomorphism from an open subset of a topological space onto an open subset of $\mathbb{C}$. p. 11
- closed point:** A point P of a k -variety whose residue field $k(P)$ is finite over k (for varieties, equivalently $\{P\}$ is closed); its degree is $[k(P) : k]$. p. 31
- compatible charts:** Two charts whose domains are disjoint, or whose transition maps are holomorphic. p. 11
- complex structure:** A family of pairwise compatible charts whose domains cover the space. p. 11
- complex torus:** The quotient $\mathbb{C}/\Lambda$ of $\mathbb{C}$ by a lattice Λ : a compact Riemann surface that is also an abelian group. p. 14
- connected:** A topological space that is not the union of two disjoint non-empty open subsets. p. 11
- cubic twist:** For $d \in k^\times$, the Selmer curve C_d , regarded as a twist of the Fermat cubic C_1 ; its isomorphism class depends only on d in $k^\times/k^{\times 3}$. p. 68
- curve:** An integral variety of dimension 1. p. 31
- cyclic isogeny:** An isogeny of complex tori whose kernel is a cyclic group. p. 16
- definite quaternion algebra:** A quaternion algebra B over $\mathbb{Q}$ with $B \otimes_{\mathbb{Q}} \mathbb{R}$ isomorphic to Hamilton's quaternions; $(a, b \mid \mathbb{Q})$ is definite exactly when $a, b < 0$. p. 62
- degree of a morphism:** For a non-constant morphism of curves, the degree $[k(C_1) : \varphi^*k(C_2)]$ of the extension of function fields. p. 32
- degree of an isogeny of complex tori:** The order of its kernel; for $z \mapsto \alpha z$ from $\mathbb{C}/\Lambda_1$ to $\mathbb{C}/\Lambda_2$ it is the index $[\Lambda_2 : \alpha\Lambda_1]$. p. 17
- differential:** An element of $\Omega_{k(C)/k}$, the $k(C)$ -vector space generated by the dx subject to the Leibniz rules and $da = 0$ for $a \in k$. p. 36
- discriminant:** The polynomial Δ in the coefficients of a Weierstrass equation that vanishes exactly when the cubic is singular; $\Delta = -16(4A^3 + 27B^2)$ for $y^2 = x^3 + Ax + B$. p. 43
- divisor:** A finite formal integer combination of closed points of a curve. p. 34
- dual isogeny:** For an isogeny φ of degree d , the unique isogeny $\hat{\varphi}$ with $\hat{\varphi} \circ \varphi = [d]$. p. 54
- dual isogeny of complex tori:** For an isogeny $z \mapsto \alpha z$ of complex tori of degree d , the isogeny $z \mapsto (d/\alpha)z$ in the opposite direction; both composites are multiplication by d . p. 17
- Eisenstein integers:** The ring $\mathbb{Z}[\omega]$, where ω is a primitive cube root of 1. p. 14
- Eisenstein series:** For a lattice Λ and even $k > 2$, the sum $G_k(\Lambda)$ of ω^{-k} over the non-zero $\omega \in \Lambda$. p. 22
- elliptic curve:** A pair (E, O) of a nice curve E of genus 1 over k and a rational point $O \in E(k)$. p. 43
- finite type:** A scheme over k covered by finitely many open subschemes $\operatorname{Spec} A$, each with A a finitely generated k -algebra. p. 25
- flex:** A smooth $\bar{k}$ -point of a plane cubic whose tangent line meets the cubic there with multiplicity 3. p. 44

- free:** An abelian group that has a $\mathbb{Z}$ -basis. p. 13
- Frobenius automorphism:** For a finite field $\mathbb{F}_q$, the automorphism $x \mapsto x^q$ of $\overline{\mathbb{F}_q}$; it topologically generates $\text{Gal}(\overline{\mathbb{F}_q}/\mathbb{F}_q)$. p. 69
- function field:** For an integral variety X , the local ring $k(X)$ at its generic point, which is a field; for $X = \text{Spec } A$ it is $\text{Frac } A$. p. 28
- fundamental group:** The group $\pi_1(X, x)$ of homotopy classes of loops in X based at x . p. 13
- fundamental parallelogram:** For a lattice with $\mathbb{Z}$ -basis $\{\omega_1, \omega_2\}$, the half-open parallelogram $\{t_1\omega_1 + t_2\omega_2 : 0 \leq t_1, t_2 < 1\}$. p. 15
- Gaussian integers:** The ring $\mathbb{Z}[i]$. p. 13
- genus:** For a nice curve, $\ell(K_C)$, the dimension of the space of regular differentials. p. 38
- geometrically integral:** A k -variety X such that $X_{\overline{k}}$ is integral. p. 28
- Hasse invariant:** For $p \geq 3$ and an elliptic curve $y^2 = f(x)$ over $\mathbb{F}_q$, the coefficient A of x^{p-1} in $f(x)^{(p-1)/2}$; it satisfies $\mathcal{C}(\omega) = A^{1/p}\omega$, and vanishes exactly for a supersingular curve. p. 77
- Hausdorff:** A topological space in which any two distinct points of the space have disjoint open neighborhoods. p. 11
- holomorphic differential:** A meromorphic differential on a Riemann surface whose local expressions $f_\phi(z) dz$ are all holomorphic. p. 19
- holomorphic map:** A continuous map between Riemann surfaces that is holomorphic when read in charts. p. 12
- homogenization:** The homogeneous polynomial $F(X_0, \dots, X_n) = X_0^{\deg f} f(X_1/X_0, \dots, X_n/X_0)$ attached to $f \in k[x_1, \dots, x_n]$. p. 26
- integral variety:** A k -variety whose rings of sections over non-empty opens are integral domains; equivalently, a reduced and irreducible one. p. 27
- intersection number:** For plane curves $C = V_+(F)$ and $D = V_+(G)$ without common component, $I_P(C, D) = \dim_k \mathcal{O}_{\mathbb{P}^2, P}/(f, g)$. p. 39
- invariant differential:** The differential $\omega = dx/(2y + a_1x + a_3)$ of a Weierstrass equation; it has no zeros or poles and is invariant under translation. p. 43
- invertible module:** A finitely generated projective module of constant rank 1 over a ring R ; equivalently, an R -module P with $P \otimes_R Q \cong R$ for some Q . p. 26
- irreducible variety:** A k -variety whose underlying topological space is non-empty and not the union of two proper closed subsets. p. 27
- isogeny:** A non-zero holomorphic group homomorphism between complex tori. Its algebraic analogue is an isogeny of elliptic curves. p. 15
- isogeny of elliptic curves:** A morphism $\varphi: E_1 \rightarrow E_2$ of elliptic curves with $\varphi(O_1) = O_2$; it is automatically a group homomorphism. p. 53
- isomorphism of complex tori:** An invertible isogeny. p. 15
- isomorphism of Riemann surfaces:** A bijective holomorphic map whose inverse is holomorphic. p. 12
- j -invariant:** The quantity $j = c_4^3/\Delta$ of a smooth Weierstrass equation; it classifies elliptic curves up to isomorphism over $\overline{k}$. p. 43
- k -variety:** A separated scheme of finite type over a field k . p. 25
- lattice:** A subgroup of $\mathbb{C}$, free of rank 2, that spans $\mathbb{C}$ as an $\mathbb{R}$ -vector space. p. 13
- lattice in an algebra:** A finitely generated $\mathbb{Z}$ -submodule of a finite-dimensional $\mathbb{Q}$ -algebra B that spans B over $\mathbb{Q}$. p. 63
- line at infinity:** The line $Z = 0$ of $\mathbb{P}_k^2$, the complement of the affine patch $Z \neq 0$. p. 43
- m -torsion subgroup:** The kernel $(\mathbb{C}/\Lambda)[m]$ of multiplication by m ; it is isomorphic to $(\mathbb{Z}/m\mathbb{Z})^2$. p. 16
- maximal order:** An order not properly contained in another order. p. 63
- meromorphic differential on a Riemann surface:** A family of expressions $f_\phi(z) dz$, one for each chart ϕ , with f_ϕ meromorphic, related on overlaps by the chain rule. p. 19
- meromorphic function:** A holomorphic map from a Riemann surface to $\mathbb{P}^1(\mathbb{C})$ that is not identically ∞ . p. 18
- multiplication by m map:** The isogeny $[m]: \mathbb{C}/\Lambda \rightarrow \mathbb{C}/\Lambda$, $z + \Lambda \mapsto mz + \Lambda$. p. 15
- multiplicative group:** The k -variety $\mathbb{G}_m = \text{Spec } k[x, x^{-1}]$, with $\mathbb{G}_m(R) = R^\times$ for every k -algebra R . p. 26
- nice variety:** A smooth, projective, geometrically integral variety. p. 29

- nonsingular polynomial:** A homogeneous $F \in \mathbb{C}[X, Y, Z]$ such that F and its three partial derivatives have no common zero in $\mathbb{P}^2(\mathbb{C})$. p. 21
- open unit disk:** The Riemann surface $\mathbb{D} = \{z \in \mathbb{C} : |z| < 1\}$. p. 12
- order:** A lattice in a finite-dimensional $\mathbb{Q}$ -algebra that is also a subring. p. 63
- ordinary:** An elliptic curve in characteristic p that is not supersingular; equivalently, one with $E[p](\bar{k}) \cong \mathbb{Z}/p\mathbb{Z}$. p. 76
- Picard group:** The group $\text{Pic } C$ of divisors on a curve C modulo principal divisors. p. 34
- point at infinity:** The rational point $O = (0 : 1 : 0)$, the only $\bar{k}$ -point of a Weierstrass cubic on the line at infinity, where it is a flex. p. 45
- principal divisor:** The divisor $\text{div}(f) = \sum_P \text{ord}_P(f)P$ of a non-zero rational function. p. 34
- projective closure:** The Zariski closure in $\mathbb{P}_k^n$ of a closed subvariety $X = V(I)$ of $\mathbb{A}_k^n = U_0$; it is cut out by the homogeneous ideal generated by the homogenizations of all elements of I . p. 26
- projective plane:** The space $\mathbb{P}^2(\mathbb{C}) = (\mathbb{C}^3 - \{(0, 0, 0)\})/\mathbb{C}^\times$, with points written $(X : Y : Z)$. p. 21
- projective plane curve:** The zero locus $V_+(F) \subseteq \mathbb{P}_k^2$ of a non-constant homogeneous polynomial with no repeated factor over $\bar{k}$. p. 39
- quadratic twist:** For $d \in k^\times$ and $\text{char } k \neq 2$, the curve $E^d: y^2 = x^3 + d^2Ax + d^3B$ attached to $E: y^2 = x^3 + Ax + B$; it has the same j -invariant, and its isomorphism class depends only on d in $k^\times/k^{\times 2}$. p. 46
- quaternion algebra:** An algebra over a field F with basis $1, i, j, ij$ and $i^2 = a, j^2 = b, ji = -ij$ for some $a, b \in F^\times$; written $(a, b | F)$. p. 62
- ramification divisor:** For a separable morphism φ , the divisor $\text{div}(\varphi^*\omega) - \varphi^*\text{div}(\omega)$. p. 38
- ramification index:** For a non-constant morphism of curves, $e_\varphi(P) = \text{ord}_P(\varphi^*t)$ with t a uniformizer at $\varphi(P)$. p. 33
- rank:** The number of elements of a $\mathbb{Z}$ -basis of a free abelian group. p. 13
- rational map:** For an integral variety X , a morphism to Y from a non-empty open subvariety of X , up to agreement on overlaps. p. 28
- rational point:** A point of a k -variety X with values in k , i.e. an element of $X(k) = \text{Hom}_k(\text{Spec } k, X)$. p. 25
- reduced norm:** The element $\text{nrd}(\alpha) = \alpha\bar{\alpha}$ of F , for α in a quaternion algebra over F ; it is multiplicative. p. 62
- reduced trace:** The element $\text{trd}(\alpha) = \alpha + \bar{\alpha}$ of F , for α in a quaternion algebra over F . p. 62
- reduced variety:** A k -variety whose rings of sections have no non-zero nilpotents. p. 27
- regular differential:** A differential ω that is zero or has $\text{div}(\omega) \geq 0$; these form a k -vector space. p. 36
- regular variety:** A variety whose local rings are all regular local rings. p. 29
- relative Frobenius morphism:** For $C \subseteq \mathbb{P}_k^n$ with q a power of $\text{char } k > 0$, the morphism $C \rightarrow C^{(q)}$ raising homogeneous coordinates to the q -th power, where $C^{(q)}$ is defined by the equations of C with coefficients raised to the q -th power. p. 33
- Riemann sphere:** The projective line $\mathbb{P}^1(\mathbb{C})$, a compact Riemann surface covered by two charts. p. 12
- Riemann surface:** A connected Hausdorff topological space with a complex structure. p. 11
- scheme:** A locally ringed space covered by open subsets isomorphic to affine schemes $\text{Spec } A$. p. 25
- second countable:** A topological space with a countable basis for its topology. p. 12
- Selmer elliptic curve:** For $d \in k^\times$ and $\text{char } k \neq 3$, the plane cubic $C_d: X^3 + Y^3 = dZ^3$ with base point $O = (1 : -1 : 0)$. p. 65
- separated:** A scheme X over S whose diagonal $X \rightarrow X \times_S X$ is a closed immersion. p. 25
- simply connected:** A connected topological space whose fundamental group is trivial. p. 13
- smooth projective plane curve:** The zero locus in $\mathbb{P}^2(\mathbb{C})$ of a nonsingular homogeneous polynomial. p. 21
- smooth variety:** A variety which, locally cut out of $\mathbb{A}_k^n$ with pure dimension d , has Jacobian matrix of rank $n - d$ at every $\bar{k}$ -point; equivalently, $X_{\bar{k}}$ is regular. p. 29
- split quaternion algebra:** A quaternion algebra over F isomorphic to $M_2(F)$; equivalently, one that is not a division ring. p. 62
- standard involution:** On a quaternion algebra, the conjugation $t + xi + yj + zij \mapsto t - xi - yj - zij$; in general, an F -linear anti-automorphism of order at most 2 with $\alpha\bar{\alpha} \in F$ for all α . p. 62
- supersingular:** An elliptic curve in characteristic p with $E[p](\bar{k}) = 0$; equivalently, one whose endomorphism ring over $\bar{k}$ is an order in a quaternion algebra. p. 76

- Tate module:** For a prime $\ell \neq \text{char } k$, the $\mathbb{Z}_\ell$ -module $T_\ell(E) = \varprojlim_n E[\ell^n](k^{\text{sep}})$, free of rank 2. p. 58
- trace of Frobenius:** For an elliptic curve $E/\mathbb{F}_q$, the integer $a = q + 1 - \#E(\mathbb{F}_q)$, which is the trace of Frobenius on $T_\ell(E)$. p. 72
- transition map:** For charts ϕ_1 and ϕ_2 , the map $\phi_2 \circ \phi_1^{-1}: \phi_1(U_1 \cap U_2) \rightarrow \phi_2(U_1 \cap U_2)$. p. 11
- translation-invariant differential:** On $\mathbb{C}^\times$ or a complex torus, a meromorphic differential ω with $\tau_a^* \omega = \omega$ for every translation τ_a . p. 19
- twisted cubic:** The affine curve $V(x_2 - x_1^2, x_3 - x_1^3) \subseteq \mathbb{A}_k^3$, the image of $t \mapsto (t, t^2, t^3)$. p. 27
- uniformizer:** A function t with $\text{ord}_P(t) = 1$ at a regular closed point P . p. 31
- upper half plane:** The Riemann surface $\mathbb{H} = \{z \in \mathbb{C} : \text{Im}(z) > 0\}$. p. 12
- Verschiebung:** In characteristic p , the dual $V^r = \widehat{F^r}$ of the p^r -th power Frobenius isogeny $F^r: E \rightarrow E^{(p^r)}$. p. 55
- Weierstrass coordinates:** Functions x, y on an elliptic curve with poles of order 2 and 3 at O only, giving an isomorphism onto a Weierstrass cubic. p. 45
- Weierstrass equation:** A projective plane cubic $Y^2Z + a_1XYZ + a_3YZ^2 = X^3 + a_2X^2Z + a_4XZ^2 + a_6Z^3$. p. 43
- Weierstrass $\wp$ -function:** For a lattice Λ , the meromorphic function $\wp_\Lambda(z) = z^{-2} + \sum_{\omega \in \Lambda - \{0\}} [(z - \omega)^{-2} - \omega^{-2}]$ on $\mathbb{C}$. p. 22
- Weil pairing:** For N not divisible by $\text{char } k$, the bilinear, alternating, non-degenerate, Galois-equivariant pairing $e_N: E[N](k^{\text{sep}}) \times E[N](k^{\text{sep}}) \rightarrow \mu_N(k^{\text{sep}})$. p. 60
- Weil reciprocity:** For non-zero rational functions f, g on a nice curve whose divisors have disjoint support, the identity $f(\text{div } g) = g(\text{div } f)$, where $f(D) = \prod_P f(P)^{n_P}$ for $D = \sum n_P P$. p. 35
- zeta function:** For a nice curve C over $\mathbb{F}_q$, the power series $Z(C/\mathbb{F}_q; T) = \exp\left(\sum_{n \geq 1} \#C(\mathbb{F}_{q^n}) T^n / n\right)$. p. 70

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